Contradiction Fields: CRT Rigidity, Prime Matrix Geometry, and Zeta-Zero Escape

Project manuscript

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Abstract

This monograph draft unifies two contradiction-field research programs. The first studies prime existence in CRT prime matrices through row, column, diagonal, anchor, and rough-composite rigidity. The second studies an off-critical zero of the Riemann zeta function through a global contradiction-field ledger for prime anomalies. The common method is to convert a hypothetical counterexample into a structured covering field, decompose it into local rigidity layers, and exclude all free escape channels by capacity, periodicity, and no-cycle bookkeeping.

This is a synthesis and dependency manuscript. It does not assert that the Riemann Hypothesis or the prime-matrix row/column theorem has already been accepted as a final unconditional theorem. Each claim is marked by status in the accompanying status table: proved in manuscript, reduced to a structured input, external, computationally certified, or requiring independent referee verification.

Contents

Chapter 1

Guiding Principle: The Contradiction Field

A contradiction field is a structured ledger attached to a hypothetical counterexample. Instead of estimating a single exceptional set, it records every possible explanation for the counterexample as a named channel. A proof closes when every channel is either absorbed, forced to descend in a finite potential, transferred with source deletion, or contradicts a global capacity lower/upper bound.

The common rigidities used throughout this manuscript are:

1. CRT skeletons and nonzero-residue balance;
2. local short-window non-reuse of large prime factors;
3. small-prime shell migration and diagonal locking;
4. rough-composite and double-anchor decompositions;
5. capacity ledgers after baseline subtraction;
6. no-cycle potentials for internal escape mechanisms.

Chapter 2

Critical-Density Feedback Bridge

This chapter records the critical-density principle used to organize the current frontier. It is a bridge from heuristic insight to proof obligations, not a replacement for the remaining row/column, two-point, or RH exit proofs.

Definition 2.1 (Self-sieving density model)**.** Let a self-sieving set have local density

$$\rho(x) \sim \frac{C}{(\log x)^{\alpha}}.$$

Its previous elements impose a first-order one-residue sieve intensity

$$S(x) = \sum_{a \le x,\ a \in A} \frac{1}{a} \sim \int_2^x \frac{\rho(t)}{t}\, dt.$$

The associated model survivor density is $\exp(-S(x))$.

Lemma 2.2 (Critical fixed point)**.** *Inside the self-sieving density model, the only logarithmic fixed-point exponent is*

$$\rho(x) \asymp \frac{1}{\log x}.$$

More precisely, if $\rho(x) \sim C/(\log x)^{\alpha}$ and the model survivor density is required to have the same logarithmic order as $\rho(x)$, then $\alpha = 1$; at that layer, exponent matching gives $C = 1$.

Proof. If $\alpha > 1$, then

$$\int_2^x \frac{C\, dt}{t(\log t)^{\alpha}}$$

converges, so the model survivor density stays bounded away from zero while $C/(\log x)^{\alpha}$ tends to zero. This is not self-consistent.

If $\alpha < 1$, then the same integral is

$$\frac{C}{1-\alpha}(\log x)^{1-\alpha} + O(1),$$

so the survivor density is

$$\exp\{-c(\log x)^{1-\alpha} + O(1)\},$$

which is smaller than any fixed negative power of $\log x$. This is again not of order $C/(\log x)^{\alpha}$.

Thus $\alpha = 1$. Then $S(x) \sim C \log\log x$, and the survivor density is $(\log x)^{-C+o(1)}$. Matching the logarithmic exponent with $C/\log x$ forces $C = 1$ at the model level. □

Remark 2.3 (Proof boundary)**.** The lemma is a fixed-point calculation for the self-sieving model. It does not by itself prove a short-interval prime, a prime pair, or RH. To become a theorem, the fixed-point pressure must be upgraded into local lower bounds, new-modulus anti-concentration estimates, or controlled-exit inequalities.

Reduction-closed Statement 2.4 (Critical feedback bridge for the three programs)**.** *The critical-density principle routes the three current frontiers as follows.*

1. *For the prime-matrix row/column program, a fully covered row or column is a local collapse below the $1/\log x$ fixed point. Such a collapse must be explained by fixed CRT concentration, by moving high-factor slots that pay Rankin/SAE mass, or by a failure of the local survivor lower bound. The active narrow gate is*

   $$\mathit{AffineTwinFillResidueArrivalBoundOrFillCatchUpMassPDECExclusion}.$$

2. *For the two-point secondary sieve, the corresponding fixed point is the two-residue product*

   $$\prod_{q \le y} \left(1 - \frac{\nu_q(w)}{q}\right) \asymp \frac{\mathfrak{S}(w)}{(\log y)^2}.$$

   The remaining theorem-level input is not a one-point density estimate but the true-residual new-modulus correlation bound in the I3-Core/TRC package.

3. *For the RH contradiction field, an off-critical zero creates a smooth prime anomaly relative to the same $1/\log x$ fixed point. The anomaly must pass through the recorded sparse, dense, tail, internal, and global exits; the manuscript can be promoted only after those controlled exits are independently verified under one load convention.*

Status and reference. This is a routing statement. The detailed version is archived as

`docs/monograph/three-claims-critical-density-feedback-frontier.md`.

It identifies the next prime-matrix input as a critical fill-arrival anti-overfilling theorem: persistent AffineTwin threshold crossing must either reuse a residue and become a reset/ColumnCRT/PDEC certificate, or introduce new fill residues whose Rankin/SAE mass exceeds the critical-density carrying capacity. That final carrying-capacity inequality is the new proof obligation; it is not asserted here as already proved. □

Reduction-closed Statement 2.5 (Critical fill-arrival pressure packet frontier)**.** *In the AffineTwin branch, write $g = q - 2$, $f = q$, and let A_g, A_f be the generator-side and fill-side residue counts. The safe integer gate is*

$$A_g A_f \le T_q := \lfloor \sqrt{gf} \rfloor.$$

For a threshold-crossing route with final counts A'_g, A'_f, crossing is equivalent to

$$A'_g A'_f > T_q, \qquad \textit{hence} \qquad \frac{(A'_g)^2}{g} \frac{(A'_f)^2}{f} > 1.$$

Thus fill catch-up is not merely a raw residue-arrival event. It is a threshold-relevant pressure packet: for fixed A'_g the minimum fill target is

$$A'_{f,\min}(A'_g) = \left\lfloor \frac{T_q}{A'_g} \right\rfloor + 1.$$

Consequently a persistent row/column counterexample in this branch must enter one of four named exits: repeated-residue reset/ColumnCRT PDEC, fixed pressure packet recurrence PDEC, moving pressure packet SAE/Rankin, or non-sparse AffineTwin pressure demand.

Status and reference. The product gate and the pressure form are algebraic equivalents. The minimum fill target follows by solving $A'_g A'_f > T_q$ in integers. The repeated-residue exit is the previous fill-catchup dichotomy; the fixed-packet exit is the AffineTwin slot phase lock, where the combined modulus $q(q-2)$ exceeds the common phase support; and the moving-packet exit is the existing moving-family SAE/ColumnCRT split. The remaining unproved input is the global pressure-packet carrying ceiling, recorded in

`docs/monograph/prime-matrix-critical-fill-arrival-anti-overfilling-router.md`.

□

Reduction-closed Statement 2.6 (Pressure-packet carrying ceiling). *Let $M_q = A_g A_f$ be the pressure-packet multiplicity for an AffineTwin modulus q, so that q and $q-2$ are prime. The packet SAE mass is*

$$\mu_q = \frac{M_q}{q(q-2)}.$$

If the square-root packet gate

$$M_q^2 \le q(q-2)$$

holds, then

$$\mu_q \le \frac{1}{\sqrt{q(q-2)}} \le \frac{1}{q-2}.$$

Consequently, any standard Brun–Selberg upper-bound sieve of the form

$$\Pi_2(X) := \#\{q \le X : q,\ q-2 \text{ prime}\} \le C\frac{X}{(\log X)^2}$$

implies by partial summation that the moving pressure-packet tail is summable. If the square-root packet gate fails, then

$$M_q^2 > q(q-2),$$

which is exactly the named SuperSqrt/PressureProduct PDEC exit.

Status and reference. The first inequalities are immediate from the square-root gate. The Brun–Selberg bound gives

$$\sum_{\substack{q \ge Q \\ q,\ q-2 \text{ prime}}} \frac{1}{q} \le \frac{\Pi_2(Q)}{Q} + \int_Q^\infty \frac{\Pi_2(t)}{t^2}\,dt \ll \frac{1}{(\log Q)^2} + \frac{1}{\log Q},$$

and $1/(q-2) \ll 1/q$ for $q \ge 5$. Thus all packets satisfying the square-root gate are absorbed by the reciprocal ceiling. The complementary case is the previous pressure-product inequality with product > 1, hence the registered PDEC/ColumnCRT exit. The detailed route is recorded in the pressure-packet carrying-ceiling router under `docs/monograph/`. This still leaves two explicit tasks before final row/column promotion. First, internalize or cite the Brun–Selberg reciprocal ceiling under the monograph's convention. Second, exclude the SuperSqrt/PressureProduct PDEC family. □

Reduction-closed Statement 2.7 (Critical-error frontier). *After the critical-density fixed point has been fixed, the remaining frontier is best measured by a normalized critical error*

$$\mathcal{E} = \frac{\textit{actual load}}{\textit{critical capacity}} - 1.$$

A positive critical error is not allowed to remain an anonymous remainder in the contradiction field. In the prime-matrix AffineTwin branch it is the pressure product

$$\frac{A_g^2}{g}\frac{A_f^2}{f} - 1;$$

in the two-point sieve it is the excess of the large-incidence mean above the Buchstab semiprime transition constant $K(\alpha)$, together with the terminal gap to the threshold 1; in the RH branch it is the off-critical zero load after zero-frequency CRT subtraction, normalized by the capacity of each controlled exit. Thus every positive critical error must become a fixed CRT/PDEC defect, a moving SAE/Rankin or analytic-average packet, a denominator-floor defect, or a named controlled exit.

Status and reference. This is a frontier reduction and audit rule, not a final theorem. The detailed ledger is recorded in

`docs/monograph/three-claims-critical-error-frontier-audit.md`.

It sharpens the current next tasks to three interfaces: a non-PDEC square-root phase-support bound for the prime-matrix pressure packets, a critical denominator-floor check for the two-point BMD-to-TLI transfer, and a common baseline-subtracted load convention for every RH controlled exit. □

Reduction-closed Statement 2.8 (Actual-packet square-root phase support). *In the prime-matrix AffineTwin branch, distinguish the formal product*

$$M_q^{\text{form}} = A_g A_f$$

from the actual non-PDEC packet count N_q. The latter counts only generator–fill packets whose two slot conditions are compatible, whose common phase support is nonempty, and which have not already become a reset or ColumnCRT/PDEC event. For a primitive AffineTwin slot pair with $q \geq 13$, the recorded depth identities give common support width

$$W_q = \frac{q+9}{2}.$$

Since

$$W_q \leq \sqrt{q(q-2)}$$

for all $q \geq 13$, and since a repeated actual packet projection inside this support is by definition a PDEC/ColumnCRT event, every non-PDEC primitive packet family satisfies

$$N_q \leq W_q \leq \sqrt{q(q-2)}.$$

Thus the square-root gate is closed for actual primitive packets. A remaining formal SuperSqrt failure must be either product-accounting looseness (M_q^{form} overcounts N_q), a projection-collision PDEC, or a primitive support escape.

Status and reference. The inequality follows by squaring:

$$4q(q-2)-(q+9)^2 = 3q^2 - 26q - 81 \geq 0$$

for $q \geq 13$. The injection of actual packets into the primitive support is the non-PDEC projection rule: two actual packets with the same primitive projection would be a repeated fixed CRT phase and therefore a reset/ColumnCRT/PDEC event. The detailed audit is archived in

`docs/monograph/prime-matrix-nonpdec-sqrt-phase-support-reduction.md`.

This does not yet prove the row/column theorem. It reduces the remaining Prime-Matrix obligation to product-accounting tightening and to excluding or routing primitive-support escape packets. □

Reduction-closed Statement 2.9 (Formal envelope versus actual load)**.** *Across the three programs, a critical contradiction can only be triggered by actual load, not by a formal envelope. If F is a formal upper envelope, A is the actual load, and C is the critical capacity, then*

$$0 \leq A \leq F.$$

The case $F \leq C$ is absorbed. The case $F > C$ but $A \leq C$ is only an accounting issue and requires tightening the envelope to the actual load. Only the case $A > C$ can force a structural exit. In the prime-matrix AffineTwin branch this is the distinction between $(A_g A_f)^2$ and N_q^2; in the two-point sieve it is the distinction between model numerator/denominator envelopes and the actual ratio

$$|U_Y(I)|^{-1} \sum_{x \in U_Y(I)} D_Y^P(x);$$

in the RH branch it is the distinction between routed atom envelopes and source-deleted final loads in the GEE ledger.

Status and reference. This is a status-discipline and reduction rule. It prevents any of the three frontiers from closing by a coarse upper envelope alone. The detailed cross-program audit is recorded in

`docs/monograph/three-claims-formal-to-actual-critical-load-frontier.md`.

The resulting active interfaces are actual packet enumeration for the prime-matrix branch, actual denominator-floor control for the two-point BMD-to-TLI transfer, and actual final-load normalization for the RH controlled exits. □

Reduction-closed Statement 2.10 (Actual-load closure contracts)**.** *The current frontier is organized into three actual-load contracts. In the prime-matrix branch, PM–ALC requires that the actual AffineTwin packet count N_q replace the formal product $A_g A_f$ in the critical load; the current machine contract records candidate $q = 31, 43, 103$, total formal product upper bound 40, but only one actual packet in the current sweep. In the two-point branch, TP–ALC requires both a BMD/KLS numerator bound and a denominator floor for $|U_Y(I)|$ in the same Buchstab convention, with*

$$K(\alpha) + \varepsilon(\alpha) < 1 - \delta(\alpha).$$

In the RH branch, RH–ALC requires every controlled exit to be written with baseline-subtracted source-deleted final load. Failure of any contract is not an anonymous error; it is a named PDEC/SAE/parity/contro exit obligation.

Status and reference. The PM current-sweep contract is generated by

`experiments/prime_matrix_affine_twin_actual_packet_contract.py`.

The unified contract ledger is archived in

`docs/monograph/three-claims-actual-load-closure-contracts.md`.

This is a reduction and evidence-organizing step. It does not close the row/column theorem, the two-point theorem, or RH. □

Reduction-closed Statement 2.11 (AffineTwin formal-pair pruning). *In the current prime-matrix AffineTwin sweep, the formal product universe has total size* 40*, while the actual packet count is* 1*. The difference is now fully explained at the current level:* 11 *formal residue pairs are removed by an exact CRT-window emptiness check, and* 28 *formal residue pairs have no matching gap-fill source materialization in the required orientation. Thus the current formal-to-actual gap is not an unaccounted critical load.*

Status and reference. The pruning audit is generated by

`experiments/prime_matrix_affine_twin_formal_pair_pruning_audit.py`.

For $q = 31$, the audit enumerates the $3 \cdot 4$ formal residue pairs and finds only the pair $(19, 8)$ with CRT representative 2687 in the support interval $[2669, 2688]$. The remaining current candidates $q = 43, 103$ have no matching gap-fill source in the required direction. The global obligation remains: future formal pairs must be routed to CRT-window emptiness, source materialization failure as PDEC/SAE, or primitive-support escape. This does not close the row/column theorem. □

Reduction-closed Statement 2.12 (AffineTwin source materialization gate). *The current source-materialization failures in the AffineTwin audit are not unclassified missing mass. An actual AffineTwin packet must have a matching gap-fill source with gap q, generator modulus $q - 2$, fill modulus q, the prescribed orientation, and affine delay*

$$p_{\text{fill}} - p_{\text{gen}} = \frac{11q - 21}{4}.$$

In the current sweep, $q = 31$ is the only candidate passing this gate. The candidate $q = 43$ has a same-gap source, but it uses generator 47*, orientation plus-to-minus, and delay* 74 *instead of generator* 41*, orientation minus-to-plus, and delay* 113*. The candidate $q = 103$ has no gap-fill source. Thus the* 28 *source-unmaterialized formal pairs from the pruning audit are fully classified at the current level.*

Status and reference. The source gate audit is generated by

`experiments/prime_matrix_affine_twin_source_materialization_gate_audit.py`.

It records 16 blocked formal pairs for the wrong-source $q = 43$ row and 12 blocked formal pairs for the no-source $q = 103$ row. The remaining global obligation is to prove that every future source-gate failure routes to SameGapWrongSource-PDEC/SAE or NoGapSource-PDEC/SAE, rather than contributing to actual load. This does not close the row/column theorem. □

Reduction-closed Statement 2.13 (AffineTwin CRT window gap). *After the source gate, the remaining current AffineTwin formal pairs are controlled by a CRT window test. For* $q = 31$*, the exact source has generator modulus* 29*, fill modulus* 31*, combined modulus* 899*, and common phase support* $[2669, 2688]$ *of width* 20*. Among the* 12 *formal residue pairs, exactly one pair,* $(19, 8)$*, has a CRT representative in the support, namely* 2687*. The other* 11 *pairs have nearest CRT representatives at positive distance from the support; the minimum such distance is* 40*.*

Status and reference. The CRT-window distance audit is generated by

`experiments/prime_matrix_affine_twin_crt_window_gap_audit.py`.

It upgrades current CRTWindowEmpty rows from a Boolean non-hit to positive phase-gap data. The global obligation remains to prove that future source-materialized non-hits have a uniform CRT window gap, or otherwise route the failure to the named PDEC/SAE exits recorded in the audit. This does not close the row/column theorem. ☐

Reduction-closed Statement 2.14 (Window edge collision). *The current CRT-window non-hits can be expressed as residue-grid displacement requirements. For* $q = 31$*, the phase window* $[2669, 2688]$ *induces* 20 *target residue pairs. Each of the* 11 *non-hit formal pairs has a nearest target pair; the least* L^1 *displacement is* 1*, realized by*

$$(19, 9) \longrightarrow (19, 8).$$

Thus the narrowest current edge-collision atom is a one-step fill-residue collision with the already realized actual pair $(19, 8)$*.*

Status and reference. The edge-collision displacement audit is generated by

`experiments/prime_matrix_affine_twin_window_edge_collision_audit.py`.

It records two non-hit pairs whose nearest target is the existing actual pair, and nine non-hit pairs whose nearest target requires unused target-residue arrival. The global obligation remains to prove that such residue movements cannot be supplied persistently by a counterexample chain, or else to route the failure to residue-collision, target-arrival, or support-motion exits. This does not close the row/column theorem. ☐

Reduction-closed Statement 2.15 (Existing actual CRT jump). *In the current AffineTwin edge-collision audit, the existing-actual branch is not a small boundary slip. The two non-hit formal pairs whose nearest target is the already realized pair* $(19, 8)$ *have CRT phase jumps* 120 *and* 435*, whereas the common phase support has width* 20*. In particular the narrowest residue atom*

$$(19, 9) \longrightarrow (19, 8)$$

has L^1 *residue displacement* 1*, but the fill-residue unit step in the combined CRT modulo* $29 \cdot 31$ *is* 435*.*

Status and reference. The CRT jump audit is generated by

`experiments/prime_matrix_affine_twin_existing_actual_collision_jump_audit.py`.

It records generator and fill CRT unit steps 465 and 435, respectively, and verifies the exact signed jumps from the empty-pair nearest representatives to the existing actual point. This closes the current-sweep existing-actual collision jump ledger. The global obligation remains to prove that such jumps cannot be reset persistently by moving support, or to route the failure to repeated-residue, ColumnCRT, PDEC, or SAE exits. This does not close the row/column theorem. ☐

Reduction-closed Statement 2.16 (Unused target arrival). *In the current AffineTwin edge-collision audit, the unused-target branch also has no hidden actual load. The* 9 *non-hit formal pairs whose nearest targets are unused collapse to* 5 *target residue pairs:*

$$(10, 30),\ (16, 5),\ (17, 6),\ (18, 7),\ (20, 9).$$

None of these pairs belongs to the current formal product. To make them available, one must introduce new generator residues

$$10, 16, 17, 18, 20$$

and new fill residues

$$5, 6, 7, 30.$$

Moreover every corresponding CRT jump has absolute size at least 59*, while the common phase support has width* 20*.*

Status and reference. The unused-target arrival audit is generated by

`experiments/prime_matrix_affine_twin_unused_target_arrival_audit.py`.

It records that all unused targets are outside the current formal product, all need a new side residue, all need a new generator residue, and all CRT jump identities exceed the support width in the current sweep. The global obligation remains to control new generator/fill residue arrival or route the failure to support-motion, PDEC, or SAE exits. This does not close the row/column theorem. □

Reduction-closed Statement 2.17 (Support motion depth). *In the current AffineTwin CRT-window audit, support motion is not a free escape. The common support is the intersection of*

$$[2669, 2693] \quad \textit{and} \quad [2659, 2688],$$

namely $[2669, 2688]$*. Therefore an empty CRT representative can enter by support motion only if the corresponding generator endpoint and shifted-fill endpoint are both released. In the current sweep all* 11 *empty pairs require such a two-endpoint release, and the common side depth required ranges from* 58 *to* 435*.*

Status and reference. The support-motion depth audit is generated by

`experiments/prime_matrix_affine_twin_support_motion_depth_audit.py`.

For the narrowest row $(19, 12)$, the nearest CRT representative is 2629. To include it, the generator left depth must increase by 40 and the fill left depth by 30, so the total endpoint release is 70. The audit records the analogous depth-inflation vector for every empty row. This closes the current-sweep support-motion depth ledger. The global obligation remains to prove non-persistence of such synchronized depth inflation, or to route it to moving-support, endpoint-coupling, PDEC, or SAE exits. This does not close the row/column theorem. □

Reduction-closed Statement 2.18 (Support motion primitive defect). *The current support-motion branch cannot be absorbed by the fixed* $q = 31$ *AffineTwin primitive key. The slot-lock identities give*

$$d_g^- = 18, \quad d_g^+ = 6, \quad d_f^- = 28, \quad d_f^+ = 1.$$

Every support-motion candidate requires a common side depth different from the corresponding generator and fill depths. The smallest defect is the row $(19, 12)$*: it requires left common depth* 58*, producing defects*

$$58 - 18 = 40, \qquad 58 - 28 = 30.$$

Thus even the narrowest support-motion atom breaks both fixed primitive depth identities.

Status and reference. The primitive-depth defect audit is generated by

`experiments/prime_matrix_affine_twin_support_motion_primitive_defect_audit.py`.

It records 11 support-motion candidates, all of which break both the generator and fill affine-depth identities for the fixed $q = 31$ primitive key. This closes the current-sweep fixed-key absorption route. The global obligation remains to exclude moving primitive keys, or to route them to moving-key, depth-inflation, endpoint-coupling, PDEC, or SAE exits. This does not close the row/column theorem. □

Reduction-closed Statement 2.19 (Moving-key depth formula)**.** *Even if the AffineTwin key is allowed to move, same-orientation depth absorption is blocked in the current support-motion rows. On the lower side, a common depth* D *would have to satisfy*

$$(q + 5)/2 = D, \qquad q - 3 = D,$$

so $q = 2D - 5$ *and* $q = D + 3$ *must agree. This occurs only at* $D = 8$*. The narrowest lower support-motion row has* $D = 58$*, giving* $q = 111$ *from the generator formula and* $q = 61$ *from the fill formula. On the upper side, the fill depth is fixed at* 1*, while the current required depths are at least* 342*.*

Status and reference. The moving-key depth formula audit is generated by

`experiments/prime_matrix_affine_twin_moving_key_depth_formula_audit.py`.

It records 11 support-motion candidates. None admits a common moving q with the same orientation and both depth formulas. This closes the current-sweep same-orientation moving-key absorption route. The global obligation remains to exclude orientation changes, source rematerialization, or other primitive-key migration, or to route them to PDEC/SAE/endpoint-coupling exits. This does not close the row/column theorem. □

Reduction-closed Statement 2.20 (Moving-key source rematerialization)**.** *The one-sided* q *values produced by the moving-key depth formulas also fail to become AffineTwin sources with the same orientation in the current sweep. Across the* 11 *support-motion rows the formulas produce* 19 *distinct one-sided candidate keys. Thirteen are composite. The six prime candidates*

$$61,\ 181,\ 293,\ 761,\ 1499,\ 1747$$

all fail the AffineTwin prime/source gate: none satisfies the full same-orientation requirement that q *and* $q - 2$ *are prime,* $q \equiv 3 \pmod 4$*, the delay* $(11q - 21)/4$ *is integral, and a matching gap-fill source with generator* $q - 2$*, fill* q*, and sides minus-to-plus exists.*

Status and reference. The moving-key source-rematerialization audit is generated by

`experiments/prime_matrix_affine_twin_moving_key_source_rematerialization_audit.py`.

It records no exact rematerialized candidate. The narrowest prior depth atom 19:12 gives $q = 111$ and $q = 61$; 111 is composite, while 61 has $61 \equiv 1 \pmod 4$ and hence a nonintegral AffineTwin delay. The nearest available gap source has gap 59, with generator 61, fill 59, and the wrong orientation for a $q = 61$ AffineTwin source. Thus the current-sweep same-orientation moving-key source-rematerialization route is closed. The global obligation remains to exclude orientation changes or persistent source rematerialization, or to route them to the named PDEC/SAE exits. This does not close the row/column theorem. □

Reduction-closed Statement 2.21 (Endpoint-release critical error)**.** *In the current AffineTwin sweep, support motion has no anonymous critical-error channel left. If an empty CRT representative is pulled into the primitive support by endpoint motion, the actual load is the total generator and shifted fill endpoint release. The critical capacity is the current common support width, which is* 20*. All* 11 *support-motion candidates are supercritical: the total endpoint-release load is* 4929 *against total capacity* 220*, and the narrowest atom* 19:12 *already has load* 70*, hence critical error*

$$\frac{70}{20} - 1 = \frac{50}{20} = 2.5.$$

Status and reference. The endpoint-release critical-error audit is generated by

`experiments/prime_matrix_affine_twin_endpoint_release_critical_error_audit.py`.

It records that every support-motion row requires both endpoints to release, that every endpoint-release load exceeds the support width, and that the fixed primitive-depth defect total matches the endpoint-release load row by row. Together with the moving-key source-rematerialization audit, this eliminates same-orientation source rematerialization as a hidden carrier for the positive local error. Thus, in the current sweep, the positive endpoint-release critical error is routed to `EndpointReleaseCoupling-PDEC`. The global obligation remains to exclude persistent endpoint-release coupling, orientation changes, and source rematerialization, or to route them to named exits. This does not close the row/column theorem. □

Reduction-closed Statement 2.22 (Endpoint-release feedback loss)**.** *Even after the current support-motion rows are given full geometric credit for the larger endpoint movement, a supercritical feedback loss remains. For each row, write*

$$L = \textit{generator release} + \textit{fill release}, \qquad G = \max(\textit{generator release}, \textit{fill release}).$$

The credited support gain is G, so the unrecovered feedback loss is

$$L - G = \min(\textit{generator release}, \textit{fill release}).$$

In the current audit, the total release load is 4929*, the total geometric credit is* 2512*, and the remaining feedback loss is* 2417 *against support-width budget* 220*. The narrowest atom* 19:12 *has load* 70*, credit* 40*, and feedback loss* 30*, still larger than the support width* 20*.*

Status and reference. The endpoint-release feedback-loss audit is generated by

`experiments/prime_matrix_affine_twin_endpoint_release_feedback_loss_audit.py`.

It records that all 11 feedback losses are positive, equal to the smaller synchronized endpoint release, and still exceed the original support width. Thus the current sweep has no self-feedback absorption channel after geometric support gain is fully credited. The remaining positive residual is routed to `EndpointReleaseFeedbackLoss-PDEC`. The global obligation is to exclude that persistent feedback-loss family, or route it to the named moving-key, orientation-change, source-rematerialization, or SAE exits. This does not close the row/column theorem. □

Reduction-closed Statement 2.23 (Endpoint-release feedback horizon). *The endpoint feedback loss has an equivalent phase-horizon form in the current sweep. A support-motion representative can be absorbed by endpoint self-feedback only if its distance from the current support window is at most*

$$\textit{support width} + |\textit{generator side depth} - \textit{fill side depth}|.$$

For all 11 *support-motion rows the representative lies outside this horizon. The narrowest row is* 19:12*: its representative is at distance* 40*, while the support width is* 20 *and the side-depth skew is* 10*, so the feedback horizon is* 30 *and the phase-horizon surplus is* 10*.*

Status and reference. The endpoint-release feedback-horizon audit is generated by

`experiments/prime_matrix_affine_twin_endpoint_release_feedback_horizon_audit.py`.

It proves, for the current sweep, the identities

$$\text{feedback loss} = \text{window distance} - \text{side-depth skew},$$

and

$$\text{post-credit overcritical units} = \text{window distance} - (\text{support width} + \text{side-depth skew}).$$

Thus the residual critical error is exactly the distance by which the CRT representative crosses the feedback horizon. All rows have positive horizon surplus, so the current self-feedback absorption route is closed and the residual is routed to `EndpointReleaseFeedbackHorizon-PDEC`. The global obligation remains to exclude persistent horizon surplus, or to route it to orientation-change, source-rematerialization, or SAE exits. This does not close the row/column theorem. □

Reduction-closed Statement 2.24 (Endpoint-release feedback-horizon slack). *In the same current sweep, the feedback-horizon surplus is exactly the missing side-depth skew needed for absorption. If a representative has distance d from the current support window and the support width is W, then absorption requires total side-depth skew at least*

$$\max(0, d - W).$$

Hence the required extra skew is

$$\max(0, d - W) - |\textit{generator side depth} - \textit{fill side depth}|,$$

which equals the phase-horizon surplus row by row. The total required skew is 2292*, while the current skew supplies only* 95*; the missing extra skew is* 2197*. The narrowest row is again* 19:12*, where distance* 40 *and support width* 20 *require total skew* 20*, but the current skew is* 10*.*

Status and reference. The endpoint-release feedback-horizon slack audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
feedback_horizon_slack_audit.py
```

It verifies that all 11 rows have positive required extra skew, that this extra skew equals the phase-horizon surplus, and that a pure orientation flip does not change the absolute skew. It also inherits from the moving-key source-rematerialization ledger that same-orientation exact rematerialization is absent. Thus the current anonymous slack absorption route is closed and the remaining obstruction is routed to `EndpointSkewGrowth-PDEC/OrientationChangingPrimitiveKey-SAE`. The global row/column theorem remains open until those named exits are excluded or absorbed. □

Reduction-closed Statement 2.25 (Endpoint-release bidirectional skew hull). *The current slack rows also have a simultaneous CRT-hull form. Each formal generator/fill residue pair gives a local alignment condition in the variable P,*

$$P \equiv r_g \pmod{q-2}, \qquad P \equiv r_f - \Delta \pmod{q},$$

where Δ is the affine delay. For the current $q = 31$ atom, the combined modulus is 899 and the primitive support interval is [2669, 2688], of width 20. The nearest representatives of all 12 formal pairs lie in the hull [2304, 3122], of width 819. Thus a single support hull would need left and right extensions 365 and 434; after the feedback horizons are credited, positive two-sided deficits 335 and 409 remain. Moreover

$$899 - 819 = 80,$$

which equals the affine delay.

Status and reference. The bidirectional skew-hull audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
bidirectional_skew_hull_audit.py
```

It imports the inverse-alignment equivalence only as a local CRT-alignment discipline: the global statement that the minimal zero-row alignment x exceeds P is not used as a theorem. The audit proves that the current formal-pair system cannot be absorbed by one-sided skew growth, because below and above representatives both remain outside the credited feedback horizons. The remaining obstruction is routed to `BidirectionalSkewHull-PDEC/ColumnCRT` or to the moving-family multiplicity exits. This is a current-sweep closure, not a final unconditional row/column theorem. □

Reduction-closed Statement 2.26 (Endpoint-release circular aperture). *The bidirectional hull also has to be read on the circle modulo $q(q-2)$, not only in the current linear cut. For the current $q = 31$ atom, the previous interval [2304, 3122] has width 819 in one chosen cut of the modulus 899. Removing the largest circular open gap instead gives the minimal circular alignment arc*

$$[3029, 3586],$$

of width 558. The largest open gap runs from the actual pair 19:8 to the pair 13:9 and has open width 341. The affine delay gap of width 80 is present, but it is only the third largest circular gap, so the previous equality $899 - 819 = 80$ is a linear-cut rigidity, not the complement of the circular minimal arc.

Status and reference. The circular-aperture audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
circular_aperture_audit.py
```

It verifies that even after the optimal circular cut, the minimal arc has width 558, whereas the primitive support has width 20. After optimally shifting the support interval to [3568, 3587], the total extension still required is 539; after crediting the most favorable one-sided feedback horizon, a conservative deficit 509 remains. Thus the current single-sided circular-aperture absorption route is closed for this sweep. The remaining obstruction is routed to `CircularAperture-PDEC/ColumnCRT`, `SAE`, or the already named moving-family multiplicity exits. This refinement corrects the linear-hull wording but does not close the global row/column theorem. □

Reduction-closed Statement 2.27 (Endpoint-release anchored circular depth). *The circular aperture can be sharpened by keeping the unique actual packet as an anchor. In the current sweep the minimal circular arc is* $[3029, 3586]$*, and its right endpoint* 3586 *is the actual pair* $19{:}8$ *after one period of the modulus* 899*. Hence an actual-anchored absorption of the whole arc requires the generator phase and the shifted-fill phase to both reach the left endpoint* 3029*. The required common left depth is*

$$D = 3586 - 3029 = 557.$$

The current generator and fill left depths are 18 *and* 28*, so the required endpoint releases are* 539 *and* 529*, for a total of* 1068*. This is* 53.4 *times the primitive support width* 20*.*

Status and reference. The anchored circular-depth audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
anchored_circular_depth_audit.py
```

It shows that the previous one-sided circular support extension 539 hides a second endpoint release 529: the intersection support moves only when both underlying phase intervals move. The same-orientation moving-key explanation also fails at this anchored depth. The lower-side AffineTwin depth formulas would require

$$q = 2D - 5 = 1109, \qquad q = D + 3 = 560,$$

so there is no common odd-prime key. Thus the current same-orientation anchored circular-depth absorption route is closed for this sweep. The remaining obstruction is routed to anchored circular-depth PDEC, ColumnCRT, SAE, orientation-changing keys, or the moving-family multiplicity exits; the global row/column theorem remains open. □

Reduction-closed Statement 2.28 (Anchored parity no-go). *For a same-orientation lower-side AffineTwin key to absorb an anchored common left depth* D*, the generator and fill depth formulas must give the same prime parameter:*

$$q_g = 2D - 5, \qquad q_f = D + 3.$$

Thus a common key forces $2D - 5 = D + 3$*, hence* $D = 8$ *and* $q = 11$*. The actual-anchored circular arc in the current sweep instead forces* $D = 557$*. It gives*

$$q_g = 1109, \qquad q_f = 560.$$

The fill-side candidate is an even integer larger than 2*, so it cannot be an odd-prime AffineTwin key.*

Status and reference. The anchored parity no-go audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
anchored_parity_nogo_audit.py
```

It records both obstructions: the forced depth is 549 away from the only common-depth solution $D = 8$, and its odd parity makes the fill-side candidate $D + 3$ even. Therefore the current same-orientation anchored parity absorption route is closed for this sweep. The remaining obstruction is a global family version of this no-go, or one of the already named exits: orientation-changing keys, ColumnCRT/PDEC, SAE, or moving-family multiplicity. The global row/column theorem remains open. □

Reduction-closed Statement 2.29 (Endpoint-release cut-anchor sweep)**.** *The anchored parity obstruction can be checked against every circular cut, not only the optimal one. In the current* $q = 31$ *atom there are* 12 *adjacent CRT cuts on the circle modulo* 899*. The unique actual pair* 19:8 *is retained in every lifted arc: it is the left endpoint for one cut, the right endpoint for one cut, and an interior point for the remaining ten cuts. For each cut we anchor the actual representative in the lifted arc and charge both the generator interval and the shifted-fill interval for the left and right depths needed to cover that arc.*

The smallest arc is still the previous cut $19{:}8 \to 13{:}9$*, with width* 558 *and actual pair at the right endpoint. Its total endpoint release is* $1068 = 53.4 \cdot 20$*, where* 20 *is the primitive support width. Across all cuts, the endpoint release ranges from* 1068 *to* 1737*, so no cut fits inside the support budget.*

Status and reference. The cut-anchor sweep audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
cut_anchor_sweep_audit.py
```

It also checks the same-orientation moving-key formulas for every cut. On the left lower side a common key can occur only at $D = 8$; the closest positive left depth in the sweep is $D = 58$. On the right above side the shifted-fill right depth is fixed at $D = 1$; the closest positive right depth in the sweep is $D = 342$. Hence changing the circular cut or moving the actual endpoint does not reopen the same-orientation absorption channel. Endpoint cuts fall into the left anchored parity no-go or the right fixed-fill no-go; interior cuts require both side releases simultaneously. The current sweep is closed, but the global theorem still requires a family version of this cut-anchor no-go, or one of the named exits: orientation-changing keys, ColumnCRT/PDEC, SAE, or moving-family multiplicity. □

Reduction-closed Statement 2.30 (Cut-anchor ColumnCRT compression)**.** *The circular cut is a gauge choice, not a new actual residue. In the current cut-anchor sweep all* 12 *cuts retain the same actual pair* 19:8*. The actual representative is either* 2687 *or* $3586 = 2687 + 899$*, and in both cases*

$$P \equiv 889 \pmod{899}.$$

Thus the fixed $q = 31$ *actual slot contributes one ColumnCRT atom of mass* $1/899$*, not twelve independent atoms of total mass* $12/899$*. The apparent extra mass*

$$\frac{11}{899}$$

is only cut multiplicity.

Status and reference. The cut-anchor ColumnCRT compression audit is generated by

```
experiments/prime_matrix_affine_twin_endpoint_release_
cut_anchor_columncrt_compression_audit.py
```

It registers the persistent fixed-residue escape as a `CutAnchorColumnCRT-PDEC` object with phase $P \equiv 889 \pmod{899}$, equivalently $P \equiv 19 \pmod{29}$ and $P \equiv 21 \pmod{31}$ after the fill shift. Therefore, after same-orientation cut-anchor absorption is closed, changing the cut cannot supply additional capacity. If q or the residues move, the case leaves this fixed ledger and returns to the existing AffineTwin moving-family SAE/ColumnCRT ledger. The current sweep closes cut multiplicity as an independent capacity source; it does not yet exclude the global ColumnCRT/PDEC or orientation-changing exits. □

Chapter 3

Prime-Matrix Contradiction Fields

Let P be an odd prime and write the $P \times P$ prime matrix entries as

$$n(r,c) = (r-1)P + c, \qquad 1 \le r, c \le P.$$

The row and column programs attempt to show that every nontrivial row or column contains a prime. The current formal appendix reduces a row or non-P column counterexample to a structured bad configuration after removing small-factor locks and tail anchors.

Reduction-closed Statement 3.1 (A/B reduction, recorded form)**.** *A nontrivial full-composite row or column, after small-factor and tail-anchor peeling, induces a Structured-EHPD bad configuration satisfying CRT balance, large-factor non-reuse, and anchor-capacity constraints.*

Reference. See `docs/row-column-reduction-formal-appendix.md` and the theorem list

`docs/monograph/prime-matrix-ab-entrance-theorem-list.md`.

This statement is a reduction interface; it does not by itself prove the Structured-EHPD exclusion theorem. □

3.1 Two-Point Rough-Pair Propositions

A natural next strengthening asks for two simultaneous nonzero-residue constraints in the back half of the prime matrix. Fix an even integer $w > 0$ and ask whether there is an entry x in the back half of the $P \times P$ matrix such that both x and $x - w$ are nonzero modulo every prime $p \le P$, or equivalently both x and $w - x$ are nonzero modulo every prime $p \le P$.

Proposition 3.2 (Equivalence and strength of the two-point rough-pair problem)**.** *For fixed even $w > 0$, the two conditions*

$$p \nmid x(x-w) \quad (p \le P)$$

and

$$p \nmid x(w-x) \quad (p \le P)$$

are identical. In the region $1 < x \le P^2$, any integer nonzero modulo every prime $p \le P$ is prime. Consequently, a proof of this two-point assertion for every sufficiently large P would imply infinitely many prime pairs of gap w; in particular $w = 2$ would imply the twin-prime conjecture.

Proof. For each $p \le P$, the condition $p \nmid x - w$ is $x \not\equiv w \pmod p$, and the condition $p \nmid w - x$ is the same congruence. Both formulations therefore exclude exactly the two residue classes 0 and w modulo each p, with possible collision when $p \mid w$.

If $1 < x \le P^2$ is nonzero modulo every prime $p \le P$ and is composite, then it has a prime factor $q \le \sqrt{x} \le P$, contradicting the nonzero-residue condition. Thus such an x is prime. If the two-point assertion holds for all sufficiently large $P > w$, then x and $x - w$ are both prime in infinitely many growing back-half matrix intervals, giving infinitely many prime pairs of gap w. □

Remark 3.3 (Status)**.** This two-point problem is therefore not a routine corollary of the one-point row/column contradiction field. It is a prime-pair-level strengthening, naturally expressible as a two-point rough Jacobsthal problem. In this monograph it is recorded as a future contradiction-field direction, not as an unconditional theorem.

3.2 Two-Point Secondary-Sieve Contradiction Field

The two-point rough-pair problem admits a more detailed contradiction-field research program. The current working manuscript is recorded in

`docs/monograph/two-point-secondary-sieve-research.md`.

It studies the secondary sieve condition

$$x \not\equiv 0, w \pmod p \qquad (p \le P)$$

inside back-half matrix windows, and tries to exclude a two-point rough-pair counterexample by a hierarchy of local rigidity fields.

Definition 3.4 (Two-point secondary-sieve field)**.** Fix an even integer $w > 0$. A two-point secondary-sieve field is the structured ledger obtained from a hypothetical interval in which every candidate x fails at least one of the two congruence conditions $p \nmid x$ or $p \nmid x - w$ for some prime $p \le P$. Its named channels are small-prime double locks, top-scale large-factor coverage, short-block zero-mass loss, multiplicative near-crossings, Fourier CRT defects, and smoothing boundary terms.

Lemma 3.5 (Fixed-order local crossing complexity)**.** *In the secondary-sieve manuscript, every fixed-order local crossing test is expanded only through truncated hard-skeleton weights*

$$d \le R, \qquad R \le (\log P)^{C_R},$$

with coefficient mass $\sum_{d \le R} |\lambda_d| \le (\log P)^{C_\lambda}$ *and short-block shifts* $|h| < L \asymp (\log\log P)^2$*. For every fixed order k, the effective historical complexity satisfies*

$$Q_{\mathrm{eff},k} \ll_k \left(\sum_{d \le R} |\lambda_d| \right)^k R^k L^{O(k)} \le (\log P)^{C_k}.$$

In particular the required $k = 2, 3$ local crossing tests have polylogarithmic effective complexity.

Proof. After expanding the k truncated weights, only moduli $d_1, \ldots, d_k \le R$ occur. For fixed new-line primes p_i, phases, and block shifts, all conditions are one-variable linear congruences. By the Chinese remainder theorem the local system is either inconsistent or occupies one residue class modulo a divisor of

$$\mathrm{lcm}(d_1, \ldots, d_k)\, \mathrm{lcm}(p_1, \ldots, p_k).$$

The primes p_i are the explicit variables being summed in the crossing estimate; they contribute the usual Y^k scale, not hidden historical complexity. The historical part is bounded by $\operatorname{lcm}(d_1, \dots, d_k) \le R^k$, the block shifts contribute $L^{O(k)}$ patterns, and the coefficient expansion contributes $(\sum |\lambda_d|)^k$. This proves the displayed bound. The true residual set U_Y is not expanded into a full primorial period. □

Lemma 3.6 (Zero-mass and smoothing discharge). *Assume the SC2/diagonal package supplies the moment bounds*

$$M_1 := \frac{\sum_B X_B R_B}{\sum_B X_B} \le 0.45, \qquad M_2 := \frac{\sum_B X_B \binom{R_B}{2}}{\sum_B X_B} \ge 0.05,$$

and

$$M_3 := \frac{\sum_B X_B \binom{R_B}{3}_+}{\sum_B X_B} \le 0.03.$$

Then the weighted zero-hit mass satisfies

$$\frac{\sum_B X_B 1_{R_B=0}}{\sum_B X_B} \ge 0.57.$$

Moreover, if the directional-balance endpoint estimate gives

$$N_{\text{endpoint}}/N_{\text{total}} \le C_{\text{end}}\Delta + o(1),$$

then choosing $\Delta \le 0.01/C_{\text{end}}$ gives a smooth-to-sharp loss < 0.03 for all sufficiently large P.

Proof. For every integer $R \ge 0$,

$$1_{R=0} \ge 1 - R + \binom{R}{2} - \binom{R}{3}_+.$$

Multiplying by $X_B \ge 0$ and summing over blocks gives

$$\frac{\sum_B X_B 1_{R_B=0}}{\sum_B X_B} \ge 1 - M_1 + M_2 - M_3 \ge 1 - 0.45 + 0.05 - 0.03 = 0.57.$$

For smoothing, choose even smooth cutoffs $W_- \le 1_{|t|<1} \le W_+$ whose difference is supported in $1 - \Delta \le |t| \le 1 + \Delta$. The sharp and smoothed counts differ only on this endpoint layer. The displayed directional-balance bound therefore gives loss at most $C_{\text{end}}\Delta + o(1)$, which is < 0.03 after the stated choice of Δ and then taking P large. □

Lemma 3.7 (Adaptive true-hit layering dichotomy). *Let U_Y be the current true residual set and define, for each new prime $p > Y$,*

$$a_p = \frac{1}{|U_Y|}\#\{x \in U_Y : x \equiv 0 \text{ or } w \pmod p\}.$$

Fix $\eta = 0.05$. Either some p has $a_p > \eta$, in which case the configuration is a Single-Prime CRTDefect, or the remaining primes can be greedily partitioned into consecutive layers $\mathcal{P}_j$ satisfying

$$\nu_j^{\text{real}} := \sum_{p \in \mathcal{P}_j} a_p \le 0.4,$$

and, except for the final layer, $\nu_j^{\text{real}} > 0.4 - \eta \ge 0.35$.

Proof. If $a_p > \eta$ for some p, then a fixed positive proportion of U_Y lies in the two residue classes $0, w$ modulo p. This is exactly the Single-Prime CRTDefect exit, to be excluded by the CRTDefect/Directional Balance part of the SC2 package. Otherwise all $a_p \le \eta$. Starting from the smallest remaining prime, add primes to the current layer until adding the next one would make the partial sum exceed 0.4. The completed layer has sum at most 0.4, and because the next summand is at most η, every non-final completed layer has sum greater than $0.4 - \eta$. This gives the claimed adaptive true-hit decomposition. □

Proposition 3.8 (Recorded conditional closure chain for the two-point secondary sieve)**.** *Using the three lemmas above, assume the following remaining input holds with the constants recorded in the secondary-sieve research note:*

1. *the I3-Core SC2/CRTDefect package controls the true-residual two-line near-crossing error, the odd-prime 45° main-synchronization peak, Single-Prime CRTDefect exits, the normalized $q = 2$ parity skeleton, the moment bounds M_1, M_2, M_3, and the endpoint Directional Balance used in the zero-mass and smoothing discharge lemma.*

Then the two-point secondary-sieve contradiction field has no remaining top-scale escape channel under the parameter choice

$$\alpha_0 = 0.75, \qquad c_0 = 1/2, \qquad \nu_j^{\mathrm{real}} \le 0.4$$

for adaptive true-hit layers, subject to the same fixed-w convention used in the research manuscript.

Proof sketch and status. The argument is a conditional synthesis, not a final unconditional theorem. A top-scale covering excess is first decomposed into adaptive true-hit layers; any single-prime concentration is routed to the Single-Prime CRTDefect exit. The remaining excess is converted by a TCA linear bridge into a positive short-block load. The block load is forced into the SMC/SC2 near-crossing ledger. The 45° main synchronization is split into large-factor reuse, shared-complement factors, and good synchronized pairs; reuse and shared factors are excluded by short-window rigidity, while good pairs are sent to an auxiliary-prime Fourier CRT-defect. Directional balance excludes the Fourier defect unless the top-scale energy is already below the capacity threshold. Finally, the weighted Bonferroni zero-mass estimate supplies enough zero-hit block mass to make the capacity comparison incompatible.

The current audit is archived under `docs/archive/monograph-reviews/` and classifies this as a research proposition conditional on the single I3-Core package above. The fixed-order complexity input, zero-mass/smoothing discharge, and adaptive layering dichotomy have been inlined as the preceding lemmas. The fact that an unconditional proof would imply fixed-gap prime-pair consequences, including the twin-prime case when $w = 2$, is not a logical obstruction. It only marks the strength of the claim. The reason this chapter remains conditional is that I3-Core has not yet been inlined as an independently verified unconditional proof. □

Proposition 3.9 (Window-compressed conditional variant)**.** *Keep the same fixed even integer $w > 0$ and the same remaining I3-Core input as in the preceding proposition. Let I be any consecutive back-half window of $H(P)$ rows in the $P \times P$ matrix, so that $|I| = H(P)P + O(P)$. By the fixed-order complexity lemma, write*

$$Q_{\mathrm{eff}} \le C_Q (\log P)^C.$$

For a prescribed middle-scale error budget $\varepsilon_{\mathrm{SC2}} > 0$, the conditional row threshold supplied by the current ledger is

$$H_{\min}^{\mathrm{cond}}(P; \varepsilon_{\mathrm{SC2}}) = \left\lceil \frac{C_Q}{\varepsilon_{\mathrm{SC2}}} P^{1/2} (\log P)^C \right\rceil.$$

Equivalently, one may use the asymptotic form

$$H(P) \geq H^{\mathrm{asym}}_{\mathrm{min}}(P) := \left\lceil P^{1/2}(\log P)^{C_*} \right\rceil, \qquad C_* > C.$$

Under either threshold, the same conditional contradiction-field chain gives an element

$$x \in I, \qquad p \nmid x(x-w) \quad (p \leq P).$$

When $x - w > 1$, both x and $x - w$ are primes.

Recorded scale check. The only scale-sensitive term in the recorded chain is the middle/high-scale $K = 2$ near-crossing error. With top layer $Y = P^{3/4}$ and $Q_{\mathrm{eff}} \leq (\log P)^C$, replacing the original back-half length $\asymp P^2$ by $|I| \asymp H(P)P$ changes the normalized error to

$$\frac{Q_{\mathrm{eff}} Y^2}{|I|} \ll \frac{P^{1/2}(\log P)^C}{H(P)}.$$

The displayed lower bound on $H(P)$ makes this at most $\varepsilon_{\mathrm{SC2}}$ in the explicit form, or $o(1)$ in the asymptotic form. Hence the constants in the TCA, SC2, zero-mass, finite-package, and smoothing ledgers are unchanged. The conclusion is therefore a conditional window-compressed version of the previous proposition. It becomes an unconditional prime-pair theorem only if I3-Core is independently proved and accepted. □

Remark 3.10 (Referee boundary)**.** The two-point secondary-sieve chapter is included to document the contradiction-field architecture and its parameter ledger. The window-compression audit is recorded in `docs/monograph/two-point-window-compression-and-unconditionality.md`. It must not be cited as an unconditional proof of prime pairs, the twin-prime conjecture, or Goldbach-type assertions unless I3-Core is independently proved and accepted.

Remark 3.11 (Current obstruction)**.** The current hard attack reduces I3-Core to a true-residual correlation problem. The local CRT and matrix rigidities do not by themselves exclude a model residual set concentrated on a new prime's bad residue class while still satisfying all previously imposed two-residue exclusions and short-window non-reuse constraints. Thus the remaining task is not another bookkeeping step but a genuine lower-bound/correlation theorem for the true residual set in new prime moduli.

Remark 3.12 (Latest TRC-1G reduction)**.** The single-prime part of the remaining TRC input has been sharpened to a heavy-atom prohibition. If more than 40% of the true residual mass lies in the two bad classes $0, w \pmod p$ for some new prime $p > Y$, then Parseval gives a non-zero Fourier coefficient modulo p of size $> 1/5$ after normalization. Equivalently, the three long-direction difference families $0, \pm w \pmod p$ carry a fixed positive proportion of the true-residual pair energy. The existing matrix rigidities remove the short-direction, 45°-locked, reuse/shared-complement, and endpoint pieces. The remaining unproved input is therefore a new-modulus long-direction balance theorem for the true residual set, not a finite-template or local CRT bookkeeping issue.

Remark 3.13 (Total large-incidence reduction)**.** The pointwise new-modulus balance can be weakened further. It is enough to prove the total large-incidence inequality

$$\sum_{P^\alpha < p \leq P} \#\{x \in U_{P^\alpha}(I) : p \mid x(x-w)\} < |U_{P^\alpha}(I)|.$$

Indeed, a point not counted on the left is unsieved by every prime up to P, and in the P^2 matrix range this forces both x and $x - w$ to be prime. The CRT model predicts the normalized left side to be $2\log(1/\alpha) + o(1)$; for $\alpha = 3/4$ this is $2\log(4/3) = 0.57536\ldots$, leaving margin $0.42463\ldots$. The unproved part is the true-residual total-incidence estimate with an error below that margin.

Remark 3.14 (RB–TLI audit boundary)**.** The direct attempt to prove the total large-incidence estimate separates cleanly into a numerator and a denominator. The numerator is an upper-sieve problem: after writing $x = pm$ or $x = w + pm$, the variable m again avoids two residue classes modulo each old prime, giving an upper bound of shape

$$A_p \le 2C_+|I|V_2(P^\alpha)/p + R_p.$$

The denominator requires a model-level lower bound $|U_{P^\alpha}(I)| \ge c_-|I|V_2(P^\alpha)$. For $\alpha = 3/4$, this route would need $C_+/c_- < 1/(2\log(4/3)) = 1.738\ldots$. This relative Buchstab total-incidence input is the remaining unproved prime-pair-strength statement.

Remark 3.15 (Local collapse when $q \mid w$)**.** If an old odd prime q divides the fixed even difference w, then the two forbidden classes $0, w \pmod q$ coalesce into a single forbidden class. The local density is therefore $1-1/q$ instead of $1-2/q$, giving the singular local gain $(q-1)/(q-2)$. This increases the absolute number of admissible residual points and weakens the small-prime synchronization peaks. In the relative RB–TLI ratio, however, the same local factor multiplies both $|U_Y|$ and the model numerator $A_p \sim 2|I|V_w(Y)/p$ for all new primes $p > Y > |w|$. Thus the main relative constant remains $2\log(1/\alpha)$; the collapse improves finite constants but does not by itself close the hardest case $w = 2$.

Remark 3.16 (Buchstab semiprime transition for $w = 2$)**.** The latest numerical audit of the hardest case $w = 2$ shows that the naive constant $2\log(1/\alpha)$ is not the correct conditional main term after the old primes have been sieved. For $\alpha > 2/3$, any new-prime hit $p \mid x$ or $p \mid x - 2$ with $Y = P^\alpha < p \le P$ forces the complementary quotient to be prime, because it is Y-rough and smaller than Y^2. Thus the large-incidence layer is a Buchstab semiprime transition. The corresponding two-sided main constant is

$$K(\alpha) = \frac{2\log((2-\alpha)/\alpha)}{1 + \log((2-\alpha)/\alpha)}.$$

Exact scans recorded in `docs/rb-tli-w2-scan-large.md` give, for example, $E_U D = 0.691696$ at $P = 10007, \alpha = 3/4$, compared with $K(3/4) = 0.676220$. The remaining target is therefore a two-point stability theorem for this Buchstab semiprime transition, not the earlier naive incidence model.

Remark 3.17 (BST error decomposition)**.** The refined scan decomposes the remaining error. At $P = 10007, \alpha = 3/4$, the shell contributions in the five exponent intervals between 0.75 and 1 track the Buchstab shell constants, and the covariance between the x-side and $(x-2)$-side large-incidence counts is approximately $-9.3\cdot 10^{-5}$. The zero-incidence set is independent of the intermediate cutoff α, as it is exactly the set unsieved by all primes up to P. Thus the remaining hard input has been reduced to a shifted biprime roughness estimate of the form

$$\#\{p, m : pm \in I,\ pm - 2 \in U_Y\} \le (B(\alpha) + o(1))|U_Y|.$$

This is the current BST–2 obstruction.

Remark 3.18 (BMD reduction)**.** The BST–2 obstruction can be written as a one-dimensional multiplicative sieve on biprime variables. After setting $x = pm$, the condition $pm - 2$ is unsieved by old primes is, for each $q \le Y$,

$$pm \not\equiv 2 \pmod q.$$

The prime $q = 2$ is peeled off deterministically: for odd primes p, m, the integer $pm-2$ is odd. Thus the multiplicative sieve is taken over odd q. For each odd $q \le Y$, since $p, m > Y \ge q$, the forbidden curve has $(q-1)$ points inside $(\mathbb{F}_q^\times)^2$, out of $(q-1)^2$ possible non-zero pairs. The local forbidden density is therefore $1/(q-1)$. The remaining unproved estimate is a biprime multiplicative dispersion bound: weighted sums of the errors in the congruences $pm \equiv 2 \pmod d$, with odd squarefree d, must be $o(|U_Y|)$ for the Buchstab/Rosser weights used in the transition. This is the current BMD input.

Remark 3.19 (Row-column input for BMD)**.** If the prime-matrix row/column theorem is accepted as a conditional input, it can be used only to remove zero-section degeneracies in complete CRT blocks: entire empty rows, columns, or unit residue cells are then not allowed to masquerade as the main BMD error. It does not by itself prove BMD, because BMD is a signed distribution statement. For odd d, expanding

$$1_{pm\equiv 2 \pmod d} = \frac{1}{\varphi(d)} + \frac{1}{\varphi(d)} \sum_{\chi\neq\chi_0} \overline{\chi}(2)\chi(p)\chi(m)$$

shows that the remaining obstruction is a bilinear character-dispersion estimate

$$\sum_{d\leq D} \frac{|\lambda_d|}{\varphi(d)} \sum_{\chi\neq\chi_0} \left| \sum_{Y<p\leq P} \chi(p) \sum_{\substack{m\in J_p \\ m \text{ prime}}} \chi(m) \right| = o(|U_Y|).$$

Thus the row/column input gives a useful BMD-Zero exclusion, but the genuine remaining hard point is BMD-Char.

Remark 3.20 (Direct BMD attack via an E_2 Bombieri–Vinogradov input)**.** Set $N \asymp P^2$ and

$$a_n = \#\{(p,m) : Y < p \leq P,\ m \text{ prime},\ n = pm,\ n \in I\}.$$

For $\alpha > 2/3$ the multiplicity of a_n is bounded. After the deterministic modulus 2 has been peeled off, the BMD remainder for odd squarefree d is exactly the arithmetic-progression discrepancy of this restricted two-prime convolution:

$$R_d = \sum_{\substack{n\in I \\ n\equiv 2 \pmod d}} a_n - \frac{1}{\varphi(d)} \sum_{\substack{n\in I \\ (n,d)=1}} a_n.$$

Thus BMD follows from the standard Bombieri–Vinogradov theorem for restricted E_2 sequences at level

$$d \leq P(\log P)^{-B} = N^{1/2}(\log N)^{-B+O(1)}.$$

Indeed, if

$$\sum_{d\leq P/\log^B P} \max_{(a,d)=1} |R_d(a)| \ll_A N(\log N)^{-A},$$

then every Rosser/Buchstab weighted remainder is $o(|U_Y|)$ since $|U_Y| \asymp N/(\log P)^2$. The condition $\alpha < 1$ ensures $Y < P/\log^B P$ for large P, so all one-prime forbidden curves $q \leq Y$ are included in the available distribution range. Therefore BMD is closed if this standard E_2 BV input is cited; a fully self-contained manuscript must instead prove that input in an appendix.

Remark 3.21 (Self-contained BV–E_2 obligation)**.** The local BV–E_2 appendix records the exact self-contained obligation. A direct multiplicative large-sieve estimate is insufficient: in the balanced dyadic block $p \sim P_1$, $m \sim M_1$, with $P_1 \asymp M_1 \asymp P$ and $Q \asymp P/\log^B P$, Cauchy plus the large sieve gives size roughly $P^3/\log^{2B} P$, while the required Bombieri–Vinogradov error is $P^2/\log^A P$. Thus the only genuinely deep remaining self-contained input is a Type-II dispersion/Kloosterman-cancellation estimate for the balanced convolution blocks. If that standard dispersion theorem is cited, the BMD step is closed as an external-input theorem; if not, this is the next appendix that must be proved line by line.

Remark 3.22 (Weighted BE2–3K core)**.** BMD does not require the full $\max_a$ Bombieri–Vinogradov statement. It only needs the fixed class $2 \bmod d$ against the well-factorable Rosser/Buchstab weights. After Cauchy, the self-contained problem is a variance bound for

$$T_r = \sum_{d \le Q} \lambda_d \left(\sum_{s \equiv 2r^{-1} \bmod d} \beta_s - \frac{1}{\varphi(d)} \sum_{(s,d)=1} \beta_s \right).$$

Opening $\sum_r |T_r|^2$, the off-diagonal conditions

$$rs_1 \equiv 2 \pmod{d_1}, \qquad rs_2 \equiv 2 \pmod{d_2}$$

give, by CRT and Fourier expansion of the interval condition, the Kloosterman phase

$$e\left(-\frac{2h\overline{s_1}\overline{d_2}}{d_1} - \frac{2h\overline{s_2}\overline{d_1}}{d_2}\right).$$

Thus the true no-black-box core is a weighted bilinear Kloosterman dispersion estimate, denoted BE2–3K in the notes. Proving BE2–3K gives BE2–3, then weighted BV–E_2, then BMD.

Remark 3.23 (Referee status of the no-black-box claim)**.** At the current stage the manuscript is not yet fully no-black-box. The BMD obstruction has been reduced to the single deep BE2–3K Kloosterman-dispersion estimate. If a BFI/Deshouillers–Iwaniec/Kuznetsov type mean-value theorem is cited, BMD is closed as an external-input theorem. If no external input is allowed, BE2–3K remains the unique missing line-by-line proof obligation.

Remark 3.24 (Further reduction of BE2–3K)**.** The BE2–3K core can be narrowed further. Pointwise Weil bounds for individual incomplete reciprocal sums give only single-modulus square-root cancellation and do not provide the arbitrary logarithmic saving required after summing over the d_1, d_2, h families. The common-divisor strata $(d_1, d_2) > 1$ are not the main obstruction: compatibility forces $s_1 \equiv s_2$ modulo the common divisor and yields only polylogarithmic loss. The remaining genuine input is a windowed Kloosterman spectral large-sieve estimate, denoted KLS-window in the notes. Its role is precisely

$$\text{KLS-window} \Rightarrow \text{BE2–3K} \Rightarrow \text{BE2–3} \Rightarrow \text{weighted BV–}E_2 \Rightarrow \text{BMD}.$$

Remark 3.25 (External-theorem closure of KLS-window)**.** The KLS-window input is tied to two classical external sources: the Deshouillers–Iwaniec spectral Kloosterman large-sieve technology, as developed in *Kloosterman sums and Fourier coefficients of cusp forms*, Invent. Math. 70(2), 219–288 (1982), and the Bombieri–Friedlander–Iwaniec dispersion framework with well-factorable weights, as in *Primes in Arithmetic Progressions to Large Moduli. II*, Math. Ann. 277, 361–394 (1987). With these external theorems allowed, the analytic chain closes as

$$\text{DI+BFI} \Rightarrow \text{KLS-window} \Rightarrow \text{BE2–3K} \Rightarrow \text{BE2–3} \Rightarrow \text{weighted BV–}E_2 \Rightarrow \text{BMD}.$$

This is an external-deep-theorem closure, not a fully self-contained reproof of Kuznetsov/dispersion theory.

Chapter 4

Zeta-Zero Contradiction Fields

Assume an off-critical zero $\rho = \beta + i\gamma$ with $\beta > 1/2$. The RH contradiction-field manuscript transfers the resulting smooth prime anomaly into a CRT rough-composite ledger, decomposes it into sparse/dense/tail/internal/global exits, and proves that no named escape channel remains under the recorded local theorems and restricted external inputs.

Theorem 4.1 (RH contradiction-field synthesis, recorded form). *Under the numbered local theorems, controlled-exit lemmas, threshold hierarchy, and restricted external inputs recorded in `paper/rh-proof/rh-contradiction-field.tex`, an off-critical zero has no free escape channel in the contradiction-field event graph.*

Reference. See the RH manuscript in `paper/rh-proof/` and the final audit in `docs/`. The submission warning in the RH manuscript remains in force until independent line-by-line referee verification and final monograph page-number checks are completed. □

Chapter 5

Unified Dependency Map

The two programs share the same logic:

counterexample $\Rightarrow$ structured covering field $\Rightarrow$ named exits $\Rightarrow$ capacity/no-cycle contradiction.

The prime-matrix program is more geometric and finite-CRT in nature; the zeta-zero program is analytic and scale-asymptotic. The monograph treats both as projections of the same contradiction-field method.

Layer	Prime-matrix projection	RH projection
Baseline	CRT nonzero class balance	smoothed CRT candidate baseline
Local rigidity	large-factor non-reuse, diagonal locks	sparse/dense/tail routing
Capacity	anchor coverage capacity	projection and square-function capacity
No-cycle	recursive peeling and shell migration	GEE transfer/no-cycle ledger
External inputs	finite verification and Structured-EHPD	explicit formula, KL, Vaaler, BG/Baker, sieve, Vaughan
Final status	reduction package, not final unconditional theorem	verification package with submission warning

Chapter 6

Directory and Theory System

The detailed working directory and theory-system map is maintained in

`docs/monograph/combined-monograph-directory-and-theory-system.md`.

This chapter records the concise version used by the merged monograph.

6.1 Proposed Book Structure

Part	Contents	Status
I	Unified contradiction-field method: entrance, local rigidity, exits, capacity, no-cycle potential	Methodological framework
II	Prime-matrix row/column program: CRT skeleton, A/B reduction, Structured-EHPD interface	Reduction package; not final theorem
III	Two-point secondary sieve: TLI, BST, BMD, WBE2, BE2–3K, KLS-window	External-deep-theorem closure for BMD
IV	Zeta-zero contradiction field: explicit-formula entrance and controlled exits	Verification package; not final RH theorem
V	Dependency map, referee checklist, optimization routes	Audit and future work

6.2 Latest Two-Point Chain

For the $w = 2$ hardest two-point branch, the current analytic chain is

$$\text{DI+BFI} \Rightarrow \text{KLS-window} \Rightarrow \text{BE2–3K} \Rightarrow \text{BE2–3} \Rightarrow \text{WBE2} \Rightarrow \text{BMD} \Rightarrow \text{BST–2} \Rightarrow \text{BST} \Rightarrow \text{TLI}.$$

The chain is closed only in the external-deep-theorem sense: DI and BFI are cited as classical analytic inputs. A fully self-contained version would require reproving the spectral Kloosterman large-sieve and dispersion machinery.

6.3 Prime-Matrix Low-Hole Closure Package

The latest prime-matrix update closes the low-hole bucket submodule as a finite-certificate plus explicit-external-theorem package. The verification is split into five non-overlapping ranges:

$$\begin{aligned} 13 \le P < 61 &\;:\; \text{low-range final certificate,}\\ 61 \le P \le 103 &\;:\; \text{narrow-band collision-energy certificate,}\\ 107 \le P \le 229 &\;:\; \text{exact } P/5 \text{ splitting recursion,}\\ 233 \le P \le 13207 &\;:\; \text{integer cross-multiplied product certificate,}\\ P \ge 13208 &\;:\; \text{Rosser–Schoenfeld explicit constants.} \end{aligned}$$

The low-range certificate checks 15414 zero buckets: 15282 close by whole-set Hall deficit, 108 close by bridge-critical deletion, and the remaining 24 exceptional phases at $P = 41$ close by exact finite set-cover dynamic programming. The narrow band $61 \le P \le 103$ covers 23100 low phases by fixed ascending high-prime residue blocks. The two finite tail certificates cover 23 and 1520 prime rows respectively, the latter by exact integer cross multiplication rather than floating-point comparison.

For the infinite tail, Rosser–Schoenfeld Corollary 1, formulas (3.5)–(3.6), supplies the prime-counting bounds, while Theorem 7, formula (3.26), supplies the Mertens product bound. Since $P \ge 13208$ implies $\lfloor P/5 \rfloor \ge 2641$, the factor $1 + 1/(2\log^2\lfloor P/5\rfloor)$ is below 1.009, hence the manuscript's conservative 1.03 Mertens constant is admissible.

This closes the BPN low-hole bucket interface in the stated finite-certificate/external-theorem sense. It does not by itself promote the whole prime-matrix row/column program to a final theorem, because the remaining global exits still require the separately named PDEC/SAE and Rankin-ledger acceptance obligations.

6.4 Prime-Matrix WSH-Hall Phase Certificate

The latest row-program compression keeps the endpoint status honest. The terminal RCI branch has been reduced to the comparison between no-tail primes and balanced two-tail semiprimes. The next local interface is

$$\text{RCI/PDEC} \Rightarrow \text{Distributed-RCI} \Rightarrow \text{WSH-Hall/PDEC}.$$

Here WSH-Hall means that every balanced two-tail semiprime in a row can be injectively matched to a same-row prime through an offset allowed by the small-prime wheel. If such a Hall matching fails, the failing interval must be routed to a named exit: persistent endpoint CRT defect, tail-anchor concentration, or endpoint prime deficit.

A finite phase certificate is archived in

`docs/monograph/prime-matrix-wsh-hall-phase-certificate.md`.

For $17 \le p \le 2000$, $y = \lfloor p/e \rfloor$, and wheel primes $2, 3, 5, 7, 11, 13$, it scans 215074 rows containing balanced two-tail semiprimes. It finds no Hall matching failure, no failure under the candidate radius $3\log^2 q$, maximum minimal matching radius 132, maximum ratio $R/\log^2 q = 2.3771082468335174$, and minimum margin #primes − #semiprimes = 1. The certificate rows list semiprime labels, matched primes, offsets, fixed-offset loads, and wheel-phase loads; each matched offset is checked against the congruence condition $d \not\equiv -b \pmod r$.

This is a computational-certificate update, not a global theorem. The remaining proof obligation is either a uniform WSH-Hall theorem or a proof that every Hall defect necessarily triggers PDEC, Tail-anchor, or Endpoint deficit.

6.5 WSH-Hall Defect Trichotomy

The corresponding defect audit is archived in

`docs/monograph/prime-matrix-wsh-hall-defect-trichotomy.md`.

It converts a possible Hall failure into a consecutive block B_0 of balanced two-tail semiprimes with

$$|N_R(B_0)| < |B_0|.$$

The same block is then examined through five projections: endpoint prime deficit, tail-factor load, small-wheel residue load, fixed-offset capacity, and the terminal mirror $n \mapsto q^2 - n$. The resulting target is the named interface

WSH-Expansion-or-Defect.

If Tail-anchor and fixed-offset/PDEC thresholds do not fire, one must prove the expansion inequality

$$|N_R(B_0)| \geq |B_0|$$

for $R = C \log^2 q$; otherwise the failing block must be absorbed by Endpoint/PDEC. A forced-radius audit on the 44 tight certificate rows produces 124 explicit Hall defects and records their tail, phase, offset, and mirror data. This audit fixes the certificate fields and exit geometry; it is not a substitute for the uniform expansion theorem.

6.6 WSH Expansion Margin Ledger

The expansion term itself is audited in

`docs/monograph/prime-matrix-wsh-expansion-margin-ledger.md`.

For $17 \leq p \leq 2000$, $R = \lceil 3 \log^2 q \rceil$, and the same $y = \lfloor p/e \rfloor$, the audit scans every consecutive balanced-semiprime block in every relevant row. It checks 8440419 blocks across 215074 rows and finds

$$\min_B \big(|N_R(B)| - |B|\big) = 1,$$

with no zero-surplus block. Only seven blocks have surplus at most 1.

This strengthens the finite certificate from “a matching exists” to “every interval-Hall block has positive margin” in the audited range. It still does not prove the uniform theorem. The remaining target is:

WSH Positive Expansion or Named Defect.

After Tail-anchor, fixed-offset/PDEC, and endpoint-mirror deficit exits are removed, one must prove the positive expansion inequality for all formal blocks.

6.7 Short Critical Block Reduction

The short-critical-block audit is archived in

`docs/monograph/prime-matrix-wsh-short-critical-block-reduction.md`.

It reruns the expansion-margin audit with tight surplus threshold 2. Among the same 8440419 consecutive semiprime blocks, only 45 have surplus at most 2, and every such block has at most three semiprimes. The distribution is:

surplus	$\|B\|$	count
1	1	4
1	2	3
2	1	24
2	2	10
2	3	4

This suggests the next reduction:

$$\text{SCB–1: } |B| \geq 4 \Rightarrow \text{positive expansion or named defect,}$$

$$\text{SCB–2: } |B| \leq 3 \Rightarrow \text{local exclusion or named defect.}$$

This is still finite evidence and a proof target, not a theorem. It is useful because SCB–2 is a small local problem involving only one, two, or three two-tail semiprimes, where tail labels, fixed offsets, wheel residues, and endpoint mirrors can be written explicitly.

6.8 SCB–2 Local Routing Certificate

The SCB–2 local certificate is archived in

`docs/monograph/prime-matrix-wsh-scb2-local-exclusion-template.md`.

It enumerates all consecutive blocks with $|B| \leq 3$ for $17 \leq p \leq 2000$. The block counts are

$\|B\|$	count	$\min(\|N_R(B)\| - \|B\|)$
1	1496400	1
2	1281326	1
3	1088870	2

There are no negative-surplus or zero-surplus short blocks. The local routing statement is:

$$|B| \leq 3, \quad |N_R(B)| < |B| \Rightarrow \text{Endpoint/PDEC deficit,}$$

with additional Tail-repeat and fixed-offset-full-load checks on tight two- and three-point blocks. Thus SCB–2 is closed as a routing lemma modulo the still-open Endpoint/PDEC exclusion.

6.9 SCB–1 Long-Block Certificate

The complementary long-block certificate is archived in

`docs/monograph/prime-matrix-wsh-scb1-long-block-expansion-template.md`.

It enumerates all consecutive balanced two-tail semiprime blocks with $|B| \geq 4$ for $17 \leq p \leq 2000$ and the same radius $R = \lceil 3 \log^2 q \rceil$. The audit checks 4573823 long blocks and finds

$$\min_{|B| \geq 4} \big(|N_R(B)| - |B|\big) = 3,$$

with no negative-surplus or zero-surplus long block. The four tight long blocks have sizes 4 or 5, and every tight block carries both Endpoint-margin and fixed-offset-full-load labels.

This is a finite certificate, not the uniform SCB–1 theorem. Its proof-theoretic role is to isolate the next target:

$$\text{long-block positive expansion} \quad \text{or} \quad \text{fixed-offset/PDEC absorption.}$$

Together with SCB–2, the present WSH chain is therefore:

$$\text{SCB–2 routing modulo Endpoint/PDEC,} \qquad \text{SCB–1 finite long-block margin,}$$

The remaining global obstructions are fixed-offset/PDEC absorption and Endpoint/PDEC exclusion.

6.10 Fixed-Offset/PDEC Absorption

The fixed-offset absorption interface is archived in

`docs/monograph/prime-matrix-wsh-fixed-offset-pdec-absorption.md`.

It proves the no-third-escape lemma behind the fixed-offset-full-load label. If $p < q$ are consecutive primes and $1 < n < q^2$ is composite, then n has a prime factor at most p. Hence, if a fixed offset d is allowed by the wheel $2, 3, 5, 7, 11, 13$ for every $b \in B$, every missing candidate $b + d$ is explained by some factor in $(13, p]$.

The finite ledger

`docs/monograph/prime-matrix-wsh-fixed-offset-pdec-ledger.md`

checks the tight SCB–1 long blocks. It finds 11 full-load offset rows and 47 candidates, among which 15 are prime and 32 are missing. Every missing candidate has an explaining factor at most p; there is no unexplained candidate.

Thus fixed-offset-full-load is no longer an unnamed escape. It routes to expansion, Tail-anchor, PDEC/low-mod CRTDefect, or SAE/Endpoint. The still-open global target is

$$\text{FO-PDEC: distributed explaining factors} \Rightarrow \text{persistent PDEC or sparse SAE/Endpoint.}$$

6.11 FO-PDEC Low-Mod Equations

The low-mod equation audit is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-hard-attack.md`.

For every missing candidate $n = b + d = (r - 1)q + c$ and every explaining factor ℓ, the exact CRT equation is

$$r \equiv 1 - cq^{-1} \pmod{\ell}.$$

If $b = uv$ is the corresponding two-tail semiprime, the same missing candidate also satisfies

$$uv + d \equiv 0 \pmod{\ell}.$$

The finite audit checks 43 such low-mod equations in the tight fixed-offset ledger. There are no CRT-equation failures and no bilinear-equation failures.

This closes the equation layer of FO-PDEC. It does not close the global theorem. The remaining target is the quantitative inequality

$$\text{low-mod defect energy} \geq \text{PDEC threshold},$$

unless the non-persistent part is absorbed by SAE/Endpoint.

6.12 FO-PDEC Energy Lower Bound

The energy ledger is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-energy-lemma.md`.

For a multiset E of explaining equations, let t_ℓ be the number of equations using ℓ and let s_ℓ be the number of occupied row residues modulo ℓ. Define

$$\mathcal{E}_\ell(E) = t_\ell - \frac{s_\ell t_\ell}{\ell}, \qquad \mathcal{E}(E) = \sum_\ell \mathcal{E}_\ell(E).$$

For every $0 < \theta < 1$, either some explaining factor has high load $t_\ell > \theta\ell$, which is a Tail/PDEC exit, or

$$\mathcal{E}(E) \geq (1-\theta)|E|.$$

Thus FO-PDEC now produces low-mod energy unconditionally, modulo the named high-load exit.

The finite ledger with $\theta = 0.25$ gives global energy per equation

$$0.9547817296210457$$

and no high-load factor. The remaining obstruction is not energy production; it is the final threshold comparison on the same bad-window set:

$$U_{\mathrm{CRT}}(E) < \mathcal{E}(E) \quad \text{or} \quad L_{\mathrm{PDEC}}(E) \leq (1-\theta)|E|.$$

6.13 FO-PDEC Explicit Threshold

The explicit threshold ledger is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-threshold-comparison.md`.

For each explaining factor ℓ, it computes the nonzero Fourier threshold

$$M_\ell(E) = \max_{1 \leq h < \ell} \left| \sum_{\rho \bmod \ell} g_\ell(\rho) e^{2\pi i h\rho/\ell} \right|.$$

The current best projection is

$$\ell = 199, \qquad h = 95, \qquad M_\ell(E) = 3.959247567099438.$$

Thus the remaining task is now concrete:

$$U_{\mathrm{CRT},199} < 3.959247567099438$$

for the same vector g_{199}. Otherwise this projection must be absorbed by SAE/Endpoint.

6.14 FO-PDEC Dual-Cluster Obstruction

The dual-cluster audit is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-dual-cluster-route.md`.

For the best projection, the mass is 4 and the Fourier threshold is 3.959247567099438, so the trivial mass bound misses closure by only

$$0.04075243290056196.$$

The best frequency $h = 95$ sends the support to a short dual arc of length 11 modulo 199:

$$40, 126, 61 \mapsto 19, 30, 24.$$

Thus the last obstruction is a concrete dual-cluster exclusion problem: this short dual arc must be ruled out by a PDEC dual certificate, or else absorbed as a non-persistent SAE/Endpoint event.

6.15 FO-PDEC Multiplicity Legitimacy

The primitive-cluster audit is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-multiplicity-legitimacy.md`.

It checks whether the best projection counts independent bad-window equations or repeated physical candidates. Under the equation or block-local convention, the projection has mass 4 and Fourier value

$$3.959247567099438.$$

Under physical deduplication, it has mass 2 and Fourier value

$$1.9699193446802263.$$

Thus the final PDEC comparison must first fix the multiplicity convention. The lower bound and the upper bound must be computed on the same multiset. Otherwise the argument must switch to the primitive threshold or absorb the repeated events by SAE/Endpoint.

6.16 FO-PDEC Formal Unit Consistency

The formal-unit audit is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-formal-unit-route.md`.

It checks a stricter requirement: the PDEC vector $g(t)$ must come from one formal bad-window set or multiset with one phase map. The current strong value

$$3.959247567099438$$

is obtained by pooling the finite certificate library. After splitting by formal unit, the audit gives

convention	best Fourier value
global library	3.959247567099438
layer deduplication	2.9698366905785227
physical deduplication	1.9997507790353146
single coordinate-deduplicated formal unit	1.0.

Thus the strong threshold cannot yet be used as a single-branch PDEC lower bound. One must prove FormalUnit-Stitching for the cross-q events and NestedBlock-Independence for exact nested-coordinate duplicates, or route the repeated events to SAE/Endpoint. This is now the narrowest FO-PDEC obstruction before any global unconditional conclusion can be claimed.

6.17 FO-PDEC Branch Separation

The branch-separation theorem and stitching audit are archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-branch-separation-theorem.md`.

They separate two issues. First, events from different matrix widths q cannot be pooled into one PDEC vector without a cross-q persistence theorem. Second, two occurrences with the same

$$(q, row, candidate\_row, column, offset, n, \ell, \rho)$$

give the same CRT equation and must be quotiented unless a weighted Hall-dual independence row is supplied. In the present finite ledger there are 7 exact nested-coordinate duplicates and 9 cross-level physical reuses. After coordinate deduplication inside a single formal unit, the best Fourier value recorded by the audit is only 1.0. Therefore the next narrow obstruction is Weighted Hall Dual Independence, or else SAE/Endpoint absorption of the exact duplicates.

The subsequent dominance audits discharge these finite FO–PDEC subgates without promoting the full theorem. The nested duplicates are laminar repeated coordinates; the fractional weighted Hall differences add one semiprime and one neighboring prime, with zero surplus, so they cannot pay for a second independent unit of mass. The cross-q reuses are coordinate-chart overlaps of the same physical candidate, not independent persistent events. Thus the raw library value 3.959247567099438 and the coordinate-cap value 2.9698366905785227 cannot be used as single-formal-unit lower bounds in the audited branch.

6.18 FO-PDEC Physical Primitive Tautology and SAE Absorption

The physical primitive audit is archived in

`docs/monograph/prime-matrix-wsh-fo-pdec-physical-primitive-tautology.md`.

After physical deduplication the current 199-factor frontier has only two physical atoms. For a prime modulus ℓ and two distinct residues a, b, choosing

$$h \equiv (b-a)^{-1} \pmod{\ell}$$

makes the dual residues adjacent. Hence every two-point support satisfies

$$\max_{h \neq 0} |e(ha/\ell) + e(hb/\ell)| = 2\cos(\pi/\ell).$$

For $\ell = 199$ this is exactly

$$2\cos(\pi/199) = 1.9997507790353146.$$

Therefore the remaining physical primitive value is a two-point Fourier tautology, not a usable PDEC defect threshold.

The two atoms are then routed to the SAE/Endpoint interface and audited in

`docs/monograph/prime-matrix-wsh-fo-pdec-sae-endpoint-absorption.md`.

Each associated fixed-offset fiber has a local prime survivor: the 250541 atom has witness 250543, and the 1664237 atom has witness 1664227. In all four source rows the 199-factor load in the same fiber is 1. Consequently the audited two-atom primitive branch is absorbed by LocalSurvivor witnesses.

This closes only the current finite $\ell = 199$ FO–PDEC sample chain:

$$\begin{gathered}\text{raw library signal} \to \text{nested/weighted rejection,}\\ \text{coordinate-cap signal} \to \text{cross-}q \text{ chart-overlap rejection,}\\ \text{physical two-point signal} \to \text{Fourier tautology,}\\ \text{two physical atoms} \to \text{LocalSurvivor witness absorption.}\end{gathered}$$

It does not close the global prime-matrix theorem. The next honest frontier is the global LocalSurvivor certificate family, or a future non-tautological primitive PDEC family with at least three physical atoms or an extra fixed-frequency constraint. The non-tautological PDEC admission audit records that the current materialized frontier has no such candidate: the raw three-point signal is not admissible as one primitive formal unit, nested and weighted duplicate recovery are blocked, cross-q reuse is only chart overlap, and the physical primitive remnant has two atoms. Those two atoms are already absorbed by LocalSurvivor/SAE witnesses. Thus any future PDEC certificate must introduce a genuinely new same-formal-unit primitive family with at least three physical atoms, or an additional constraint defeating the two-point tautology.

6.19 Materialized LocalSurvivor Packet Ledger

The current LocalSurvivor packet ledger is archived in

`docs/monograph/prime-matrix-local-survivor-materialized-packet-ledger.md`.

It combines the Triad-A1 SparseCap ledger, the FO–PDEC two-atom SAE/Endpoint absorption ledger, and the RPZ Endpoint-SAE finite ledger. The combined current ledger records

quantity	value
materialized packets	9
materialized phase atoms	32
local survivor witnesses	5
finite PDEC atoms	25
open materialized obligations	0.

Thus every LocalSurvivor/SAE packet currently materialized by the proof frontier is either witnessed, vacuous in the current ledger, or routed to a finite PDEC packet beyond the early $P \times P$ rows.

The packet-generation contract is archived in

`docs/monograph/prime-matrix-local-survivor-packet-generation-contract.md`.

It records the structural dichotomy for any future sparse escape. If the window data can be extracted, it is a finite LocalSurvivor packet and must be checked by a witness or blocker-deficit certificate. If the same finite signature persists along an infinite counterexample family, it is no longer a sparse terminal: according to the repeated signature it returns to PDEC, ColumnCRT, endpoint PDEC, TailAnchor, or CofactorAnchor. If the finite signatures do not persist and the layers keep escaping, the branch enters CleanKLS/DLS admission.

Consequently the next narrow frontier is

$$\mathrm{LocalSurvivorPacketGenerationOrNonTautologicalPDEC}.$$

This is still not a final unconditional theorem. It says that the current materialized LocalSurvivor obligations are exhausted and that no unnamed LocalSurvivor terminal remains; the remaining work is packet-extractor completeness, a genuinely non-tautological primitive PDEC family, or the CleanKLS/DLS and referee inputs.

The known-entry extractor audit is archived in

`docs/monograph/prime-matrix-local-survivor-packet-extractor-coverage.md`.

It checks eight currently named LocalSurvivor entry types. Four are covered by machine extractors: Triad-A1 SparseCap, FO–PDEC two-atom SAE/Endpoint, RPZ Endpoint-SAE, and the materialized packet aggregator. Four are covered by contract routes: generic SAE, persistent finite signatures, layer escape to CleanKLS/DLS, and descent/seam return. The audit has no missing or open known entry.

The new-entry admission audit is archived in

`docs/monograph/prime-matrix-new-sparse-entry-admission-audit.md`.

It checks the possible sources of a future sparse entrance: short-window or endpoint sparse packets, PDEC dual sparse caps, endpoint seams, Bohr caps, column-tail/cofactor sparse packets, descent seams, persistent signatures, and layer escape. Each is already admitted to a named route: SAE/LocalSurvivor, PDEC, ColumnCRT, TailAnchor/CofactorAnchor, or CleanKLS/DLS. Hence there is no additional unnamed LocalSurvivor entry route in the current contract system.

After these two audits the LocalSurvivor branch is conditionally exhausted: any future explicitly introduced sparse route must bring its own extractor schema and finite ledger. The current narrow frontier is therefore

$$\mathrm{NonTautologicalPDECOrCleanKLS}.$$

This remains a reduction boundary, not the final prime-matrix theorem.

The clean KLS side is no longer an unnamed black-box exit in the current routing table. The A1 clean-admission router checks that K1–K9 failures return to PDEC/SAE/Multiplicity/Promotion, while a fully admitted clean residual enters the Kuznetsov–LS atom SC-9. The SC-9 frontier router expands this atom into KZ-A–KZ-E: KZ-A–KZ-D are routed by smoothing, trace specialization, Bessel decay, and the spectral-large-sieve/pretrace chain; KZ-E remains the actual WFD block non-concentration input NC-BLK, or else a precisely matched external DI/BFI original dispersion theorem. Thus the updated narrow frontier is

$$\mathrm{NonTautologicalPDECOrNCBLK}.$$

This also remains a reduction boundary, not a final unconditional row/column theorem. The NC-BLK boundary reconciliation audit records the exact reading of this label. On the canonical RIW/Buchstab source branch, the NC-BLK clean-KLS chain is already absorbed by the existing same-set capacity boundary. On the broader full-S non-AP generic branch, NC-BLK remains an exact-source-entropy or external DI/BFI/Kuznetsov route and is not imported into the self-contained theorem. Thus the label no longer denotes an unnamed CleanKLS exit, but it also does not promote the whole row/column theorem.

The subsequent global-terminal-family boundary router updates the frontier one more time. After the NC-BLK reconciliation and the non-tautological PDEC admission audit, the current materialized frontier has no local terminal instance left to eliminate: the non-tautological primitive PDEC candidate count is zero, the materialized LocalSurvivor ledger has no open packet, all known sparse entry routes are admitted, and NC-BLK is reconciled with the canonical boundary. The router records the new narrow label

`CurrentMaterializedFrontierExhausted_GlobalTerminalFamiliesOpen`.

Equivalently, the next required theorem is no longer a fixed numerical improvement for the current $\ell = 199$ sample or a new reading of NC-BLK. It is the global terminal-family exclusion package: every terminal object emitted by a minimal counterexample must be certified by a PDEC-family exclusion, a LocalSurvivor witness/deficit certificate, a CleanKLS/DLS internal large-sieve certificate, or an explicitly accepted external/referee input. This is again a boundary closure, not a final unconditional row/column theorem.

The split router for this global terminal family then removes the remaining ambiguity in the word "family". The non-final reductions are closed: there is no fourth terminal route, the current LocalSurvivor and NC-BLK ledgers are not independent blockers, and the finite-projection terminal dichotomy sends positive-limsup mass to PDEC-CAP and diffuse mass to CleanKLS/DLS. Thus the self-contained mathematical remainder is

`PDEC_CAP_OR_INTERNAL_CleanKLS_LargeSieve`,

while final promotion also keeps the D-structure/Rankin referee gate. In particular, the frontier is not an unnamed local sample problem; it is the pair of named analytic terminal estimates plus the referee interface.

The self-contained terminal bottleneck router then removes one more apparent branch. Inside the current canonical-source boundary, InternalCleanKLS is not an independent minimal blocker: a clean large-sieve failure returns a dual concentration object to PDEC/SAE, the canonical NC-BLK/CleanKLS branch is already absorbed by the same-set capacity boundary, and the unrestricted generic full-S WFD self-contained theorem is refuted and not claimed. Hence the independent self-contained mathematical bottleneck is

`PDEC_CAP_SameSetGlobalDualCertificate`,

namely a same-bad-window global dual capacity certificate $U_{\mathrm{CRT}} < L_{\mathrm{PDEC}}$ for the PDEC families. The D-structure/Rankin referee gate remains a separate final-promotion condition, so this is still not a final unconditional row/column theorem.

The next PDEC-CAP router then opens this named bottleneck rather than treating it as a black box. All currently materialized PDEC intermediate gates are routed: the DualCap families have same-M_Q mass sources and closed early P-row exits; the current PersistentCaps route through promotion deletion or NoDeletion–KL/PDEC/CleanKLS; the current ForcedCaps route into multi-bucket ActualPaymentStitching; and LHB projection-stitching is removed by the common full-period completion set. The profinite ActualPaymentStitching router then closes the remaining APS logic by the finite-projection pigeonhole dichotomy. The PDEC-CAP theorem is still not proved. Its current narrow internal frontier is now the single terminal estimate

`PrimitiveMultiAtomSameFormalUnitPDECCertificate`.

Persistent real payment Γ and fixed-shell low-mod persistence are now treated as the same finite-signature formal-unit problem: the former is a phase-bucket/tail-column signature, and the latter is a shell/displacement signature. ColumnCRT displacement is absorbed as displacement PDEC, PDEC dual failure is absorbed as cap refinement, and formal-unit mismatch is absorbed by the multiplicity-stitching normalization. Thus the single persistent terminal is PersistentFiniteSignaturePDECColumnCRT. The flat-clean SC–9 branch has been reconciled with the canonical-source boundary: clean failure returns to PDEC/SAE, canonical NC-BLK is absorbed by the same-set boundary, and generic WFD is not imported into the self-contained claim. Finally, the persistent terminal admission router removes raw ColumnCRT, dual-failure, multiplicity, and two-point-tautology pseudo-terminals. Thus the remaining admitted object must be a primitive multi-atom same-formal-unit PDEC certificate. The intermediate diffuse split is now routed: persistent KL or phase-residue mutual information returns to refined/new-layer PDEC, while only the KL-flat branch may enter CleanKLS/DLS; divergent deletion already implies support exhaustion or a named sparse/PDEC return, and HRO reduces deletion non-divergence to occupancy saturation or TailIndependence. The latter is already NoDeletion/CleanKLS. The occupancy-kernel router then applies the HRO injection bound $|\mathrm{Occ}_t| \le \min(|H_Q(t)|, r)$: sparse old-hole or residue-deficient phases cannot saturate, and any remaining saturation must contain a near-full old-hole selector kernel. The common-variable router writes $c_b = \rho_b + rk_b$, turning every old-prime ban into one forbidden residue for the same shell variable k_b. Hence no legal shell is a capacity/Hall deletion, fixed-shell mass is PDEC/ColumnCRT persistence, and shell-dispersed flat mass is the SC–9 atom. External KLS input closes the flat branch only as an external-theorem version.

6.20 Triad-A1 Self-Contained Theorem-Boundary Closure

The final Triad-A1 review separates the exact theorem that is now closed from a stronger statement that is not true under the recorded hypotheses. In plain terms, the closed result is the following. Once the source of the A1 payment measure is fixed to the canonical RIW/Buchstab decision-tree source, every remaining same-set capacity gate in the full-S terminal chain is internally accounted for. The proof no longer needs to borrow the external FullS–KLS theorem for this canonical-source branch.

Proved-in-text Statement 6.1 (Canonical-source self-contained boundary)**.** *The Triad-A1 same-set capacity / full-S terminal is self-contained on the canonical RIW/Buchstab source branch.*

Recorded proof certificate. The certificate is recorded in

`docs/monograph/prime-matrix-triad-a1-self-contained-theorem-boundary-review.md`.

It checks seven gates: exact theorem statement boundary, final closure certificate, absorption by the same-set frontier, canonical-source provenance, branch coverage without silent generic upgrade, no false generic claim, and separation of the external FullS–KLS contract. The review verdict is

`APPROVE_CANONICAL_SOURCE_SELF_CONTAINED_BOUNDARY`,

with no open review gate. □

Not-claimed Statement 6.2 (Unrestricted generic WFD self-contained theorem)**.** The manuscript does not claim an unrestricted generic full-S well-factorable WFD self-contained theorem.

Reason for non-claim. The generic anti-atom input needed for that broader statement is refuted by the moving-delta model recorded in the A1 no-go audit. Hence the generic WFD version is not an open missing lemma of the canonical-source proof; it is a different and stronger statement that is false under the current formal hypotheses. The external FullS–KLS contract remains available as an external-theorem route, but it is kept separate from the self-contained canonical-source claim. □

Remark 6.3 (Boundary in everyday language)**.** The row-program terminal proof has reached a closed theorem-boundary for the canonical source: the bookkeeping, provenance, and branch split now agree on what is proved. What remains outside this boundary is not a hidden gap in that theorem. It is either a broader generic WFD theorem that the current model refutes, or one of the separate global prime-matrix terminal certificate obligations listed below for promoting the whole row/column program to a final unconditional theorem.

6.21 Claim Status Legend

The monograph uses the following status labels.

Label	Meaning
Proved-in-text	The statement is proved inside the manuscript under the stated hypotheses.
Reduction-closed	The statement has been reduced to a smaller named interface.
External-theorem closed	The statement is closed after citing explicitly named external theorems.
Computational-certificate	The statement is closed over a finite range or certified submodule by reproducible scripts and archived outputs.
Referee-block	A proof package exists, but it still requires line-by-line referee verification.
Not claimed	The statement is not asserted as a theorem of the manuscript.

The status discipline and promotion rules are maintained in

`docs/monograph/claim-status-discipline.md`.

The manuscript provides theorem-like environments with matching names for future editing. A statement may be promoted only when its required proof, reduction endpoint, external theorem template, or certificate package is present.

6.22 External Theorem Index

The external-input index is maintained in

`docs/monograph/external-theorem-index.md`.

For the prime-matrix low-hole tail, the current explicit package is Rosser–Schoenfeld 1962: Corollary 1, formulas (3.5)–(3.6), supplies the required $\pi(x)$ bounds, and Theorem 7, formula (3.26), supplies the required Mertens product bound. These constants close the $P \geq 13208$ tail in the external-theorem sense once the finite certificates below that threshold are accepted.

For the two-point sieve, the critical external package is DI plus BFI. The index records the required checks for the KLS-window input: Kloosterman phase normalization, modulus range, frequency range, inverse-variable support, well-factorable weights, gcd strata, smoothing and sawtooth losses, and the final choice of the logarithmic-loss parameter $B(A)$. The standalone adaptation template is maintained in

`docs/monograph/kls-window-di-bfi-adaptation-template.md`.

It closes the BMD distribution input only in the external-deep-theorem sense; the separate BMD-to-TLI transfer remains a hard obligation.

6.23 Critical Chain Audit

The detailed chain audit is maintained in

`docs/monograph/key-proof-chain-optimization-audit.md`.

Its main purpose is to prevent status mixing. Prime-matrix existence rigidity may support zero-section exclusion, but it does not replace signed spectral distribution estimates such as BMD-Char. Numerical experiments may guide constants and structures, but they are not proof inputs. The RH branch remains a verification package and is not claimed as an unconditional proof.

6.24 Optimization Priorities

1. Separate theorem environments for proved-in-text, reduction-closed, external-theorem closed, and referee-block statements.
2. Replace the overly strong $\max_a$ BV–E_2 formulation by the precise WBE2 well-factorable weighted statement in the main line.
3. Rewrite KLS-window directly in the notation of the cited DI/BFI theorems to reduce translation risk.
4. Compress the prime-matrix program around the single Structured-EHPD exclusion interface.
5. Preserve the RH warning language until every controlled exit has independent referee verification.

6.25 Pre-External-Referee Hard Obligations

The current hard-obligation ledger is maintained in

`docs/monograph/pre-external-referee-remaining-obligations.md`.

It records the author-side status of every terminal interface that cannot honestly be promoted to a final theorem without further proof or independent verification. The decisive interfaces are:

Program	Hard interface	Required closure before theorem promotion

Prime matrix	Structured-EHPD / D-structure exclusion	Prove the exclusion theorem directly, or replace it by a fully proved RHI-type large-prime incidence inequality.
Prime matrix	PDEC / SAE / Rankin exits	Prove persistent endpoint CRT defect exclusion, local isolated-escape exclusion, and complete the formal Rankin certificate ledger.
Prime matrix	RRD / OSPC / Selberg-uniform constants	Put all constants in one convention and prove each budget item over the full asymptotic range, not only in sampled windows.
Two-point sieve	I3-Core true-residual package	Prove true-residual new-modulus balance, TCA, SC2, 45° synchronization peak control, Directional Balance, Zero Mass, and endpoint smoothing.
Two-point sieve	DI/BFI to KLS-window adaptation	State the external theorems in their standard variables and verify phase, modulus, frequency, weights, gcd strata, and logarithmic-loss absorption.
Two-point sieve	BMD to TLI transfer	Prove the Buchstab transfer and total-large-incidence comparison without using a hidden prime-pair lower bound.
RH branch	controlled exits	Prove every sparse, dense, tail, internal, analytic, and global exit under a single load convention.

For the RRD/OSPC constants, the current acceptance matrix is maintained in

`docs/monograph/h5-rrd-ospc-proof-obligation-matrix.md`.

It splits the remaining constant obligation into six independently checkable inequalities and does not claim that those inequalities have already been proved. The first item, RRD-low routing, is recorded in

`docs/monograph/h5-1-rrd-low-exit-theorem.md`.

It proves that an RRD-low excess enters either OSPC* or a weighted CRT-defect exit; excluding those exits remains part of the PDEC/SAE and tail-anchor obligations. The absorption of these two exits into the unified PDEC-or-SAE interface is recorded in

`docs/monograph/h5-4-ospc-weighted-crtdefect-absorption.md`.

This absorption is not an exclusion theorem; the required PDEC and SAE certificates remain open hard obligations. The PDEC certificate format itself is now recorded in

`docs/monograph/h4-pdec-certificate-template.md`.

It fixes the data, Fourier lower bound, admissible CRT upper-bound certificates, and failure-routing rules for persistent defects. This is a certificate-format closure only; the actual PDEC certificates for the formal bad-window families remain to be supplied. The first source lemmas for admissible PDEC constraint rows are recorded in

`docs/monograph/h4-pdec-constraint-source-lemmas.md`.

They justify mass, phase-capacity, mirror-pair, low-hole-bucket, and conditional-routing rows when their stated hypotheses hold. They do not by themselves provide the full constraint table or the final dual upper bound. The first admissibility table for those rows is recorded in

`docs/monograph/h4-pdec-admissible-constraint-table.md`.

It separates tautological, finite-certificate, symbolic, conditional-routing, still-unproved, and rejected rows. In particular, full-cycle CRT balance is not allowed to control an arbitrary bad-window subset. The column-cap source rule is recorded in

`docs/monograph/h4-pdec-column-cap-source-lemma.md`.

It allows column information to enter only through finite column-projection capacity, symbolic column-capacity theorems, or conditional routing to ColumnRadius/ColumnCRT/Tail-anchor exits. The coefficient ledger for those rows remains open. The first coefficient ledger is recorded in

`docs/monograph/h4-pdec-column-cap-coefficient-ledger.md`.

It registers finite column-witness, RCI/CDB, LHB column-residue, and conditional ColumnDefect rows. Most rows still require machine-readable phase blocks before they can become actual rows of the dual certificate matrix. The first materialized phase-block file is

`docs/monograph/h4-pdec-lhb-column-phase-blocks.json`.

It gives finite $Q = 2310$ LHB column rows for $13 \leq p \leq 47$. Empty anomaly blocks are machine-readable finite rows; diagnostic support blocks still require a projection-capacity proof before entering the final dual system. The support-to-capacity transfer conditions are recorded in

`docs/monograph/h4-pdec-lhb-support-to-capacity-transfer.md`.

They make explicit that support size is not a multiplicity bound for the persistent count vector $g(t)$; a phase-indicator reduction, a multiplicity cap, or an allowed-set projection bound is still required. The multiplicity route is made explicit in

`docs/monograph/h4-pdec-lhb-multiplicity-cap-route.md`.

It states the precise acceptance contract: for the same (p, Q, S, τ) one must prove $g(t) \leq M(t)$, and only then may a diagnostic block C be promoted to the capacity row $\sum_{t \in C} g(t) \leq \sum_{t \in C} M(t)$. The first finite multiplicity-cap certificate is

`docs/monograph/h4-pdec-lhb-multiplicity-cap-certificate.json`.

It takes $M(t)$ to be the high-layer CRT completion count $C_P(t; Q)$ and verifies, for $Q = 2310$ and $p = 13, 17, 19, 23, 29, 31, 37, 43, 47$, that the WHOLEDEF and BRIDGED support blocks have $\sum_{t \in C} M(t) = 0$. This is an allowed-set projection certificate: it can be used in the LHB branch after proving $S \subset Z_{\mathrm{LHB}}(p, Q)$, but it is not by itself a global PDEC exclusion theorem. The attachment lemma for that branch is

`docs/monograph/h4-pdec-lhb-attachment-lemma.md`.

It proves that an LHB-typed full-row bad window is indeed an element of $Z_{\mathrm{LHB}}(p, Q)$. Thus the remaining global task is no longer this attachment step, but the classification step: every formal PDEC bad window must either be LHB-typed or be routed to a named SAE, ColumnCRT/ColumnRadius, TailAnchor, or Rankin exit. The classification step is recorded in

`docs/monograph/h4-pdec-bad-window-classification-lemma.md`.

It combines the unified PDEC-or-SAE dichotomy, column-conflict routing, tail-anchor persistence, and Rankin access rules to split each named low-mod bad-window family into SAE, LHB-PDEC, or Routed-PDEC. This is still a routing theorem: the non-LHB exits and the final dual $U_{\mathrm{CRT}} < L_{\mathrm{PDEC}}$ certificate remain open. The homogeneous-splitting rule is recorded in

`docs/monograph/h4-pdec-homogeneous-splitting-lemma.md`.

It closes the bookkeeping case in which one attempted certificate mixes different (Q, τ, F, κ, p) types: such a family must be partitioned into homogeneous subfamilies before applying UPS-1 or the bad-window classification. Thus mixed type is not an additional mathematical exit; the remaining exits are SAE, ColumnCRT/ColumnRadius, TailAnchor, and Rankin. The column-defect routing contract is recorded in

`docs/monograph/h4-pdec-column-defect-routing-contract.md`.

It defines the conditional certificate objects for ColumnRadius and ColumnCRT rows: radius rows use a threshold D_0, a column-witness selector, and phase-compatible weights $W_D(t)$, while displacement rows use a label ℓ, a non-zero displacement residue a, a threshold L_D, and phase-compatible weights $W_{\ell,a}(t)$. Violating such a row routes to the named ColumnRadius or ColumnCRT exit after the zero displacement class has been separated by the same-column prime-witness rigidity. This is a routing-contract closure only; the exits themselves still require materialized weights, thresholds, and exclusion certificates. The first finite materialization of those weights is recorded in

`docs/monograph/h4-pdec-column-defect-weight-certificate.json`.

It uses the refined finite phase $\tau_{\mathrm{fin}} = (p, q, \mathrm{row})$ for $p \leq 1000$ and the five tightest RCI rows for each p. In this finite domain, the blocks $D_{\mathrm{col}} > 81$ and displacement-residue load > 2 are both empty, over 835 phases. This is a finite refined-phase certificate; it does not provide a global D_0, L_D theorem or exclude the ColumnRadius/ColumnCRT exits. The recursive zero-row peeling route is isolated in

`docs/monograph/prime-matrix-recursive-peeling-zero-row-hardpoint.md`.

It proves the exact layer-peeling identity: after a p-sieve zero window is peeled to the previous prime level, any resurrected survivor must be a multiple of the peeled prime with rough cofactor. Thus the recursive object is a punctured zero window, not automatically a lower-level zero row. The accompanying finite audit finds no forced consecutive lower zero rows in the known first-zero-row samples. The useful remaining target is therefore an absorption theorem routing those resurrected points to ColumnCRT, ColumnRadius, or TailAnchor, rather than a direct short-period contradiction. The adjacent-shell refinement is recorded in

`docs/monograph/prime-matrix-adjacent-shell-recursive-descent-route.md`.

For adjacent primes $p < q$, it proves that the only composite old p-sieve survivor in $[1, q^2]$ is q^2. Hence a non-first q-zero row first descends to an old p-sieve q-window, with only the terminal q^2 puncture in the last row. That window either contains a full p-aligned zero row or is a seam window whose two guard intervals have lengths a and $p - (q - p) - a$. The resulting narrow target is SeamGuard-Elimination: guards cannot absorb all forced survivors without entering SAE, PDEC, or ColumnCRT. The multi-level seam descent audit is recorded in

`docs/monograph/prime-matrix-seam-multilevel-descent-route.md`.

In this conditional model, if a seam interval is already an old p-sieve zero window, then after descending to a lower prime $h < p$ the only resurrected points are those with $P^-(n) \in (h, p]$, together with the terminal q^2 puncture in the last q-row. Any full h-aligned row avoiding these resurrected points is therefore a forced h-zero row. The finite ledger checks all 21339 seam windows for $p \le 500$ and a deterministic $p \le 2000$ sample of 11488 seam windows, with no persistent blocking cases. This is not a global theorem; it sharpens the remaining gate to the SMD-Global Inequality, or else a proof that persistent revival blocking enters SAE, PDEC, or ColumnCRT. The tail-mirror refinement is recorded in

`docs/monograph/prime-matrix-seam-tail-mirror-descent-route.md`.

For a forced lower h-zero row with row number R, set $N_h = M_h/h$ and $\rho = ((R-1) \bmod N_h) + 1$. If $\rho \le h$ or $N_h - \rho + 1 \le h$, then periodicity or CRT reflection places a conditional zero row inside the lower $h \times h$ matrix. The finite ledgers above again show full success in the same domains. This motivates the sharper TailMirror-SMD gate: every genuine seam counterexample must hit such a lower head/tail phase, or its persistent non-hit phases must be routed to SAE, PDEC, or ColumnCRT. The all-branch version is recorded in

`docs/monograph/prime-matrix-total-zero-row-recursive-descent-route.md`.

It treats aligned containment, seam windows, and the terminal q^2 puncture by the same descent rule. In the $p \le 500$ ledger all 21936 non-first q-rows, including 597 aligned rows and 21339 seam rows, force a lower matrix head/tail phase. In the deterministic $p \le 2000$ sample all 11583 checked rows do the same. The remaining theorem-level statement is TotalDescent-TM: every genuine higher zero row must force a lower $h \times h$ zero row by periodicity or tail reflection, or else its non-hit phase set must enter SAE, PDEC, or ColumnCRT. The fixed $h = 3$ refinement is recorded in

`docs/monograph/prime-matrix-total-descent-h3-margin-route.md`.

Each complete 3-row contains a unique candidate c_R coprime to 6. If $5 \le P^-(c_R) \le p$, that row is blocked by a resurrected point when descending from the p-sieve to the 3-sieve; if $P^-(c_R) > p$, then c_R is already a survivor of the old p-sieve and contradicts the assumed p-sieve zero window. Thus the narrowest remaining form of TotalDescent is the six-wheel inequality

$$\#\{R : J_R \subset I_s\} > \#\{R : J_R \subset I_s,\ 5 \le P^-(c_R) \le p\}.$$

The audit

`docs/monograph/prime-matrix-total-descent-h3-margin-audit.md` checks this for all 1,552,462 non-first q-rows with $p \le 5000$, with minimum margin 1. This is a finite certificate and a sharpened interface, not a global theorem: a final proof must establish the six-wheel inequality uniformly, or route full blocking to SAE/PDEC/ColumnCRT. The hard-attack reduction

`docs/monograph/prime-matrix-h3-sixwheel-hard-attack.md`

clarifies the remaining obstruction. The direct six-wheel inequality sits at the Buchstab/linear-sieve boundary $u = 2$, i.e. at square-root interval length near $x = q^2$. Therefore a proof cannot be obtained by a generic short-interval theorem or a finite template. If the six-wheel candidates are all blocked, they admit the first-factor partition $n = \ell m$ with $5 \le \ell \le p$, m in an interval of length q/ℓ, and $P^-(m) \ge \ell$. The next theorem-level gate is consequently an H3 full-blocking defect theorem: high first-factor load enters Tail/PDEC, distributed low-load coverage produces low-mod PDEC energy, and endpoint residue concentration is SAE or ColumnCRT. The near-miss audit

`docs/monograph/prime-matrix-h3-full-blocking-defect-audit.md`

checks the same ledger for all $p \le 5000$. Among 1,552,462 non-first q-rows, only 4015 have margin at most 20, and these occur only up to $p = 317$ and $q = 331$. Their aggregate first-factor load is led by the small-prime skeleton $5, 7, 11, 13, \ldots$. This data does not prove the theorem, but it identifies the correct next split: small-first-factor overload should be absorbed by Tail/PDEC, while low-load many-label blocking should produce low-mod PDEC/ColumnCRT energy. The small-factor envelope route

`docs/monograph/prime-matrix-h3-small-factor-envelope-route.md`

makes this split deterministic. For a cutoff y, let C_y be the load carried by first factors at most y, let $R_y = A - C_y$, and let $\ell_+(y)$ be the next prime. If full blocking holds, then either C_y is already a small-skeleton overload, or at least

$$K_y = \left\lceil \frac{R_y}{\lfloor q/\ell_+(y) \rfloor + 1} \right\rceil$$

middle-tail labels are active. The finite ledger for the near-miss rows gives maximum forced values $K_y = 7$ for $y = 31$ and $K_y = 9$ for $y = 43$. The remaining proof obligation is to convert small-skeleton overload into Tail/PDEC, or many active labels into H3-PDEC energy, with endpoint losses routed to SAE/ColumnCRT. The global scaling audit

`docs/monograph/prime-matrix-h3-global-scaling-law-route.md`

records the underlying scale. The natural six-wheel rough count is

$$\#A_s \prod_{5 \le \ell \le p} \left(1 - \frac{1}{\ell}\right) \asymp \frac{q}{\log q}.$$

In the finite ledger, from $p = 331$ onward every prime layer has minimum H3 margin at least 21, and bucket averages track this Mertens scale. Thus a formal counterexample must annihilate a growing $q/\log q$ rough-residue scale. The current proof task is to show that such annihilation forces small-skeleton Tail/PDEC overload, many-label H3-PDEC energy, or endpoint SAE/ColumnCRT. The constant-level formula is recorded in

`docs/monograph/prime-matrix-h3-universal-scaling-inequality.md`.

In the finite ledger, $M_{H3}(p,s) \geq 0.30\,q/\log q$ holds from $p = 113$ onward, and $0.40\,q/\log q$ holds from $p = 2011$ onward. These are not asserted as unconditional short-interval prime theorems. They are target constants for a defect inequality: if the H3 margin falls below $cq/\log q$, then SmallSkeletonOverload, ManyLabel-PDEC, or Endpoint-SAE/ColumnCRT must fire. The global tail-energy lemma

`docs/monograph/prime-matrix-h3-global-tail-energy-lemma.md`

proves the deterministic core of this defect inequality. For any cutoff y, target B, threshold L, and modulus

$$d \geq 2\left(\left\lfloor \frac{q}{\ell_+(y)} \right\rfloor + 1\right),$$

where $\ell_+(y)$ is the next prime after y, the implication

$$M_{H3}(p,s) < B \quad \Longrightarrow \quad C_y > \#A_s - B - 2L \quad \text{or} \quad E_y(d) > L$$

holds for every adjacent pair $p < q$ and every row window $2 \leq s \leq q$. Substituting $B = cq/\log q$ and $L = \lambda q/\log q$ gives a uniform infinite-scale defect formula. This is still not the final H3 lower bound: the remaining theorem-level exits are SmallSkeletonOverload $\Rightarrow$ Tail/PDEC and TailEnergy $\Rightarrow$ H3-PDEC/ColumnCRT. The first exit is bridged in

`docs/monograph/prime-matrix-h3-small-skeleton-pdec-bridge.md`.

Let

$$V_y = \#A_s \prod_{5 \leq r \leq y} \left(1 - \frac{1}{r}\right), \qquad D_y = V_y - R_y,$$

where $R_y = \#\{n \in A_s : P^-(n) > y\}$. The overload inequality $C_y > \#A_s - B - 2L$ gives $R_y < B + 2L$, hence

$$D_y > V_y - B - 2L.$$

Thus, whenever $V_y \geq (c + 2\lambda + \eta)q/\log q$, small-skeleton overload forces a PDEC defect of size $D_y > \eta q/\log q$. The remaining issue is not this bridge, but the exclusion of such persistent PDEC defects and of the tail-energy alternative. The tail alternative is put in Fourier form in

`docs/monograph/prime-matrix-h3-tail-energy-fourier-bridge.md`,

where Parseval gives $\mathcal{F}_y(d) = dE_y(d)$ for the non-zero residue frequencies of the tail first-factor labels. The synthesis

`docs/monograph/prime-matrix-h3-unified-defect-criterion.md`

therefore proves the conditional closure criterion

$$D_y \leq V_y - B - 2L \quad \text{and} \quad \mathcal{F}_y(d) \leq dL \quad \Longrightarrow \quad M_{H3}(p,s) \geq B.$$

With $B = cq/\log q$ and $L = \lambda q/\log q$, this is the current global scale theorem: the empirical $q/\log q$ law is explained by two explicit low-mod defect exclusions. The unconditional row theorem still requires proving those exclusions uniformly. The source of the $q/\log q$ scale is clarified in

`docs/monograph/prime-matrix-h3-first-row-scale-bridge.md`.

Let $\Pi_1(q) = \pi(q)$ be the number of primes in the first row up to q. By the prime number theorem, $\Pi_1(q) \sim q/\log q$. On the other hand, adjacent-shell singleton rigidity says that below q^2 every old p-sieve survivor, apart from the endpoint q^2, is prime. Hence the average H3 margin over the non-first q-rows satisfies

$$\frac{1}{q-1}\sum_{s=2}^{q} M_{H3}(p,s) \sim \frac{q}{2\log q} \sim \frac{1}{2}\Pi_1(q).$$

The factor $1/2$ is simply $\log q/\log(q^2)$. Thus H3 is the square-shell projection of the first-row prime scale; a zero H3 row would have to remove a full first-row-scale mass from one row and therefore must be accounted for by the low-mod defects above. The pointwise boundary is recorded in

`docs/monograph/prime-matrix-h3-pointwise-closure-boundary.md`.

The average identity above is rigorous, but it does not imply pointwise positivity. The missing statement is a pointwise scale-transfer theorem, for example

$$M_{H3}(p,s) \geq \kappa\,\pi(q) \qquad (2 \leq s \leq q)$$

for some fixed $\kappa > 0$, or equivalently the uniform exclusion of the D_y and $\mathcal{F}_y(d)$ defects in the H3 unified criterion. The analytic boundary is recorded in

`docs/monograph/prime-matrix-h3-square-root-short-interval-barrier.md`.

It identifies this pointwise transfer with a prime lower bound in aligned intervals of length $q = \sqrt{q^2}$ near $x = q^2$. This is exactly the square-root short-interval scale, where ordinary average PNT input and the linear sieve at $u = 2$ do not supply a pointwise positive lower bound. The remaining theorem is therefore a genuine square-root defect exclusion, not a bookkeeping consequence of the first-row average. The direct hard attack is archived in

`docs/monograph/prime-matrix-h3-square-root-defect-exclusion-hard-attack.md`.

It shows that the D_y branch can be controlled at a small sieve level by the standard one-dimensional interval-sieve fundamental lemma. The obstruction is the tail branch: for a natural low modulus d, the tail Fourier energy satisfies

$$\frac{1}{2}T_y \leq E_y(d) \leq T_y,$$

so it is essentially the tail-label count. The final hard core is therefore to exclude tail labels and double-rough points as an exact full-row filler, or to force them into ColumnCRT, endpoint phase, or cofactor-structure defects. The local rigidity of such a filler is recorded in

`docs/monograph/prime-matrix-h3-tail-filler-rigidity-hardcore.md`.

Adjacent H3 candidates differ by 2 or 4, and candidates two steps apart differ by 6. If both endpoints are tail-filled, all their large factor families are disjoint. A repeated tail label must be separated by at least $y/4$ candidate steps, and any block of diameter less than y consumes at least two fresh large prime factors per candidate. Thus the remaining obstruction is the global concatenation of these local 2/4/6 rough short-difference units into a whole row. The global concatenation interface is sharpened in

`docs/monograph/prime-matrix-h3-tail-filler-global-chain-capacity.md`.

For any continuous tail-filled block B, full concatenation forces

$$\mathrm{EdgeCap}(B;y)=|B|-1,\qquad \mathrm{TriCap}(B;y)=|B|-2.$$

Here EdgeCap counts compatible adjacent 2/4 tail edges and TriCap counts compatible 2/4/6 triples. If $y>\sqrt{q+6}$, a fixed ordered tail-label pair contributes at most once in the row; if $y>q^{2/3}$, every tail-filled point is a double-rough semiprime. Hence the final H3 hard input can now be stated as an explicit capacity exclusion: prove that these edge or triple capacities are strictly below the full-chain requirements, or that full capacity forces PDEC, ColumnCRT, endpoint phase, or cofactor-structure defects. This is a necessary-capacity reduction, not yet an unconditional proof of H3. The macroscopic full-tail branch is then attacked in

`docs/monograph/prime-matrix-h3-tail-edge-selberg-exclusion.md`.

For a fixed even shift $\delta\in\{2,4,6\}$ and an interval I of length H, the classical two-dimensional Selberg upper-bound sieve gives

$$\#\{n\in I:(n(n+\delta),P(y))=1\}\le A_\delta(\theta)\frac{H}{(\log y)^2}\qquad(3\le y\le H^\theta).$$

Every filled adjacent edge gives a y-rough pair with shift 2 or 4, and every filled triple gives a y-rough endpoint pair with shift 6. Hence a continuous tail-filled block B must satisfy

$$|B|\le 2+A_*(\theta)\frac{q+6}{(\log y)^2}.$$

Thus a macroscopic block of length comparable with q, in particular a whole H3 row of about $q/3$ candidates, cannot be filled only by tail labels at sufficiently large scale. The remaining H3 obstruction is consequently narrower: any surviving bad row must split tail points into many short blocks, and the small-factor cuts or short-block endpoint phases must be routed to PDEC, ColumnCRT, endpoint, or cofactor defects. The short-block routing is made exact in

`docs/monograph/prime-matrix-h3-shortblock-singleton-barrier.md`.

If the tail points are decomposed into maximal consecutive tail blocks, with total tail mass T, number of blocks J, and internal adjacent-tail edges E, then

$$T=J+E.$$

The two-dimensional sieve controls E, so a bad row with T still of order $q/\log y$ would have almost all tail mass in singleton tail blocks. The small-factor cuts themselves are not yet a PDEC contradiction, because the small skeleton has natural size comparable with q. The final local obstruction is therefore the singleton-tail system: a point $a_j=\ell_j m_j$ with $\ell_j>y$ is isolated between two small-skeleton neighbours, so $a_j\equiv 0 \pmod{\ell_j}$ and simultaneously satisfies two non-zero congruences modulo small labels $r_-,r_+\le y$. One must prove that many such squeezed singleton phases force PDEC, ColumnCRT, endpoint, or cofactor defects. The squeezed singleton phase is converted into a CRT capacity criterion in

`docs/monograph/prime-matrix-h3-singleton-clamp-defect-criterion.md`.

For a fixed clamp datum $c=(\delta_-,\delta_+,r_-,r_+)$, the two small-skeleton congruences are either incompatible or define one class $\rho(c) \pmod{\mathrm{lcm}(r_-,r_+)}$. Adding the tail label $\ell>y$ gives one class modulo $\ell\,\mathrm{lcm}(r_-,r_+)$, hence one row contains at most

$$1+\left\lfloor\frac{q+O(1)}{\ell\,\mathrm{lcm}(r_-,r_+)}\right\rfloor$$

points from that unit. Consequently many singleton tails must enter one of three branches: tail-label concentration, clamp low-mod concentration, or distributed singleton capacity. The first two are named defect entrances. The final unresolved branch is the distributed signed excess of double-rough semiprimes over the prime survivor channel. This distributed branch is made bilinear in

`docs/monograph/prime-matrix-h3-distributed-singleton-bilinear-obstruction.md`.

For $y > q^{2/3}$, every singleton tail is a product ℓm with $y < \ell \le p$ and m prime. The clamp class $\rho(c) \pmod{R(c)}$ is equivalent to the moving cofactor congruence

$$m \equiv \ell^{-1}\rho(c) \pmod{R(c)}.$$

Thus the final distributed obstruction is a signed bilinear prime-counting defect over the short cofactor intervals I/ℓ. Without tail-label concentration, clamp concentration, endpoint accumulation, or ColumnCRT/cofactor bias, one must prove that this signed bilinear error is only $O(q/\log^2 y)$; this is the current last H3 input. The non-zero-frequency bridge is recorded in

`docs/monograph/prime-matrix-h3-bilinear-large-sieve-defect-bridge.md`.

Expanding the cofactor residue condition modulo $R(c)$ into non-principal characters gives

$$\chi(\ell^{-1}\rho(c)) = \chi(\rho(c))\overline{\chi(\ell)}.$$

Hence a large distributed singleton error forces a non-principal bilinear character sum over the short cofactor intervals. A plain large-sieve bound is still too weak at this short window scale; the final input is a dispersion/Kloosterman-type estimate for this family, or a proof that the corresponding non-zero frequency is already a PDEC, ColumnCRT, or cofactor defect. The additive Kloosterman reduction is

`docs/monograph/prime-matrix-h3-dsb-kloosterman-window-reduction.md`.

The congruence $m \equiv \rho(c)\bar{\ell} \pmod{R(c)}$ gives, after finite additive Fourier expansion, the inverse phase

$$e\left(-\frac{h\rho(c)\bar{\ell}}{R(c)}\right).$$

Thus the final H3 frequency input is a short-window Kloosterman sum with variable ℓ, cofactor window I/ℓ, modulus $R(c)$, and frequency h. The remaining audit splits into four branches: KLS-window coverage of the active parameters, high-lcm clamp escape, high-frequency endpoint tail, and coefficient concentration. The high-lcm branch is routed further in

`docs/monograph/prime-matrix-h3-dsb-high-lcm-clamp-routing.md`.

If $R(c) > R_0$, then one row contributes at most

$$1 + \left\lfloor \frac{q + O(1)}{R_0} \right\rfloor$$

points from a fixed clamp unit. Hence any high-lcm mass of order $q/\log y$ must activate many almost-singleton high-lcm units. Across a persistent family of bad rows this gives a non-zero CRT/Fourier defect, hence a PDEC/ColumnCRT exit; if it is not persistent, it is a sparse single-window escape. This does not yet exclude the branch, but it removes high-lcm mass from the low-modulus KLS-window estimate and routes it to the existing final exits. The common energy source of the two high-lcm exits is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-fourier-energy-clamp.md`.

For a homogeneous high-lcm modulus block, with residue-counting measure μ and total mass U, finite Plancherel gives

$$\sum_{1\le h<R} |\widehat{\mu}(h)|^2 = R \sum_{a \bmod R} \mu(a)^2 - U^2.$$

In particular, when $U \le R/2$, this non-zero energy is at least $RU/2$. Hence high-lcm escape necessarily injects Fourier energy: persistent energy must be excluded by PDEC/ColumnCRT, and isolated energy by SAE/endpoint analysis. The persistent branch is converted to a PDEC threshold in

`docs/monograph/prime-matrix-h3-dsb-hlc-pdec-threshold-bridge.md`.

For one homogeneous formal unit B with phase-counting vector g_B,

$$L_{\mathrm{HLC}}(B) = \left(\frac{R\sum_a g_B(a)^2 - U_B^2}{R-1}\right)^{1/2}$$

is a forced non-zero-frequency lower bound. Thus a persistent high-lcm unit is excluded by a same-unit certificate $U_{\mathrm{CRT}}(B) < L_{\mathrm{HLC}}(B)$. The case $U_B > R/2$ is not a new escape; it is dense effective-modulus concentration and routes back to KLS-window, PDEC/ColumnCRT, or SAE/endpoint. If this PDEC upper certificate fails, the failure is localized in

`docs/monograph/prime-matrix-h3-dsb-hlc-pdec-failure-localization.md`.

For a failing non-zero frequency h and direction ζ, every cap

$$\mathcal{C}_\alpha = \{a : \Re(\zeta e(ha/R)) \ge \alpha\}, \qquad 0 \le \alpha < L/U,$$

has mass at least $(L-\alpha U)/(1-\alpha)$. Hence failure is a concrete Bohr-cap concentration, which must become a same-unit PDEC constraint or a sparse SAE/endpoint certificate. The cap is further decomposed in

`docs/monograph/prime-matrix-h3-dsb-hlc-bohr-cap-component-route.md`.

With $d = (h, R)$ and $R' = R/d$, the cap is the d-fold lift of one arc in $\mathbb{Z}/R'\mathbb{Z}$. Each lifted component has length at most $d+\omega(\alpha)R$, where $\omega(\alpha) = \arccos(\alpha)/\pi$, and one component carries mass at least $(L-\alpha U)/(d(1-\alpha))$. Thus the remaining cap obstruction is either a large-gcd low-effective-modulus concentration or a short-arc component concentration. The large-gcd branch is descended in

`docs/monograph/prime-matrix-h3-dsb-hlc-high-gcd-descent.md`.

If $d = (h, R) > 1$ and $R' = R/d$, the projected vector $g'(b) = \sum_{a\equiv b \pmod{R'}} g(a)$ satisfies

$$\widehat{g}_R(h) = \widehat{g}_{R'}(h/d),$$

and the Bohr-cap mass is preserved. Hence each high-gcd step strictly lowers the effective modulus and must terminate in a low-effective-modulus PDEC/KLS exit or in the short-arc case. The short-arc case is quantified in

`docs/monograph/prime-matrix-h3-dsb-hlc-short-arc-density-pressure.md`.

If a short arc has length at most $D_0 + \omega(\alpha)R$ and mass forced by the Bohr-cap lower bound, then its density relative to the global density U/R is at least

$$\frac{R(L-\alpha U)}{D_0(1-\alpha)U(D_0+\omega(\alpha)R)}.$$

If this pressure exceeds $1+\epsilon$, a local-density PDEC row or SAE/endpoint certificate is forced; otherwise only the explicit parameter residual remains. The low-pressure residual is optimized in

`docs/monograph/prime-matrix-h3-dsb-hlc-short-arc-pressure-optimizer.md`.

Writing $t = L/U$ and

$$\Psi_{D_0,R}(t) = \sup_{0\le\alpha<t} \frac{t-\alpha}{D_0(1-\alpha)(D_0/R+\omega(\alpha))},$$

the residual is the one-dimensional condition $\Psi_{D_0,R}(t) \le 1+\epsilon$. This implies the same-unit flatness estimate

$$\sum_a g(a)^2 \le \frac{1+(R-1)\tau^2}{R} U^2$$

and the corresponding support lower bound. Thus low-pressure short-arc mass is routed back to the KLS-window coefficient-norm check or to coefficient concentration. The KLS admission step is separated in

`docs/monograph/prime-matrix-h3-dsb-hlc-l2-flat-kls-admission.md`.

It lists six required checks for an HLC L2-flat residual: effective modulus, effective frequency, endpoint smoothing, coefficient L^2 norm, gcd/unit strata, and dyadic or tail-label splitting. The L2-flat estimate supplies the coefficient-norm check; any other failed item is a named exit, and only if all six pass may the KLS-window input be invoked. Clean admission is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-clean-kls-reduction.md`.

A clean HLC formal unit is one in which the six named exits do not occur; then the K1–K6 admission checks pass. The remaining object is the clean HLC Kloosterman window

$$\mathcal{K}_{\mathrm{HLC}}(B),$$

and the remaining external/self-contained input is an estimate $\mathcal{K}_{\mathrm{HLC}}(B) = O(q/\log^2 y)$, denoted HLC–KLS-ext in the notes. The external-theorem adaptation is now separated in

`docs/monograph/prime-matrix-h3-dsb-hlc-kls-external-adaptation.md`.

It matches the clean HLC object to a DI/BFI/Kuznetsov-type windowed Kloosterman input: the inverse phase is $e(-h\rho(c)\overline{\ell}/R(c))$, the modulus and frequency checks are supplied by K1–K2, endpoint smoothing by K3, coefficient L^2 control by K4, gcd/unit strata by K5, and dyadic or tail-label losses by K6. Thus the clean HLC branch is closed in the external-deep-theorem version. A fully self-contained version would still require reproving the corresponding spectral/dispersion Kloosterman theorem inside this manuscript. The self-contained obligation is further compressed in

`docs/monograph/prime-matrix-h3-dsb-hlc-kls-core-reduction.md`.

There the clean HLC window is decomposed into dyadic Type-I/II blocks with standard kernel $\mathfrak{S}(\mathcal{D})$, and the proof of HLC–KLS-ext is reduced to one core estimate $|\mathfrak{S}(\mathcal{D})| \leq \mathcal{N}(\mathcal{D})/\log^A y$. Thus the remaining no-black-box line is this single windowed Kloosterman spectral mean estimate. The no-black-box spine is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-kls-core-self-contained-spine.md`.

It completes the smooth completion, Kloosterman quadratic-form reduction, coefficient L^2 bookkeeping, and the implication from the Kuznetsov large-sieve atom to the core estimate. It also records why pointwise Weil bounds alone do not yield the required arbitrary logarithmic saving. Hence a fully self-contained version now has one precise remaining analytic line: prove the specialized Kuznetsov large-sieve atom for this window family. That line is expanded in

`docs/monograph/prime-matrix-h3-dsb-hlc-kuznetsov-ls-atom-expansion.md`.

The atom is decomposed into smoothing, the specialized Kuznetsov trace formula, Bessel transform decay, the spectral large sieve including oldforms and Eisenstein spectrum, and the well-factorable dispersion step that supplies the logarithmic saving. The manuscript therefore records the exact no-black-box obligations rather than hiding them under the single phrase "spectral theory". The Bessel-transform part is now treated in

`docs/monograph/prime-matrix-h3-dsb-hlc-kz-c-bessel-transform-decay.md`.

It uses the Bessel differential equation, the self-adjoint Bessel operator, and repeated integration by parts to obtain the required rapid transform decay. The remaining no-black-box analytic atoms at this point are the specialized Kuznetsov formula, the spectral large sieve, and the well-factorable dispersion logarithmic saving. The spectral-large-sieve atom is further reduced in

`docs/monograph/prime-matrix-h3-dsb-hlc-kz-d-spectral-large-sieve-spine.md`.

This dualizes the spectral large sieve to a spectral projection kernel, accounts for oldforms, Eisenstein spectrum and holomorphic forms as polylogarithmic losses, and reduces KZ–D to a single pre-trace kernel bound with diagonal size T^2 and off-diagonal Schur size N_0. The pre-trace kernel is further decomposed in

`docs/monograph/prime-matrix-h3-dsb-hlc-ptk-d-pretrace-kernel-bound.md`.

There the diagonal term is recorded as the local Weyl volume, and the remaining off-diagonal part is reduced to a spatial lattice-point/correlation row-column bound, denoted LPC–D. The spatial kernel is split in

`docs/monograph/prime-matrix-h3-dsb-hlc-lpc-d-spatial-correlation-split.md`.

The far tail is absorbed by kernel decay, the identity-near sector is empty or parabolic, and the parabolic cusp sector is controlled by divisor bookkeeping. The remaining spatial line is the generic hyperbolic local-correlation bound GHLC–D. That last spatial line is now treated in

`docs/monograph/prime-matrix-h3-dsb-hlc-ghlc-d-local-schur-closure.md`.

The key correction is to estimate the pre-trace object as an integral kernel, not by pointwise hyperbolic neighbour counting. On the local radius $r_* = T^{-1} \log^B y$ one has

$$T^2 \int_0^{r_*} (1 + Tr)^{-A} \sinh r \, dr = O_A(1),$$

so the local ball area cancels the T^2 height of the kernel. Combining this local L^1 kernel mass with the Poincare-packet Schur normalization gives GHLC–D, hence LPC–D, PTK–D, and KZ–D. At that point the remaining analytic atoms were KZ–B, the specialized Kuznetsov trace formula, and KZ–E, the well-factorable dispersion logarithmic saving. The KZ–B specialization is treated in

`docs/monograph/prime-matrix-h3-dsb-hlc-kz-b-kuznetsov-trace-specialization.md`.

It derives the needed trace formula from the automorphic kernel, Poincare-packet unfolding, the double-coset Kloosterman geometric side, spectral Plancherel, and Bessel-transform normalization. This closes the formula-conversion part; it does not supply the final logarithmic saving. Hence the remaining no-black-box analytic atom in the HLC branch is KZ–E, the well-factorable dispersion logarithmic-saving step. The current KZ–E attack is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-kz-e-well-factorable-dispersion-spine.md`.

It internalizes the well-factorable convolution algebra, the dispersion variance identity, the CRT reduction to a standard Kloosterman phase, and the gcd/endpoint polylogarithmic ledger. This proves that KZ–E would follow from one remaining estimate, denoted WFD-core: a windowed well-factorable Kloosterman dispersion mean estimate. That core is further balanced in

`docs/monograph/prime-matrix-h3-dsb-hlc-wfd-core-balanced-factor-reduction.md`.

Using the square-root well-factorable decomposition $\lambda_c = \sum_{c=uv} \alpha_u \delta_v$, the condition $c \asymp C$ forces $u, v \asymp C^{1/2}$ up to polylogarithmic boundary blocks. The gcd strata are polylogarithmic, and CRT factors the phase modulo uv into coupled phases modulo u and v. At this intermediate stage the remaining atom is BWFD-core, a balanced two-modulus well-factorable Kloosterman dispersion mean estimate. The subsequent completion attack is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-bwfd-core-spectral-completion-attack.md`.

It performs exact finite Fourier completion in the s variable and uses complete Kloosterman multiplicativity modulo uv. This proves that BWFD-core follows from one still narrower atom, BSC-core: balanced complete Kloosterman bilinear correlation with arbitrary logarithmic saving. Ordinary KZ–D spectral large sieve recovers only the raw quadratic norm scale, and pointwise Weil bounds give only single-sum square-root cancellation; neither supplies the required $\log^{-A}$ saving. The current direct attack on BSC-core is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-bsc-core-kloosterman-fraction-attack.md`.

It expands the complete Kloosterman sums term by term and exposes the reciprocal-fraction phase $e(\bar{v}R/u+\bar{u}T/v)$. One-sided degeneracy is localized to quadratic congruences such as $(a_h+\ell)x^2+b_h \equiv 0 \pmod{u}$; the remaining no-black-box atom is KFLS-core, a balanced Kloosterman-fraction large-sieve estimate with arbitrary logarithmic saving. The next square-kernel attack is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-kfls-core-square-kernel-attack.md`.

It expands the square of the reciprocal-fraction sum into a four-modulus phase kernel. This also rules out an absolute Schur proof of the full kernel: the exact diagonal recovers only the raw quadratic norm scale. The remaining atom is therefore CFQK-core, a centered four-modulus Kloosterman-fraction correlation estimate covering the semi-diagonal and true off-diagonal layers. The block-centering correction is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-cfqk-core-block-centering-attack.md`.

It shows that the entire same-(u, v) block diagonal is local variance and cannot be forced to be logarithmically small for arbitrary admissible coefficients. The remaining obligations are therefore BD-CEN, the block-centering identity, OSQK-core for one shared modulus, and TFQK-core for the true four-modulus layer. The BD-CEN audit is recorded in

`docs/monograph/prime-matrix-h3-dsb-hlc-bd-cen-dispersion-centering-audit.md`.

It checks the current KZ–E dispersion spine and finds only the $h = 0$ main-term subtraction, not the same-(u, v) block projection required by BD-CEN. Thus the first blocking obligation is the identity (BDC-5); OSQK-core and TFQK-core are conditional downstream obligations until that identity is proved. The subsequent no-go note is

`docs/monograph/prime-matrix-h3-dsb-hlc-bd-cen-no-go-route-fork.md`.

It shows that, for the current unblocked WFD/KZ–E object, BD-CEN is false rather than merely unproved: a single nonzero block with nonzero frequency survives the $h = 0$ subtraction but has no cross-block counterpart. The remaining choices are therefore SOURCE-CEN, BLK-energy-core, or an explicit external DI/BFI input. The SOURCE-CEN no-go note

`docs/monograph/prime-matrix-h3-dsb-hlc-source-cen-no-go.md`

checks the first of these choices. It shows that source block-centering would replace the unblocked linear WFD target in (KE-13) by a different centered object and would leave a block mean term which still has to be estimated. Thus SOURCE-CEN is not an identity for the present target. The subsequent BLK-energy audit

`docs/monograph/prime-matrix-h3-dsb-hlc-blk-energy-core-obstruction.md`

checks the second choice. As an arbitrary coefficient-array theorem at the square-kernel level, a one-block one-atom test makes $\sum_b |S_b|^2$ comparable with the raw norm, so a bare arbitrary logarithmic saving for BLK-energy-core is false. The remaining self-contained obligation is therefore the sharper NC-BLK input: prove block non-concentration for the actual WFD coefficients produced by the Type-I/II, Fourier, and well-factorable structure. Without that new input, this branch is only closed in an external-theorem version using a precisely matched DI/BFI dispersion theorem. The CRT lift-parity variant is audited in

`docs/monograph/prime-matrix-crt-lift-parity-audit.md`.

For adjacent primes $p < q$, the non-trivial q-columns are not covered by the new prime q; therefore a hypothetical q-row zero does descend to an old p-sieve zero window of length q. However, the old p-sieve row period already has an even number of zero rows by reflection symmetry, so multiplying the period by the odd factor q preserves evenness and does not create a parity contradiction. The retained conclusion is only the early-window dichotomy: an aligned lower zero row or a seam window routed to PDEC/SAE. The attempted recursive continuation is audited in

`docs/monograph/prime-matrix-recursive-mirror-descent-audit.md`.

It separates three objects: zero rows in a full CRT period at fixed width q, zero rows after peeling the largest base prime, and zero rows in an actual smaller-width matrix. These objects do not imply one another. In the finite audit, full-period zero rows occur for several adjacent pairs, but no early q^2 zero row and no actual smaller matrix zero row is forced. The next usable gate remains seam-window exclusion, not recursive parity. The annulus-mirror localization audit

`docs/monograph/prime-matrix-annulus-mirror-localization-audit.md`

separates two reflections. The CRT reflection preserving zero residue classes is $n \mapsto -n \pmod{M_p}$ and sends a small row to the tail of the CRT period. The terminal reflection $n \mapsto q^2 - n$ does land in an early interval, but it changes the forbidden classes from 0 to $q^2 \bmod \ell$. It is therefore a terminal non-zero-class block, not a smaller prime-matrix zero row. The scaled-row and half-width version is audited in

`docs/monograph/prime-matrix-scaled-peeling-halfwidth-audit.md`.

It shows that the scaled height nP/p is only an approximate locator: a lower aligned zero row appears only when the upper window fully contains a lower row. In the same samples, passing to a prime near half the upper width gives no automatic pair of consecutive lower zero rows, because the half-level sieve resurrects points whose least factor lies above the half-width level. The resurrected-point absorption route is now recorded in

`docs/monograph/prime-matrix-rpz-absorption-defect-route.md`

with the finite audit

`docs/monograph/prime-matrix-rpz-absorption-defect-audit.md`.

It routes complete absorption into five alternatives: a survivor remains, TailAnchor load, ColumnCRT displacement load, ColumnRadius, or a residual Distributed-RPZ branch. In the known first-zero-row samples the half-level has 15 resurrected points, no missing absorber, and maximum one-window label and top-column loads equal to 1. Thus the route does not justify a single-window TailAnchor exclusion; the next theorem-level gate is the Distributed-RPZ capacity barrier, or a proof that distributed absorption necessarily returns to one of the named column or tail exits. The first attack on this gate is recorded in

`docs/monograph/prime-matrix-rpz-sliding-plateau-barrier.md`.

It proves a sliding-platform multiplicity formula: if a zero window remains zero under a consecutive block of shifts, then a resurrected source in the common core repeats with the full platform multiplicity. Hence distributed absorption on such a platform is either a persistent TailAnchor load or, after source deletion, a boundary-compressed branch. The audit in

`docs/monograph/prime-matrix-rpz-sliding-plateau-audit.md`

finds platform length 5 in the known samples and TailAnchor triggering at $T_0 = 4$ in all five cases. The remaining gate is therefore the boundary-compressed RPZ branch, not unrestricted distributed absorption. The boundary-compressed branch is further routed in

`docs/monograph/prime-matrix-rpz-boundary-compressed-core-route.md`.

It proves that, if TailAnchor is excluded, the center interval $J_{T_0} = [L + A + T_0, R + B - T_0]$ is an h-sieve zero interval. The audit

`docs/monograph/prime-matrix-rpz-bcb-core-audit.md` finds that all 11 center survivors in the known samples are TailAnchor core sources; after conditional TailAnchor deletion, all five center intervals are clean and contain an aligned half-width row. The next gate is therefore the BCB grid/endpoint exclusion: either the center contains an aligned lower zero row, or the endpoint seam enters SAE, PDEC, or ColumnCRT. The exact grid criterion is recorded in

`docs/monograph/prime-matrix-rpz-bcb-grid-endpoint-criterion.md`.

For $J = [u, v]$, $N = v - u + 1$, and $\delta_h(u) \equiv 1 - u \pmod{h}$, the interval contains an aligned h-row if and only if $N \geq \delta_h(u) + h$. Failure is therefore a finite endpoint seam phase. The audit in

`docs/monograph/prime-matrix-rpz-bcb-grid-endpoint-audit.md`

matches this criterion in all five samples and finds no endpoint-defect sample. The remaining gate is endpoint persistence exclusion: a failed grid phase must be isolated as SAE or persistent as PDEC/ColumnCRT. This endpoint gate is routed in

`docs/monograph/prime-matrix-rpz-bcb-endpoint-persistence-route.md`

and audited in

`docs/monograph/prime-matrix-rpz-bcb-endpoint-phase-ledger.md`.

For fixed (h, N), endpoint failure is a finite residue ledger in u mod h. In the known samples there are no actual endpoint failures and only two possible failure phases. Sparse loads are SAE obligations; persistent same-phase loads are PDEC/ColumnCRT objects. Thus the RPZ endpoint branch is now named, not closed: the remaining work is either certificate closure for SAE/PDEC/ColumnCRT or recursive descent from the lower h-zero row. The two remaining RPZ attacks are now advanced together in

`docs/monograph/prime-matrix-rpz-dual-track-closure-route.md`.

Track A fixes the certificate interfaces for RPZ-SAE, RPZ-PDEC, and RPZ-ColumnCRT. Track B uses the lower-zero descent audit

`docs/monograph/prime-matrix-rpz-lower-zero-descent-audit.md`

which shows that, in the known BCB samples, all six conditional lower zero rows descend to a $p = 2$ impossible zero row with no endpoint/grid obstruction. This is still a sample-level descent audit; the global task is to prove descent-grid persistence or route every obstruction to the named certificate interfaces. The obstruction side is now made explicit in

`docs/monograph/prime-matrix-rpz-lower-descent-obstruction-ledger.md`

and

`docs/monograph/prime-matrix-rpz-certificate-materialization-interface.md`.

For each adjacent descent $p \to r$, obstruction is a finite phase condition modulo $\prod_{\ell \leq r} \ell$ and is classified as either grid failure or endpoint-puncture blocking. The current descent tree has twenty actual transition nodes, all successful, with no actual blocked node. This does not close the global theorem; it reduces any future obstruction to the SAE finite package, PDEC endpoint phase row, or ColumnCRT displacement row. The certificate skeleton builder

`docs/monograph/prime-matrix-rpz-certificate-skeleton-package.md`

materializes these rows: two endpoint SAE candidates, two endpoint PDEC/ColumnCRT rows, and three lower-descent grid-failure PDEC/ColumnCRT rows. These are certificate obligations, not completed exclusions. The endpoint SAE finite certificate

`docs/monograph/prime-matrix-rpz-endpoint-sae-finite-certificate.md`

checks that the two endpoint candidates have possible load two but actual load zero in the current BCB ledger. Hence the current finite endpoint SAE line is vacuously closed; global SAE exclusion is still not proved. The lower grid-failure certificate

`docs/monograph/prime-matrix-rpz-lower-grid-fail-avoidance-certificate.md`

reduces every adjacent descent $p \to r$ to the exact test $\delta = -(a-1)(p-r) \bmod r$ and grid success iff $\delta \le p-r$. The current descent tree has twenty actual transition nodes and none hits grid failure. The remaining global task is to prove this phase inequality for formal descent paths, or route its failure to PDEC/ColumnCRT. The formal path note

`docs/monograph/prime-matrix-rpz-formal-descent-phase-inequality.md`

sharpens this statement: once a formal descent path exists, the phase inequality is automatic from interval containment. The true remaining task is path existence for every formal counterexample branch, or exclusion of the first obstruction phase through SAE/PDEC/ColumnCRT. The first-obstruction dichotomy

`docs/monograph/prime-matrix-rpz-first-obstruction-dichotomy.md`

removes endpoint puncture as a genuine obstruction: a contained lower row containing the endpoint ap would force $r \mid a$, contradicting the roughness condition needed for the endpoint to revive. Hence every non-descending branch has a first grid-failure seam phase, which must be discharged by SAE/PDEC/ColumnCRT certificates. The first-grid-failure seam certificate

`docs/monograph/prime-matrix-rpz-first-grid-fail-seam-certificate.md`

puts this obstruction in normal form. If $g = p - r$ and $g < \delta < r$, the bad row is the union of two adjacent lower-row caps of lengths δ and $r + g - \delta$, with conserved missing mass $r - g$ and at most the right endpoint ap as an r-rough survivor. The certificate produces twelve seam phase rows covering all 1752 grid-failure phases in the current ledger. It now also records the exact PDEC support row for each seam:

$$S_\tau = \{a \bmod Q : \ a \equiv \rho \pmod r\}, \qquad F_\tau = \mathbf{1}_{a\equiv\rho \pmod r} - 1/r,$$

whose non-zero Fourier support is contained in the frequencies jQ/r, $1 \le j < r$. The same certificate splits the endpoint ap into 1348 phases already killed by lower labels and 404 Q-unit endpoint phases. Thus the next narrow gate is no longer support extraction, but either an endpoint-PDEC upper bound $U_{\mathrm{CRT}} < L_{\mathrm{PDEC}}$ for these same bad-window families or a ColumnCRT displacement certificate with explicit selectors Π, λ and threshold L_D. The unit-endpoint gate certificate

`docs/monograph/prime-matrix-rpz-unit-endpoint-columncrt-gate.md` pushes this one step further. For a unit endpoint seam,

$$c \equiv ap \equiv p\rho \equiv r + (p - r) - \delta \pmod{r},$$

so the endpoint lies in a fixed non-trivial lower column. If $\pi = hr + c$ is a same-column prime witness, then the endpoint lower row satisfies $H \equiv -cr^{-1} \pmod{p}$, and the displacement $d = h - H \pmod{p}$ is independent of the row phase and non-zero. In the current ledger all twelve gate rows have explicit witnesses and non-zero displacements, covering all 404 unit endpoint phases. This routes persistent unit endpoint seams to fixed non-zero ColumnCRT candidates; it still does not exclude the ColumnCRT threshold L_D or prove that formal counterexample families avoid these gates. The threshold obstruction audit

`docs/monograph/prime-matrix-rpz-columncrt-threshold-obstruction.md`

then shows why this cannot be closed by merely lowering L_D. Inside a fixed unit gate, all unit endpoint residues already have the same label p and the same non-zero displacement class $d \bmod p$, so the intrinsic load of that class is

$$R_{p,d} \geq \prod_{\ell<r}(\ell - 1).$$

In the present ledger the maximum intrinsic load is 48, and the test value $L_D = 2$ would send ten gate rows, covering 400 phases, into ColumnCRTDefect rather than exclude them. Thus the remaining legitimate closures are a formal-family avoidance theorem, an endpoint-PDEC upper bound, or an independent exclusion theorem for the resulting ColumnCRTDefect. The three-route audit

`docs/monograph/prime-matrix-rpz-three-route-closure-audit.md`

compares these options under the same ledger. It finds no closed route yet, but ranks formal-family avoidance first: the present lower-descent ledger has twenty actual transition nodes and no grid-failure node, while the endpoint-PDEC route still lacks an admissible U_{CRT} upper bound and the ColumnCRT route has been reduced to an independent defect-exclusion theorem. The formal phase automaton

`docs/monograph/prime-matrix-rpz-formal-phase-automaton.md`

turns the first route into an explicit start-phase admission problem. For each prime level p it defines an accepted set $A_p \bmod P(p)$: phases in A_p descend canonically through successive prime levels to $p = 2$, while phases outside A_p hit a first-grid-fail seam and return to the already materialized exits. In the current ledger all six BCB-Core start rows belong to their accepted sets. The global missing theorem is that every formal counterexample-family start phase lies in A_p, or else is captured by the seam exits. The rejected-phase absorption certificate

`docs/monograph/prime-matrix-rpz-rejected-phase-absorption.md`

removes the second ambiguity within the present automaton range. It checks all 27924 rejected phases, finds twelve distinct first-fail seam rows, and verifies that every one of them is already materialized by the first-grid-fail seam certificate. Hence the RPZ formal route is now a strict two-way routing statement: an accepted start phase descends to $p = 2$, while a rejected start phase enters the named seam/PDEC/ColumnCRT certificate chain. This is an absorption theorem, not an exclusion theorem; the remaining hard point is the actual discharge of those seam/PDEC/ColumnCRT exits, or a proof that formal counterexample starts always lie in A_p. The seam exit pressure ledger

`docs/monograph/prime-matrix-rpz-seam-exit-pressure-ledger.md` then compresses this remaining exit. The 27924 rejected phases reduce to twelve seam rows; within those rows 1348 endpoint phases are already killed by lower labels, while the remaining 404 unit endpoint phases fall into ten fixed non-zero ColumnCRT displacement classes, with maximum aggregated load 96. Thus the next gate is no longer a phase search: one must either prove formal-family avoidance of the twelve seam rows, provide $U_{\mathrm{CRT}} < L_{\mathrm{PDEC}}$ for their single-residue PDEC supports, or prove an independent exclusion theorem for the ten ColumnCRT displacement classes. The symbolic ladder certificate

`docs/monograph/prime-matrix-rpz-symbolic-ladder-certificate.md`

rewrites the preferred avoidance route as local adjacent-prime digits. For each adjacent pair $p > r$, with $g = p - r$, it uses

$$\delta_p(a) = -(a-1)g \pmod{r}, \qquad \delta_p(a) \le g$$

as the exact success digit. Failure of this inequality is precisely the first-grid-fail seam already materialized above. The certificate verifies the resulting counting formula by full enumeration through modulus $P(19) = 9699690$ and symbolically records the same ladder through $p = 97$. The remaining theorem is now sharply stated: derive these digit inequalities from the formal counterexample construction itself, not from a finite phase enumeration. The BCB start-digit ledger

`docs/monograph/prime-matrix-rpz-bcb-start-digit-ledger.md`

extracts these digits for the current BCB-Core starts. All twenty digit nodes are safe, but ten of them have zero margin $\delta_p(a) = p - r$. This is a useful negative result for the proof strategy: the desired avoidance theorem cannot be closed by a coarse positive margin estimate. It must use exact congruence information for the start row modulo each lower prime r. The accepted lower-row selector ledger

`docs/monograph/prime-matrix-rpz-bcb-accepted-row-selector.md`

weakens the required positive statement to the exact selector condition

$$C_h(J) \cap A_h \neq \varnothing, \qquad C_h(J) = \{m : \ J \supset [(m-1)h + 1, mh]\}.$$

For the current BCB-Core samples all five core intervals have such a selector, and all six candidate lower rows are accepted. This is still only a current-ledger certificate; the global theorem must prove selector existence for every formal BCB core interval, or else route the all-rejected selector failure back to the seam/PDEC/ColumnCRT exits. Finally, the selector gap ledger

`docs/monograph/prime-matrix-rpz-selector-gap-threshold.md`

splits selector existence into two verifiable mechanisms. If the number of complete candidate rows in $C_h(J)$ exceeds the maximum cyclic rejected gap of A_h, length alone forces a selector. In the current samples this explains two of the five BCB cores; the remaining three are short-candidate cases and require exact endpoint phase information. Thus the next proof obligation is not merely to enlarge J, but to control the endpoint phase of short cores relative to the rejected gaps of A_h. The short-candidate endpoint phase ledger

`docs/monograph/prime-matrix-rpz-short-candidate-phase-ledger.md`

performs this refinement at modulus $hP(h)$ rather than just modulo h. In the current ledger the three short-candidate BCB records all hit an accepted selector. However, the full (h, ℓ) phase families still contain all-rejected endpoint phases: 588 for $(h, \ell) = (7, 13)$, 11880 for $(11, 19)$, and 344760 for $(13, 25)$. These rejected phases are not unnamed escapes; each carries a first-failure seam key and therefore returns to the seam/PDEC/ColumnCRT exits. The remaining theorem-level gate is consequently exact: prove that the formal BCB construction avoids these all-rejected $hP(h)$ endpoint phases, or close the corresponding seam/PDEC/ColumnCRT certificates. The follow-up forbidden-block ledger

`docs/monograph/prime-matrix-rpz-short-phase-block-formula.md`

removes another layer of enumeration. Since the relevant short cores have length $\ell < 2h$, a starting endpoint u contains at most one complete h-row. For a lower row m, the endpoint interval that contains it is exactly

$$mh - \ell + 1 \leq u \leq (m - 1)h + 1,$$

of width $\ell - h + 1$. Hence the all-rejected endpoint set is the disjoint union of such blocks over rejected row phases $m \bmod P(h)$. The formula matches enumeration in all three short families, and the current endpoints have positive distance from the rejected endpoint union. The remaining non-enumerative input is therefore narrower still: derive from the formal BCB construction that the induced candidate row phase belongs to A_h, or else close its named first-failure exit. The BCB candidate phase identity ledger

`docs/monograph/prime-matrix-rpz-bcb-candidate-phase-identity.md`

then makes the induced row phase explicit. If the top zero row is R, the top prime is P, the half-prime is h, the plateau extremes are s_-, s_+, and the tail threshold is T, the forced core is

$$L = (R - 1)P + 1 + s_- + T, \qquad U = RP + s_+ - T.$$

The complete lower rows inside this core are exactly

$$m_{\min} = \left\lfloor \frac{L + h - 2}{h} \right\rfloor + 1, \qquad m_{\max} = \left\lfloor \frac{U}{h} \right\rfloor.$$

Equivalently, after writing $P = Qh + d$, these phases depend only on $R \bmod hP(h)$ and the plateau parameters. In the current ledger the formula matches all five BCB samples and all six induced lower row phases lie in A_h. This is the sharp remaining input: prove the same residue implication for every formal BCB counterexample, or send any rejected induced phase to its first-failure exit. The accepted-preimage ledger

`docs/monograph/prime-matrix-rpz-bcb-accepted-preimage-ledger.md`

enumerates this final residue condition for the current BCB parameter families. The $P = 13, h = 5$ family has no bad residue. The other four families still have bad residues, split between no-candidate endpoint failures and all-rejected candidate phases. The actual top-row residues in the ledger all lie in the selector preimage, with positive distance from the bad set where the bad set is nonempty. Thus the residue preimage computation is now explicit, but the proof is still not global: the formal counterexample construction must be shown to land in the selector preimage, or the bad residue must be discharged through its named BCB endpoint or first-failure seam/PDEC/ColumnCRT exit. The core-run obstruction ledger

`docs/monograph/prime-matrix-rpz-bcb-core-run-obstruction.md` adds a stronger upstream test for the no-TailAnchor branch. In that branch BCB-Core forces J_T itself to be an h-sieve zero interval, so its length cannot exceed the maximum cyclic run of integers divisible by primes $\leq h$ modulo $P(h)$. For the current layers these exact maxima are

$$h = 5 : 5, \qquad h = 7 : 9, \qquad h = 11 : 13, \qquad h = 13 : 21.$$

The five current BCB cores have lengths $9, 13, 15, 19, 25$, respectively, so every one of them exceeds the corresponding low-sieve run bound, with minimum margin 4. Thus the current finite BCB no-TailAnchor samples are obstructed before the accepted-preimage question is reached. The global theorem-level input is now a Jacobsthal-type low-sieve run bound for the formal layers; without such a bound, the remaining bad cases still route to the named endpoint and first-failure exits. This input is isolated in

`docs/monograph/prime-matrix-rpz-bcb-jacobsthal-closure-interface.md`.

Writing $G(h)$ for the maximum length of a consecutive interval all of whose points are divisible by some prime $\leq h$, the exact BCB closure condition is

$$G(h) < P + m - 1 - 2T,$$

where m is the sliding platform length. Under this inequality the no-TailAnchor BCB branch closes immediately; if it fails, the proof must keep the TailAnchor, endpoint, first-failure, PDEC, and ColumnCRT exits active. The risk scan

`docs/monograph/prime-matrix-rpz-bcb-jacobsthal-risk-scan.md`

checks this linear closure route against the Ziller–Morack primorial Jacobsthal data. It shows that the current finite samples are safe, but the naive global specialization $m = 5$, $T = 4$, and only $P > 2h$ fails already at $h = 43$: there $G(h) = 89$, the minimal top prime is $P = 89$, and the core length is only 85. Thus the finite core-run obstruction cannot be promoted to a global theorem by a blanket linear bound. A global proof must either force longer platforms m, prove stronger restrictions on the formal BCB parameter set, or close the TailAnchor/endpoint/first-failure exits. The platform-threshold ledger

`docs/monograph/prime-matrix-rpz-bcb-platform-length-threshold.md`

rewrites this requirement as the exact integer condition

$$m \geq G(h) - P + 2 + 2T.$$

For $T = 4$, the first substantive unsafe layer with $h \geq 5$ is again $h = 43$, where the required platform length is 10 rather than 5. The next attack must therefore prove platform-length growth or prove that short platforms trigger one of the named exits. The short-platform embedding route

`docs/monograph/prime-matrix-rpz-bcb-short-platform-embedding-route.md`

handles the second alternative. If $N = P + m - 1 - 2T \leq G(h)$ and no TailAnchor is present, then the forced core J_T is an h-sieve zero interval of length N and hence must be embedded inside an extremal low-sieve covered block modulo $P(h)$. Therefore its left endpoint lies in a finite Jacobsthal endpoint set $E_{h,N}$. Persistent occurrence of such an endpoint phase is no longer an unnamed escape: it is routed to SAE if sparse, or to PDEC/ColumnCRT if persistent. The accompanying certificate

`docs/monograph/prime-matrix-rpz-bcb-short-platform-embedding-certificate.md` materializes the max-block subfamily $E_{h,N}^{\max}$ by reconstructing CRT left endpoints from the Ziller–Morack modulus representation. For the current $m = 5, T = 4$ interface, the first non-trivial short platform is $h = 43$, $P = 89$, $N = 85$, with 240 distinct max-block endpoint phases; across the scanned table, 1197 reconstructed longest blocks pass the coverage and boundary checks. This remains a certificate for the longest-block subfamily, not a proof that the full endpoint set $E_{h,N}$ has been exhausted.

These obligations are not typographical tasks. They are the remaining theorem-level gates. Until they are closed, the monograph may be submitted only as a synthesis, dependency, and verification manuscript, not as a final unconditional proof of the prime-matrix theorem, prime-pair assertions, or RH.

Chapter 7

Referee Closure Audit

This chapter records the strict closure standard used for the whole monograph. A contradiction-field chain is considered closed only if its entrance, decomposition, exits, and global contradiction are all verified under the same load convention.

Definition 7.1 (Closure certificate)**.** A closure certificate for a contradiction field consists of four data:

1. an entrance theorem sending every counterexample into the field;
2. a disjoint or source-deleting decomposition into named exits;
3. for each exit, an absorption, descent, transfer, external theorem, or finite certificate;
4. a global comparison showing that final load is simultaneously bounded below and above in incompatible ways.

If any item is replaced by an unverified structural input, the conclusion is conditional on that input.

Theorem 7.2 (Current monograph closure)**.** *The present manuscript closes the audit and dependency structure of the two contradiction-field programs: all known entrances, decompositions, exits, external inputs, finite certificates, and referee obligations are explicitly named. It does not close the final unconditional theorem status of either program.*

Proof. For the prime-matrix program, the A/B appendix gives the entrance reduction from a row or column counterexample to a Structured-EHPD bad configuration. Inside the BPN branch, the low-hole bucket submodule is now closed by the low-range certificate, the narrow-band collision certificate, the two finite tail certificates, and the Rosser–Schoenfeld explicit tail package recorded above. Inside Triad-A1, the same-set capacity / full-S terminal is now closed as a canonical-source self-contained theorem-boundary: the source provenance is canonical RIW/Buchstab, the final closure certificate has no open gates, and the unrestricted generic WFD self-contained statement is explicitly refuted rather than claimed. These are genuine submodule and theorem-boundary closures. The D-structure exclusion and the remaining global PDEC/SAE/Rankin or LocalSurvivor/CleanKLS certificates remain decisive inputs requiring independent line-by-line review. Therefore the whole chain is closed as a reduction, certificate, and boundary package but not as a final unconditional prime-matrix theorem.

For the RH program, the RH manuscript has completed U1–U5: statement wording, AEX acceptance, EXT localization, manuscript wording, BibTeX/PDF compilation, and warning-free LaTeX logs. The final audit still requires independent referee verification of every local theorem and controlled exit. Therefore the chain is closed as a verification package but not as a final unconditional RH theorem.

The status table and the referee audit in `docs/monograph/` record exactly which pieces are proved, reduced, external, computational, or awaiting review. Thus the monograph's logical boundary is explicit and closed, while promotion to final theorem status remains conditional on external verification. □

Theorem 7.3 (Prime-matrix frontier)**.** *The current prime-matrix row/column unconditional theorem is routed to named terminal gates but is not closed. The canonical-source Triad-A1 self-contained boundary is closed, and the unrestricted generic WFD self-contained theorem is refuted and not claimed. After the global terminal-family split and the self-contained bottleneck router, the independent self-contained mathematical bottleneck is the same-set PDEC capacity certificate*

$$\mathtt{PDEC\_CAP\_SameSetGlobalDualCertificate}.$$

Inside that bottleneck, the transverse formal-unit embedding and the downstream canonical layer admission/nonzero-transfer/thin-return gate are now closed on the canonical-source self-contained branch. The remaining materialized open gate is the generic/external original DI/BFI route

$$\begin{matrix}\mathtt{DIBFIQuantifiedNoProjection} \\ \mathtt{WindowCertificate}\end{matrix},$$

after the APS profinite dichotomy, the persistent finite-signature unification, and the SC–9 boundary reconciliation are closed as routing laws, and after the persistent terminal admission and finite-arc routers remove raw, low-rank, continuous-cap, and unnamed finite-arc pseudo-terminals. The previous transverse clean large-sieve atom is now routed to the A1 CleanKLS/SC–9 frontier, and its self-contained source-support branch is further embedded into the canonical A1/KZ-E source; the canonical layer closure then returns to the already closed A1 selector/decision-tree/source-provenance boundary. The PDEC-CAP theorem itself remains open in the broader global/external sense. Final theorem promotion still separately requires the D-structure/Tail-log4/finite Rankin referee interface. The boundary-lift router records the canonical-source branch as

$$\mathtt{NoFurtherCanonicalSourceSelfContainedPDECCapGap},$$

while keeping the generic/external DI/BFI route and the final D-structure/Rankin referee interface outside that self-contained closure. The subsequent terminal-promotion closure router sharpens the canonical-source self-contained boundary to

$$\mathtt{NoFurtherCanonicalSourceTerminalPromotionGap}.$$

Proof. The frontier router

`docs/monograph/prime-matrix-row-column-unconditional-frontier-router.md`

checks the merged boundary audit, the Triad-A1 terminal confluence router, the terminal triad contract, the formal-unit audit, the stitching feasibility audit, the nested-duplicate dominance audit, the line-by-line referee matrix, and the claim-status table. It records

`row_column_unconditional_closed=false`,

while preserving the closed canonical-source boundary and the refuted generic branch. The terminal triad has no fourth exit, so any final counterexample must be absorbed by PDEC, LocalSurvivor, or CleanKLS/DLS certificates, together with the still guarded D-structure/Tail-log4/Rankin interfaces.

The global terminal-family boundary router and its split router then show that the current materialized frontier is exhausted and that all non-final reductions are closed. The open terminal family gates reduce to PDEC-CAP, KLS/external-or-internal large sieve, and the D-structure/Rankin referee interface. For the fully self-contained route, the remaining choice was

`PDEC_CAP_OR_INTERNAL_CleanKLS_LargeSieve`.

The self-contained bottleneck router closes the non-final CleanKLS ambiguity. CleanKLS failure returns to PDEC/SAE, the canonical clean branch is absorbed by the closed same-set boundary, the generic full-S self-contained strengthening is refuted and not claimed, and the SC-9 route has already been accounted for. Therefore InternalCleanKLS is no longer an independent self-contained blocker. What remains is exactly the global same-set PDEC dual capacity certificate $U_{\mathrm{CRT}} < L_{\mathrm{PDEC}}$, plus the independent referee-promotion gate. This proves the stated frontier and also explains why it is not yet a final unconditional theorem.

The PDEC-CAP same-set global-dual router refines that frontier once more. It records

`closed_current_materialized_pdec_gates=true`

and

`pdec_cap_same_set_global_dual_closed=false`.

Thus the current local PDEC materialization is exhausted, but the global proof is not. The profinite APS router closes the missing dichotomy: along the inverse tower of signatures, the real payment graph Γ_n either has a positive-limsup persistent finite signature, which becomes a multi-bucket PDEC formal unit, or avoids every finite signature, which is a distributed CleanKLS/DLS input unless FiberDeletion or NoDeletion–KL has already routed it back to PDEC. The first persistent narrow hardpoint is therefore the terminal estimate

`PrimitiveMultiAtomSameFormalUnitPDECCertificate`.

The diffuse terminal split router first sharpens the nonpersistent side. No unnamed FiberDeletion, NoDeletion–KL, or CleanKLS exit remains; the side is reduced to either SC–9 or a finite-shell low-mod persistence object. The latter is a PDEC/ColumnCRT input. The persistent-signature unification router then identifies persistent MFU and fixed-shell persistence as the same finite-signature formal-unit grammar; ColumnCRT displacement, PDEC cap failure, and formal-unit mismatch all route back into the PDEC/ColumnCRT family rather than creating a parallel terminal. The SC–9 boundary reconciliation then removes the flat-clean SC–9 branch as an independent canonical-source blocker: clean failure returns to PDEC/SAE, canonical NC-BLK is absorbed by the same-set boundary, and generic WFD is not claimed as self-contained. The persistent terminal admission router then forces any remaining PDEC object to be primitive, multi-atom, same-formal-unit, non-tautological, and not already absorbed by SAE/Endpoint. The deletion support-exhaustion bridge closes the implication from divergent deletion to exhausted support or named sparse/PDEC return; the HRO router then shows that if deletion does not diverge, either TailIndependence returns to NoDeletion/CleanKLS or old-hole residue occupancy saturates. The occupancy-kernel router uses $|\mathrm{Occ}_t| \le \min(|H_Q(t)|, r)$ to remove the sparse-old-hole subcase. Therefore the deletion-side hardpoint first becomes the dense old-hole selector kernel. The common-variable router then rewrites each selected column as $c_b = \rho_b + rk_b$; every old-prime condition forbids exactly one residue of the same shell variable k_b. The resulting alternatives are named: no legal shell gives capacity/Hall deletion, positive mass on a fixed shell or finite shell packet gives PDEC/ColumnCRT persistence, and genuine multishell flatness enters the already reconciled SC–9/NC-BLK boundary. An external KLS/DI/BFI/Kuznetsov citation would close only the external-input version.

The primitive terminal is then reduced further. The primitive multi-atom rank router removes rank-zero and rank-one pseudo-terminals: after physical deduplication, such remnants are reuse defects, two-point Fourier tautologies, fixed-shell PDEC/ColumnCRT objects, or SAE objects. What remains is a rank-at-least-two cap-stable primitive kernel. The cap-stable kernel router rewrites its inequality by the contrapositive of cap localization: if a direction reaches L_{PDEC}, then a direction cap has mass at least $(L_{\mathrm{PDEC}} - \alpha M)/(1-\alpha)$ and must return to SAE, refined PDEC, ColumnCRT, or multiplicity normalization. Hence a genuinely cap-stable kernel only has to certify that all legal direction caps stay below their thresholds. The uniform-cap finite-basis router then removes the continuum in (ζ, α): for a finite formal unit, every such cap is the preimage of a finite cyclic arc under a nontrivial character. Finally, the finite-arc transverse router splits a high arc into low transverse support, persistent transverse bias, or transverse-flat dispersion. These route respectively to SAE/ColumnCRT/fixed-shell PDEC, refined PDEC, or CleanKLS/DLS or an external large-sieve input. The current narrow PDEC-CAP hardpoint is thereby reduced to transverse clean analysis. The transverse clean reduction router observes that a finite arc fixes only one character direction; a rank-at-least-two primitive kernel still leaves a transverse quotient. Nonflat transverse defects are named returns to SAE, refined PDEC, or ColumnCRT. If none occurs, the residual is exactly a transverse L^2-flat clean large-sieve atom. The current narrow PDEC-CAP hardpoint is therefore first reduced to

`TransverseQuotientCleanLargeSieveAtom`.

The transverse clean-atom frontier router then imports the already registered A1 CleanKLS/SC–9 grammar. K1–K9 admission failure returns to PDEC, SAE, ColumnCRT, or multiplicity normalization. If admission succeeds, the object enters the SC–9 frontier: self-contained actual-coefficient NC-BLK or external DI/BFI. The canonical NC-BLK absorption cannot be used for a generic transverse quotient until the quotient coefficients are shown to have the canonical RIW/Buchstab source support, or until actual transverse block non-concentration is proved directly. The naive factor-residue incidence bridge is blocked by the internal-fiber obstruction. Thus the current self-contained hardpoint is

`TransverseSourceSupportNonconcentrationCertificate`.

This certificate is then split once more. The A1/KZ-E actual-source provenance ledger is closed for the canonical RIW/Buchstab pre-Cauchy source, the canonical RIW support route is reduced to Buchstab-layer support, and the raw thick squarefree support count is closed. The transverse embedding router then proves that the transverse quotient formal unit is a legal restriction, quotient, or conditioning of that canonical A1/KZ-E source without changing its coefficients: actual payment is a deterministic pushforward, finite signatures are finite projections, direction arcs are preimage restrictions, and the transverse quotient is a finite factor. The canonical layer closure router then observes that the downstream layer-admission/nonzero-transfer/thin-return gate is exactly the already routed A1 chain: selector retention, finite signatures, RIW/Buchstab decision tree, actual source provenance, and canonical branch boundary. Hence the canonical-source self-contained PDEC-CAP branch has no remaining layer-transfer gate. The original external DI/BFI route is still separate and requires

`DIBFIQuantifiedNoProjectionWindowCertificate`,

unless one explicitly accepts a windowed KLS external theorem as an external input. This does not promote the full row/column statement to an unconditional theorem.

Finally, the boundary-lift router

`prime-matrix-self-contained-pdec-cap-boundary-lift-router`

propagates this conclusion back to the upper self-contained bottleneck. It records

`canonical_source_self_contained_pdec_bottleneck_closed=true`

and

`narrowest_self_contained_boundary=NoFurtherCanonicalSourceSelfContainedPDECCapGap`.

The same audit keeps

`pdec_cap_same_set_global_dual_closed=false`

and

`row_column_unconditional_closed=false`.

Thus the PDEC-CAP gap has been removed only from the canonical-source self-contained branch. The broader generic/external branch still needs the quantified no-projection DI/BFI certificate, and the whole prime-matrix theorem still needs the global terminal family promotion review and the D-structure/Rankin referee interface.

The canonical terminal-promotion closure router then reconciles the older global split with the boundary lift. It observes that the current materialized terminal frontier is exhausted, the terminal triad has no fourth exit, PDEC-CAP promotion is closed on the canonical-source branch, canonical CleanKLS/NC-BLK is absorbed or returns to PDEC/SAE, and all known LocalSurvivor entrances are covered by witness, extractor, or admission contracts. It records

`latest_self_contained_hardpoint_closed=true`

and

`open_self_contained_gates=[]`.

Thus the latest self-contained boundary is

`NoFurtherCanonicalSourceTerminalPromotionGap`.

This is still a canonical-source theorem-boundary closure, not an unrestricted global terminal-family exclusion theorem. The external DI/BFI route and the D-structure/Rankin referee gate remain outside the closure.

The final self-contained theorem router then packages the exact statement boundary. The approved self-contained statement is the Prime Matrix canonical-source terminal theorem boundary: the Triad-A1 same-set/full-S terminal together with canonical terminal promotion on the canonical RIW/Buchstab source branch. Under that exact boundary it records

`canonical_source_self_contained_theorem_closed=true`

and

`open_self_contained_gates=[]`.

Its terminal is

`NoFurtherCanonicalSourceSelfContainedTheoremBoundaryGap`.

The router also records the non-claims: the unrestricted generic full-S WFD self-contained theorem is refuted and not claimed, and the unrestricted/global Prime Matrix row-column unconditional theorem is not claimed. The external DI/BFI route and the D-structure/Rankin referee promotion gate remain outside this self-contained theorem boundary.

The global-unconditional obstruction router makes the last point explicit. It records

`current_corpus_global_unconditional_self_contained_closure_possible=false`

and

`row_column_unconditional_closed=false`.

The named blockers are the refuted unrestricted generic WFD route, the still open actual full-S source bridge or strengthened source anti-atom theorem, the DI/BFI no-projection window certificate, and the D-structure/Tail-log4/finite Rankin promotion gate. Thus the current manuscript closes the exact canonical-source self-contained theorem boundary, but does not close the unrestricted global row-column theorem without new inputs.

The actual-source bridge reconciliation router sharpens this last obstruction list. It checks the canonical provenance ledger, the source-lock branch split, and the final canonical-source theorem certificate together. On the canonical RIW/Buchstab branch it records

`actual_source_bridge_closed_for_canonical_branch=true`.

Hence the former broad blocker

`ActualFullSSourceBridge`

is superseded inside the canonical self-contained theorem boundary. The global complement is still open, however:

`actual_source_bridge_closes_global_unrestricted=false`.

The updated global blockers are

`UnrestrictedGenericWFD`,
`NoncanonicalFullSComplementAntiAtomOrExternalDIBFI`,
`FullSNonAPStrengthenedSourceAntiAtom`,
`DIBFIQuantifiedNoProjectionWindowCertificate`,
`DStructureTailLog4FiniteRankinPromotion`.

Thus the actual-source bridge is closed only for the canonical branch. Full global unrestricted closure still requires a strengthened anti-atom for the noncanonical full-S complement, or an external quantified DI/BFI no-projection certificate, together with the final D-structure/Rankin promotion gate.

The noncanonical-complement input-contract router then fixes the remaining input boundary. It records

`contract_boundary_closed=true`, `canonical_branch_removed_from_remainder=true`,

but also

`generic_wfd_template_available=false`,
`noncanonical_complement_closed=false` for the current corpus.

Thus the remaining full-S complement is not an unrestricted generic WFD lemma. The moving-delta separation blocks that template, while the canonical branch has already been subtracted. Under the current terminal map there is no hidden fourth route: one must prove actual source identity with the canonical RIW/Buchstab source, prove a strengthened anti-atom for the actual noncanonical source, or provide the quantified no-projection DI/BFI route. Final unconditional row-column promotion still additionally requires the D-structure/Tail-log4/Rankin acceptance gate.

The noncanonical-complement trilemma router sharpens this boundary. It records

`trilemma_boundary_closed=true`,

but also

`self_contained_noncanonical_package_closed=false`,
`external_contract_package_closed_if_fulls_kls_ext_accepted=true`.

The router separates the only legal closure modes after the canonical branch has been subtracted:

actual source identity,
strengthened actual-source anti-atom,
or acceptance/proof of FullS-KLS-ext.

Its negative content is important: the generic full-S self-contained anti-atom route is refuted by the moving-delta capacity model under the recorded formal WFD, Type/Fourier, K4/K6, and naive-incidence hypotheses. Hence the noncanonical package cannot be closed by returning to the generic WFD template. Without the external FullS-KLS-ext input, one must prove actual source identity or a strengthened actual-source anti-atom; with the external input, the remaining primary-source issue is the line-by-line `DIBFIPrimarySourceSpecializationProof`.

The closure-input atlas finally reviews the explored models together. The CRT/MinRep equivalence, cylindrical line covering, the P-column anchor wheel, the layered $P^2 \pm k$ wheel, far-tail cofactor inversion, SN recursive peeling, named-exit absorption, and PDEC cap refinement no longer point to independent unnamed branches. They all feed the same terminal certificate package:

`SAE-Cert`, `PDEC-Cert`, `ColumnCRT-Cert`,
`CleanMultishellKLS`, `TotalDescent`.

Together with the noncanonical full-S complement package and the D-structure/Rankin promotion package, these are the minimal remaining input basis. Thus future work should prove these packages directly rather than search for another generic WFD template or a fixed numerical constant.

The D-structure/Rankin promotion package is now fixed as an acceptance boundary rather than a hidden terminal. The corresponding router records

`promotion_package_boundary_closed=true`,
`promotion_package_independently_accepted=false`,
`rankin_sample_pass=true`.

It checks that the package is listed in the minimal input basis, that it is separated from the canonical-source self-contained boundary, that PM-16 remains a BLOCK-REFEREE item in the line-by-line matrix, and that the Rankin ledger has a checkable certificate format. The current sample has exact smooth-core count 39, allowed budget 40.0, and both the Rankin and exact budget tests passing.

This is a format and sample verification, not a global acceptance of the whole promotion package. Final promotion still requires independent acceptance of the D-structure/Structured-EHPD entrance reductions, Tail-log4 BG/RKS theorem-number and adaptation checks, reproducible finite-verification hashes, and Rankin certificates for all formal colored corridors, with failures routed to PDEC/SAE.

The terminal-certificate package has a further compression. The no-unnamed-escape machine and the named-exit absorption contract are already closed at the routing level. The canonical-source clean KLS branch is absorbed by the final canonical boundary, and the noncanonical clean branch belongs to the noncanonical complement or to an external KLS route. TotalDescent is also not a separate terminal: a formal descent path reaches $p = 2$, while a first failure is a first-grid-fail seam already routed to PDEC, ColumnCRT, or SAE. Therefore the independent terminal certificate work is reduced to

$$\texttt{SAE-Cert}, \quad \texttt{PDEC-Cert}, \quad \texttt{ColumnCRT-Cert}.$$

This is a structural compression of the first package, not yet a proof of those three certificate families.

The ColumnCRT branch is then absorbed once more. A persistent nonzero column displacement overload is a displacement PDEC after adjoining the displacement signature to the finite phase space; a sparse displacement overload is an SAE/endpoint event; and balanced displacement loads become admissible constraints in the PDEC dual problem. RPZ unit endpoint gates show that tuning the displacement threshold alone cannot prove exclusion, but they do not create a third terminal family. Hence the independent terminal remainder is reduced to

$$\texttt{SAE-Cert}, \quad \texttt{PDEC-Cert} \quad \text{including endpoint, displacement, cofactor, and primitive variants.}$$

This is still not a proof of the terminal certificate package; it is the removal of ColumnCRT as an independent terminal.

The SAE branch is also removed as an independent terminal by the SAE absorption router. The router checks the SAE local-reduction lemma, the materialized LocalSurvivor packet ledger, the known extractor coverage audit, the new-sparse-entry admission audit, the packet-generation dichotomy, and the persistent-terminal admission boundary. It records

$$\texttt{sae\_independent\_terminal\_removed=true}.$$

The absorption law is the following. A sparse or single-window escape must either emit a finite LocalSurvivor packet with a witness or blocker-deficit certificate, or else a finite signature persists and returns to PDEC, ColumnCRT, TailAnchor, or CofactorAnchor; if the signature layers keep escaping, the branch enters CleanKLS/DLS admission or an external input package. The currently materialized LocalSurvivor/SAE packets are exhausted, the known sparse entries have extractors or contract fallbacks, and no additional unnamed sparse entry is admitted by the current contract system. Hence SAE is not a third terminal family. The first terminal package is now reduced to

$$\texttt{PDEC-Cert} \quad \text{including endpoint, displacement, cofactor, primitive, and} \\ \text{non-tautological variants,}$$

together with the hygiene rule that any genuinely new sparse route must bring an explicit packet-extractor schema. This still does not prove the terminal package or the unrestricted row-column theorem; it deletes SAE as an independent terminal and pushes the remaining structural burden back to PDEC or to a future explicit sparse schema.

The broad PDEC family is then turned into an explicit input boundary. The PDEC-family router records

```
pdec_family_explicit_input_boundary_closed=true,
current_materialized_pdec_frontier_closed=true,
canonical_source_pdec_cap_closed=true,
global_pdec_family_unconditional_closed=false.
```

It checks the persistent-terminal admission gate, the non-tautological PDEC admission audit, the primitive-rank boundary, the rank-two cap-stable inverse step, the finite cyclic-arc reduction, the transverse clean reduction, the clean-frontier route, and the canonical-layer closure. Thus the currently materialized non-tautological primitive PDEC candidate count is zero, and the canonical-source PDEC-CAP route has returned to the final self-contained boundary. A future PDEC obstruction is not admitted as a label: it must come with a same-formal-unit phase map, at least three physical primitive atoms after deduplication, no two-point tautology, rank at least two after shell and column quotients, cap stability under finite cyclic-arc localization, and no sparse, persistent-biased, or clean-externalized transverse return. Hence the first package is now reduced to future explicit primitive PDEC schemas and future explicit sparse packet schemas, while generic branches remain in the noncanonical trilemma and final theorem promotion remains in the D-structure/Rankin acceptance package.

The future sparse branch is also turned from a hygiene rule into an explicit admission boundary. The sparse-schema router records

```
future_sparse_packet_schema_boundary_closed=true,
current_materialized_sparse_frontier_closed=true,
global_sparse_family_unconditional_closed=false.
```

It uses the materialized LocalSurvivor ledger, the known packet-extractor coverage audit, the new-sparse-entry admission audit, the SAE absorption router, the packet-generation dichotomy, and the PDEC-family boundary. Thus a future sparse obstruction is not admitted as a bare terminal label. It must specify the source class, a finite window or fixed-offset fiber, the candidate set, the blocker families and their incidence projection, a witness or a strict blocker-deficit inequality, phase and formal-unit normalization with deduplication, a finite-signature persistence test, a layer-escape test, and a reproducible ledger with no open local obligation. If the finite signature persists, the branch returns to explicit PDEC/ColumnCRT/Tail/Cofactor schemas; if the packet layers escape, it enters CleanKLS/DLS or an explicit external large-sieve input. Hence the current materialized sparse frontier is zero, but the global sparse family is not unconditionally excluded.

Combining these items gives the final input firewall. The firewall router records

```
final_input_firewall_boundary_closed=true,
current_materialized_terminal_frontier_closed=true,
no_hidden_terminal_remaining=true,
all_final_inputs_independently_accepted=false,
row_column_unconditional_closed=false.
```

The remaining named inputs are exactly future explicit primitive PDEC schemas, future explicit sparse packet-extractor schemas, one legal branch of the noncanonical full-S trilemma, and independent D-structure/Rankin promotion acceptance. Thus the author-side boundary and naming problem is closed: there is no hidden terminal left in the current material. The unrestricted row-column theorem is still not promoted, because the named inputs have not all been proved or independently accepted.

The four named inputs are then attacked one by one. The attack router records

`current_materialized_frontier_zero=true`, `no_hidden_terminal_remaining=true`, `conditional_closure_chain_complete=true`, `all_current_obligations_closed=false`, `unconditional_closure_possible_from_current_corpus=false`.

The future primitive PDEC and future sparse packet-extractor inputs have no present materialized obligation: they are admission firewalls for future routes. The real current obligations are the noncanonical full-S complement and D-structure/Rankin promotion. The canonical actual-source branch is closed, but the unrestricted noncanonical complement is not; the recorded primary-source audit rejects deriving the required full-S non-AP WFD Kloosterman large-sieve estimate from the existing DI/BFI sources. The D-structure/Rankin boundary is named and the sample Rankin certificate passes, but the full ledger and independent acceptance are not complete. Therefore the logical chain is conditionally complete: if future PDEC/sparse routes are discharged by their schemas, one legal noncanonical branch is proved or accepted, and the D-structure/Rankin package is independently accepted, then the no-hidden-terminal chain promotes the theorem. Without those new inputs, the present corpus cannot honestly claim the unrestricted unconditional row-column theorem.

The noncanonical input is finally narrowed further. The noncanonical-narrowing router records

`noncanonical_narrowing_boundary_closed=true`, `ap_source_lift_rejected=true`, `generic_self_contained_antiatom_refuted=true`, `exact_source_entropy_closed=false`, `external_full_s_contract_closed_if_accepted=true`, `self_contained_noncanonical_closed=false`.

The AP-source lift is rejected by the AP/non-AP branch definition and by the object ledger; the generic self-contained anti-atom is refuted by the moving-delta capacity model; the canonical branch has already been removed. Therefore the noncanonical complement has only two real remaining inputs: an internal theorem proving exact source entropy, equivalently a strengthened source anti-atom for the actual full-S non-AP source, or an external/new proof of the full-S non-AP WFD Kloosterman large-sieve input.

The last-remaining-atoms router packages this boundary into the final open atoms. It records

`last_remaining_atom_boundaries_closed=true`,
`conditional_logic_chain_complete=true`,
`all_last_atoms_proved_or_accepted=false`,
`row_column_unconditional_closed=false`.

The final atoms are

`ActualFullSNonAPExactSupportAtom`,
`ModulusDependentCompletedFullSKLSInput`,
`DStructureRankinIndependentAcceptance`.

The first two are alternative closures for the noncanonical full-S complement: either an internal exact support/source-entropy theorem for the actual full-S non-AP source, or an external/new modulus-dependent completed full-S Kloosterman large-sieve input. The third is the independent promotion acceptance for the D-structure/Tail-log4/finite Rankin package. Thus the no-hidden-terminal logic chain is conditionally complete: one of the first two atoms, together with the D-structure/Rankin acceptance atom, promotes the row-column theorem. Since the present corpus has not proved or independently accepted these atoms, the unrestricted unconditional row-column theorem remains open.

The three-final-atoms hard-attack router then pushes this boundary one step further. It records

`three_atom_attack_boundary_closed=true`,

`conditional_logic_chain_complete=true`,

`all_three_atoms_proved_or_accepted=false`,

`row_column_unconditional_closed=false`.

The final mathematical choice is:

```
ActualFullSNonAPSourceCapacityAntiAtomForActualSource
or
CDependentResidueWeightSpectralCancellationInput
```

The first input is a theorem that the actual full-S non-AP source capacity measure has no moving same-(u, v) atom. The generic version is refuted by the moving-delta model, and it is not forced by K4/K6 or naive incidence. The second input is a spectral/dispersion cancellation theorem for the completed c-dependent residue weights $B_{c,x} = \sum_k \beta_{x+kc}$; pointwise Weil, L^2 budgeting, the ordinary large sieve, and flat-residue shortcuts do not supply it. The final promotion input is:

```
DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance
```

Thus the minimal unconditional input basis is

```
(ActualFullSNonAPSourceCapacityAntiAtomForActualSource
or CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance
```

The current corpus has not proved this mathematical alternative and has not completed the independent promotion acceptance, so the unrestricted unconditional row-column theorem is not yet a proved theorem of the manuscript.

The final unconditional-closure attempt router then checks whether the mathematical alternative can be closed from the current corpus. It records

`final_attempt_boundary_closed=true`,

`math_lanes_collapsed_to_common_core=true`,

`internal_math_proof_found_in_current_corpus=false`,

`external_math_match_found_in_current_corpus=false`,

`independent_promotion_acceptance_completed=false`,

`row_column_unconditional_closed=false`.

The c-dependent completed-residue route is not an independent terminal: finite Fourier inversion connects it to the existing BWFD–BSC–KFLS chain, whose self-contained remaining duty is actual same-(u, v) block non-concentration, unless a precisely matched external DI/BFI/Kuznetsov dispersion theorem is supplied. The actual-source anti-atom is the same moving-block spread/source-entropy duty. Therefore the irreducible final input basis is

```
(MovingBlockSpreadNCBLKForActualFullSNonAPWFDCoefficients
OR PreciselyMatchedExternalDIBFIKuznetsovDispersionTheorem)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance
```

The current corpus does not prove moving-block spread, does not contain a line-by-line external theorem match for the current full-S, non-AP, uncentered, no-projection object, and has not completed independent D-structure/Rankin promotion acceptance. Thus the final input boundary is closed, but the unrestricted unconditional theorem is still not proved.

The irreducible-input refinement router then sharpens the external label. The phrase "precisely matched external DI/BFI/Kuznetsov dispersion theorem" cannot be read as a claim that the existing primary DI/BFI sources have already been matched to this object: the BFI AP theorem, the DI/Maynard J-scale, and the AP-source lift route are all blocked for the present full-S, non-AP, uncentered, no-projection WFD window. The refined basis is

```
(MovingBlockSpreadNCBLKForActualFullSNonAPWFDCoefficients
OR FullSNonAPWFDKLSTheoremInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Here FullSNonAPWFDKLSTheoremInput means a new proof or an independently cited theorem which directly estimates the current full-S non-AP WFD Kloosterman/dispersion object. The router records

`refinement_boundary_closed=true`,

`row_column_unconditional_closed=false`.

The subsequent joint attack compares the two remaining mathematical lanes directly. The canonical RIW/Buchstab source branch has already been closed and subtracted from the noncanonical full-S complement, so its support theorem cannot be imported as a proof of the noncanonical moving-block anti-atom. The internal lane is therefore narrowed to an exact support package for the actual noncanonical source:

```
FullSNonAPExactFactorSupportPackageForActualNoncanonicalSource.
```

The package has four subclauses:

exact factor-support lower bound,

balanced-range thresholds,

Type/Fourier capacity compatibility,

factor-residue incidence bridge or direct support proof.

The external lane remains

```
FullSNonAPWFDKLSTheoremInput.
```

The BFI, DI, and Maynard sources provide the AP/dispersion/Kuznetsov technology class, but the current audit finds no ready theorem covering the present c-dependent completed residue weights, uncentered no-projection full-S non-AP object, with arbitrary logarithmic saving. Thus the latest basis is

```
(FullSNonAPExactFactorSupportPackageForActualNoncanonicalSource
OR FullSNonAPWFDKLSTheoremInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The two lanes can be narrowed once more. On the internal side, the full-S balanced-range threshold is closed, since $C \asymp P/\log^{O(1)} P$ and the balanced factors of $c = uv$ dominate every fixed logarithmic threshold. The remaining exact-support and Type/Fourier-capacity clauses are therefore a single source anti-atom contract:

```
FullSNonAPStrengthenedSourceAntiAtomContractForActualNoncanonicalSource.
```

On the external side, the full-S window may be completed modulo c; the real residue is the c-dependent completed weight

$$B_{c,x} = \sum_k \beta_{x+kc}.$$

Thus the external theorem input is narrowed to

```
CDependentResidueWeightSpectralCancellationInput.
```

The refined terminal basis is

```
(FullSNonAPStrengthenedSourceAntiAtomContractForActualNoncanonicalSource
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The dual-lane common-core reconciliation router then checks whether the internal and external lanes are still independent self-contained obstacles. It records

```
common_core_reconciliation_closed=true,
self_contained_common_core_proved=false,
external_contract_accepted_as_final_input=false,
dstructure_rankin_independent_acceptance_completed=false,
row_column_unconditional_closed=false.
```

The reconciliation law is this: if `CDependentResidueWeightSpectralCancellationInput` is accepted as a FullS–KLS or c-dependent spectral theorem, it is an external theorem input. If it is proved internally from the present corpus instead, finite Fourier completion sends it back to the BWFD–BSC–KFLS chain, and then to actual same-(u, v) block non-concentration. That is the same moving-block source-entropy core as the internal source anti-atom lane. Consequently the reconciled basis is

```
((ActualA1FullSSourceLockOrNewFullSNonAPSourceAntiAtomTheoremInput)
OR AcceptedFullSKLSExtOrCDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The fully self-contained basis, without an external black box, is

```
ActualA1FullSSourceLockOrNewFullSNonAPSourceAntiAtomTheoremInput
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Thus the independence gap between the two mathematical lanes is closed. What remains is not two separate self-contained obstacles, but one actual-source anti-atom/source-lock input plus the independent D-structure/Rankin promotion acceptance. The unrestricted unconditional row-column theorem is therefore still not proved.

The self-contained narrowest-core router then removes one more ambiguity from the no-external-black-box basis. It records

`narrowest_core_reduction_closed=true`,
`canonical_source_lock_absorbed_for_canonical_branch=true`,
`source_lock_option_removed_from_global_remainder=true`,
`external_black_box_used=false`,
`generic_self_contained_antiatom_available=false`,
`noncanonical_actual_source_core_proved=false`,
`dstructure_rankin_independent_acceptance_completed=false`,
`row_column_unconditional_closed=false`.

The source-lock option has already been absorbed by the canonical RIW/Buchstab branch and the actual-source provenance ledger. It closes the canonical-source self-contained theorem, but it does not close the unrestricted global noncanonical full-S complement. The generic self-contained anti-atom is also unavailable, since the moving-delta model refutes it under the recorded formal WFD/Type/Fourier hypotheses. Therefore the latest fully self-contained global basis is

```
ActualNoncanonicalFullSSourceEntropyOrStrengthenedAntiAtomTheoremInput
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The first input means exact source entropy, equivalently no moving same-(u, v) atom, for the actual noncanonical full-S non-AP WFD coefficients. It is not an unrestricted generic WFD template. This is a sharper boundary, not a proof of the remaining source core or of the independent D-structure/Rankin promotion gate.

The noncanonical source-core atomization router then removes the remaining naming duality between "source entropy" and "strengthened anti-atom." It records

`source_core_atomization_closed=true`,
`entropy_antiatom_duality_removed=true`,
`balanced_range_threshold_closed=true`,
`k4_k6_or_naive_incidence_suffices=false`,
`canonical_import_allowed=false`,
`actual_support_capacity_core_proved=false`,
`dstructure_rankin_independent_acceptance_completed=false`,
`row_column_unconditional_closed=false`.

Exact source entropy and the strengthened anti-atom are two formulations of the same actual noncanonical source-capacity core. The balanced range threshold is already closed in the full-S regime. K4/K6 residue and dyadic controls do not imply moving factor-pair support, naive factor-residue incidence is blocked by the large internal fiber inside one (u, v) block, and canonical RIW/Buchstab support cannot be imported into the noncanonical complement. Hence the latest self-contained basis is

```
ActualNoncanonicalFullSFactorSupportCapacityTheoremInput
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The first input asks for log-power lower absolute support of the exact u- and v-factors in every surviving actual noncanonical full-S non-AP balanced block, together with Type/Fourier capacity compatibility so that no moving (u, v) pair receives an unrecorded capacity multiplier. This closes a naming and shortcut boundary, but it is not yet a proof of that support-capacity theorem.

The actual capacity-ledger microatom router then checks one last possible shortcut inside that support-capacity statement. It records

`microatom_boundary_closed=true`,

`support_only_suffices=false`,

`registered_multiplier_discipline_would_suffice_with_support=true`,

`exact_uv_support_proved=false`,

`registered_multiplier_discipline_proved=false`,

`actual_final_capacity_antiatom_proved=false`,

`dstructure_rankin_independent_acceptance_completed=false`,

`row_column_unconditional_closed=false`.

The correction is that raw support width is not itself a source anti-atom. A model with many raw (u, v) pairs but one unrecorded Type/Fourier/fiber multiplier can still concentrate the final capacity measure on a single moving pair. Conversely, if all such multipliers are registered in the same formal unit and bounded by a fixed logarithmic power, then exact support does imply the final anti-atom: with $L = \log y$, $|\alpha_u|, |\delta_v| \le L^C$, registered multipliers $W_{u,v} \le L^E$, and $S_u S_v \ge L^{2A+4C+E}$, one obtains

$$\max_{u,v} M_{u,v} / \sum M_{u,v} \le L^{-2A}.$$

Thus the sharp self-contained input is best stated as

```
ActualFinalCapacityAntiAtomLedgerForNoncanonicalFullS
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

A proof may supply the package

```
ActualNoncanonicalExactUVSupportLowerBound
AND ActualTypeFourierRegisteredCapacityMultiplierDiscipline.
```

This closes the support-only shortcut and fixes the conditional implication, but it still does not prove the final actual capacity anti-atom ledger or the independent D-structure/Rankin acceptance.

The registered capacity-multiplier discipline router then attacks the second micro-input in that package. It records

`registered_capacity_multiplier_discipline_closed=true`,

`exact_uv_support_proved=false`,

`actual_final_capacity_antiatom_proved=false`,

`dstructure_rankin_independent_acceptance_completed=false`,

`row_column_unconditional_closed=false`.

This is a bookkeeping closure, not a new cancellation theorem. The Type decomposition, dyadic summation, CRT phase normalization, the Fourier h-window and tail, the coefficient, gcd, endpoint and smoothing losses, the full-S completion fiber, and tail-label bookkeeping are all registered inside the same actual formal unit and cost only fixed logarithmic powers. These costs are absorbed into the final capacity measure $M_{u,v}$, or equivalently into a registered multiplier $W_{u,v} \le L^E$. The router does not use the external DI/BFI no-projection theorem-match branch and does not prove factor support.

Consequently the fully self-contained source-side basis is reduced from

```
ActualNoncanonicalExactUVSupportLowerBound
AND ActualTypeFourierRegisteredCapacityMultiplierDiscipline
```

to the single source micro-input

```
ActualNoncanonicalExactUVSupportLowerBound.
```

Together with the independent promotion gate, the latest basis is

```
ActualNoncanonicalExactUVSupportLowerBound
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This closes the multiplier-discipline item, but the exact u, v-support theorem and the independent D-structure/Rankin acceptance remain open.

The ExactUVSupport terminal-attack router then audits the remaining source micro-input. It records

$$\texttt{exact\_uv\_support\_terminal\_boundary\_closed=true},$$
$$\texttt{registered\_capacity\_multiplier\_discipline\_closed=true},$$
$$\texttt{exact\_uv\_support\_proved=false},$$
$$\texttt{actual\_final\_capacity\_antiatom\_proved=false},$$
$$\texttt{dstructure\_rankin\_independent\_acceptance\_completed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

The audit removes the remaining shortcut routes. Formal WFD hypotheses allow point-supported factors, so they do not force actual u, v-support. K4/K6 and naive factor-residue incidence still live on the wrong projections. Raw Buchstab/Mertens counting gives enough squarefree products only after the canonical layer is admitted, and the canonical RIW/Buchstab support chain closes only the canonical-source branch. It cannot be imported into the noncanonical full-S complement.

Thus the current fully self-contained basis is still

```
ActualNoncanonicalExactUVSupportLowerBound
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

If the exact-support theorem is supplied, the already closed multiplier discipline gives the final capacity anti-atom by the recorded inequality. Without that theorem, and without independent D-structure/Rankin acceptance, the unrestricted row-column theorem cannot yet be stated as unconditional.

The ExactUVSupport failure-packetization router then rewrites the remaining source input in its contrapositive finite form. It records

$$\texttt{failure\_packetization\_closed=true},$$
$$\texttt{exact\_uv\_support\_proved=false},$$
$$\texttt{actual\_support\_failure\_packet\_exclusion\_proved=false},$$
$$\texttt{actual\_final\_capacity\_antiatom\_proved=false},$$
$$\texttt{dstructure\_rankin\_independent\_acceptance\_completed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

After the actual noncanonical clean block, exact u, v-support notion, and registered multiplier threshold are fixed, the positive lower-bound input

```
ActualNoncanonicalExactUVSupportLowerBound
```

is equivalent to the absence of a positive-mass support-failure packet:

```
ActualNoncanonicalSupportFailurePacketExclusion.
```

Such a packet must carry the source class, formal-unit id, block key, exact u- and v-support sets, the $L^{2A+4C+E}$ threshold, the registered capacity profile, return tests to the named exits, and a reproducible certificate. Hence support failure can no longer remain an unnamed obstruction. This packetization is an equivalent boundary reformulation, not a proof of packet exclusion. The latest self-contained input basis can therefore be written as

```
ActualNoncanonicalSupportFailurePacketExclusion
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The manuscript still lacks a proof that all such packets are absent or must return to the PDEC, SAE, ColumnCRT, CleanKLS, or external KLS exits, and it still lacks independent D-structure/Rankin acceptance.

The support-failure packet return-dichotomy router then closes the return alphabet for these packets. It records

$$\texttt{support\_failure\_packet\_return\_dichotomy\_closed=true},$$
$$\texttt{clean\_core\_packet\_exclusion\_proved=false},$$
$$\texttt{actual\_final\_capacity\_antiatom\_proved=false},$$
$$\texttt{dstructure\_rankin\_independent\_acceptance\_completed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

An actual noncanonical support-failure packet cannot be a fifth terminal family. If it is not a clean-core packet, then it must return through one of the existing named interfaces: an isolated finite packet gives a LocalSurvivor/SAE certificate; a persistent finite signature gives an explicit primitive PDEC schema; a column, displacement, endpoint or cofactor load returns through the ColumnCRT-to-PDEC/SAE absorption; and a drifting moving block enters CleanKLS/DLS, exact source entropy, or an explicit external KLS input. The canonical and generic-template escapes have already been blocked by the ExactUVSupport audit.

Thus the latest source-side micro-input is

```
ActualNoncanonicalCleanCoreSupportFailurePacketExclusion.
```

A clean-core packet is a positive-mass actual noncanonical full-S non-AP balanced block, in the same formal unit, below the exact u, v-support threshold, with no canonical import, no finite sparse witness, no persistent PDEC signature, no column or displacement defect, and no external or generic CleanKLS exit. The latest self-contained input basis is therefore

```
ActualNoncanonicalCleanCoreSupportFailurePacketExclusion
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This closes return completeness, not clean-core exclusion. The manuscript still lacks a proof that such clean-core packets are impossible, or a direct proof of the final capacity anti-atom.

The clean-core moving-atom sharp-input router then corrects one final over-strong formulation. It records

$$\texttt{clean\_core\_moving\_atom\_sharp\_boundary\_closed=true},$$
$$\texttt{clean\_core\_moving\_atom\_exclusion\_proved=false},$$
$$\texttt{actual\_final\_capacity\_antiatom\_proved=false},$$
$$\texttt{dstructure\_rankin\_independent\_acceptance\_completed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

Excluding every clean-core low-support packet is sufficient, but it is stronger than the final capacity anti-atom requires. Once the registered multiplier discipline is closed, the contrapositive of the recorded capacity inequality says that any final $M_{u,v}$-atom produces a low-support packet. The converse is not needed: a low-support packet that does not concentrate final capacity is not a terminal obstruction. Therefore the sharp source-side input is

```
ActualNoncanonicalCleanCoreMovingAtomExclusion.
```

Here a clean-core moving atom is a positive-mass actual noncanonical full-S non-AP balanced block, after all return tests, with some pair (u, v) satisfying

$$M_{u,v}/\sum_{u,v} M_{u,v} > \log^{-2A}.$$

The latest self-contained input basis is

```
ActualNoncanonicalCleanCoreMovingAtomExclusion
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This is only a sharp reformulation. It is not implied by formal WFD, Type decomposition, Fourier smoothing, or fixed-projection diffuse; the moving-delta obstruction remains.

The clean-core terminal normal-form router then removes the last naming ambiguity around that sharp input. It records

```
clean_core_terminal_normal_form_closed=true,
internal_exact_entropy_proved=false,
external_completed_kls_accepted=false,
dstructure_rankin_independent_acceptance_completed=false,
row_column_unconditional_closed=false.
```

The internal normal form of `ActualNoncanonicalCleanCoreMovingAtomExclusion` is exact clean-core source entropy: for every post-return clean-core moving block $b = (u, v)$,

$$\max_b M_b/\sum_b M_b \le \log^{-2A}.$$

The external alternative is not a generic DI/BFI or Kuznetsov citation; it is the completed, modulus-dependent full-S input `ModulusDependentCompletedFullSKLSInput`. Thus the latest conditional terminal input basis is

```
(ExactCleanCoreFullSNonAPWFDSourceEntropy
OR ModulusDependentCompletedFullSKLSInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The fully self-contained basis is

```
ExactCleanCoreFullSNonAPWFDSourceEntropy
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This closes the terminal input normal form only. It does not prove exact entropy, accept the completed KLS input, or complete the independent D-structure/Rankin gate.

Finally, the clean-core exact-entropy atom router rewrites the failure mode of that self-contained input. It records

$$\texttt{clean\_core\_exact\_entropy\_atom\_boundary\_closed=true},$$
$$\texttt{clean\_core\_terminal\_support\_incidence\_proved=false},$$
$$\texttt{exact\_clean\_core\_entropy\_proved=false},$$
$$\texttt{dstructure\_rankin\_independent\_acceptance\_completed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

If exact clean-core entropy fails, there is a clean-core moving atom. The registered multiplier discipline removes unrecorded Type/Fourier/fiber capacity as an explanation; packetization forces any support failure to be reproducible; and the return dichotomy routes every non-clean-core packet back to the named exits. The only remaining self-contained failure mode is a clean-core terminal support atom. A sufficient actionable proof package is therefore

```
CleanCoreTerminalSupportIncidenceTheorem
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Here `CleanCoreTerminalSupportIncidenceTheorem` means that every positive-mass post-return actual clean-core full-S non-AP WFD block has enough exact u/v support, inside the same formal unit, to absorb the divisor bound and all registered capacity multipliers. This theorem is not proved in the current corpus.

The clean-core support-incidence attack router then audits whether this theorem follows from the already closed ingredients. It records

$$\texttt{clean\_core\_support\_incidence\_attack\_boundary\_closed=true},$$
$$\texttt{clean\_core\_exact\_layer\_transfer\_proved=false},$$
$$\texttt{clean\_core\_terminal\_support\_incidence\_proved=false},$$
$$\texttt{external\_completed\_kls\_accepted=false},$$
$$\texttt{dstructure\_rankin\_independent\_acceptance\_completed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

The range threshold, registered capacity budget, and raw squarefree/Buchstab counting in thick blocks are no longer the obstruction. The naive factor-residue incidence bridge is still blocked by the internal h, ℓ, x, z fiber, and canonical RIW/Buchstab support cannot be imported into the noncanonical clean-core branch. The remaining internal input is

```
CleanCoreExactLayerAdmissionNonzeroTransferAndThinReturn.
```

The latest conditional basis is therefore

```
(CleanCoreExactLayerAdmissionNonzeroTransferAndThinReturn
OR ModulusDependentCompletedFullSKLSInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The fully self-contained basis replaces the parenthesized alternative by

```
CleanCoreExactLayerAdmissionNonzeroTransferAndThinReturn.
```

This input requires exact clean-core layer admission, nonzero transfer to the actual α/δ coefficients, and return of thin or rejected blocks to the named exits. It is still open.

The clean-core layer-transfer path router then separates the available canonical template from what is needed for the noncanonical clean-core residual. It records

```
clean_core_layer_transfer_path_boundary_closed=true,
clean_core_path_partition_proved=false,
clean_core_exact_layer_transfer_proved=false,
external_completed_kls_accepted=false,
dstructure_rankin_independent_acceptance_completed=false,
row_column_unconditional_closed=false.
```

The canonical RIW/Buchstab decision tree and source-provenance ledger are closed only on the canonical-source branch. The noncanonical clean-core residual must supply its own actual coefficient path partition, or leave through the completed KLS or named-return interfaces. The newest internal input is

```
CleanCoreExactCoefficientPathPartitionNoCancellationAndThinReturn.
```

It asks for a polylogarithmic partition of the actual clean-core α/δ coefficients into exact path signatures, nonzero same-path contribution without cancellation, and return of thin, over-budget, cancelling, or unidentified-source blocks to the named exits.

The clean-core path-source firewall router then moves this input to the coefficient source level. It records

```
clean_core_path_source_firewall_boundary_closed=true,
clean_core_precauchy_source_law_proved=false,
clean_core_path_partition_proved=false,
external_spectral_atom_accepted=false,
dstructure_rankin_independent_acceptance_completed=false,
row_column_unconditional_closed=false.
```

The path partition must be built on the pre-Cauchy actual clean-core α/δ coefficient formula. The canonical RIW/Buchstab decision tree is only a canonical-source template, and generic WFD well-factorability is blocked by the moving-delta model. The latest conditional basis is

```
(CleanCorePreCauchyCoefficientSourceLawAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The self-contained source-side input is

```
CleanCorePreCauchyCoefficientSourceLawAndReturn.
```

It requires an exact source formula before Cauchy, Type/Fourier and completion, plus polylog path signatures, same-path nonzero contribution, and named return for source-law failure, over-budget paths, or thin/rejected blocks.

The clean-core pre-Cauchy source-law atom router then compresses this source-law input to an origin-generation ledger. It records

$$\texttt{clean\_core\_precauchy\_source\_law\_atom\_boundary\_closed=true},$$
$$\texttt{origin\_generation\_ledger\_implication\_closed=true},$$
$$\texttt{clean\_core\_original\_coefficient\_generation\_ledger\_proved=false},$$
$$\texttt{clean\_core\_precauchy\_source\_law\_proved=false},$$
$$\texttt{external\_spectral\_atom\_accepted=false},$$
$$\texttt{dstructure\_rankin\_independent\_acceptance\_completed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

The compression theorem is that a pre-Cauchy source law is exactly an origin ledger for the actual clean-core α/δ coefficients in one formal unit before Cauchy, dispersion, Type/Fourier and completion. If that ledger lists every summand, branch key, u/v map, sign and local factor, then the branch keys give the exact path signatures, the registered K6/tail-label costs give the polylog path budget, and same-key nonzero or sign-split refinement gives no cancellation. Missing source entries, over-budget paths, thin/rejected blocks or unresolved cancellation must return to the named exits or to the external spectral input. The latest conditional basis is

```
(CleanCoreOriginalCoefficientGenerationLedgerAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The self-contained source-side input is

```
CleanCoreOriginalCoefficientGenerationLedgerAndReturn.
```

This input is not proved here. The router only shows that the former source-law input has no smaller honest self-contained evidence than the actual origin ledger; the canonical RIW/Buchstab ledger remains scoped to the canonical-source branch, and generic WFD well-factorability cannot replace it.

The clean-core origin-source admission router then compresses the origin ledger to a primitive source-constructor admission gate. It records

$$\texttt{clean\_core\_origin\_source\_admission\_boundary\_closed=true},$$
$$\texttt{constructor\_admission\_implies\_origin\_ledger=true},$$
$$\texttt{unregistered\_source\_return\_absorbed=false},$$
$$\texttt{clean\_core\_primitive\_source\_constructor\_admission\_proved=false},$$
$$\texttt{clean\_core\_original\_coefficient\_generation\_ledger\_proved=false},$$
$$\texttt{external\_spectral\_atom\_accepted=false},$$
$$\texttt{dstructure\_rankin\_independent\_acceptance\_completed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

An origin ledger is not an estimate; it first needs a source constructor. If the actual clean-core α/δ coefficients are emitted by a pre-Cauchy primitive constructor inside one formal unit, then the emitted-summand schema expands into the origin ledger. If no such constructor is registered, the object must return as an unregistered-source failure rather than remain a clean-core terminal. The latest conditional basis is

```
(CleanCorePrimitiveSourceConstructorAdmissionAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The self-contained source-side input is

```
CleanCorePrimitiveSourceConstructorAdmissionAndReturn.
```

This input is still open: the current material proves the canonical constructor only on the canonical-source branch and blocks generic WFD as a constructor, but it does not prove noncanonical clean-core constructor admission or absorption of every unregistered source.

The clean-core constructor source-class firewall then separates this admission gate by source class. It records

$$\texttt{constructor\_source\_class\_firewall\_boundary\_closed=true},$$
$$\texttt{source\_class\_partition\_closed=true},$$
$$\texttt{unregistered\_source\_return\_absorbed=true},$$
$$\texttt{actual\_noncanonical\_primitive\_constructor\_formula\_proved=false},$$
$$\texttt{clean\_core\_primitive\_source\_constructor\_admission\_proved=false},$$
$$\texttt{external\_spectral\_atom\_accepted=false},$$
$$\texttt{dstructure\_rankin\_independent\_acceptance\_completed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

The firewall has five classes. The canonical constructor is already closed but scoped to the canonical-source branch. A generic well-factorable template is not a constructor. Unregistered or mixed formal-unit sources return through Multiplicity/Stitching or the K7 formal-unit failure. The external spectral class remains the completed c-dependent residue-weight input. Thus the only self-contained source-side atom is

```
ActualNoncanonicalPrimitiveSourceConstructorFormulaAndReturn.
```

Equivalently, the latest conditional basis is

```
(ActualNoncanonicalPrimitiveSourceConstructorFormulaAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The source-side input asks for a pre-Cauchy definition of the actual noncanonical clean-core coefficients, a deterministic summand emitter for α/δ, branch keys, u/v maps, signs and local factors, and compatibility with the same formal unit used by the support, capacity and return ledgers. This formula is not proved here.

Finally, the external-lemma parameter-match router compares this remaining self-contained atom with the available DI/BFI/Kuznetsov tools. It records

$$\texttt{external\_lemma\_parameter\_match\_boundary\_closed=true},$$
$$\texttt{external\_lemmas\_match\_constructor\_formula=false},$$
$$\texttt{external\_lemmas\_close\_self\_contained\_remainder=false},$$
$$\texttt{external\_spectral\_atom\_accepted=false},$$
$$\texttt{actual\_noncanonical\_primitive\_constructor\_formula\_proved=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

The comparison is structural. DI/Kuznetsov and BFI act after coefficients, residue weights, or Kloosterman phases have already been produced: they give spectral or AP-distribution control. The current self-contained atom is earlier in the chain: it asks for the pre-Cauchy actual noncanonical source constructor and summand emitter. Therefore those external lemmas cannot replace the constructor formula. They remain relevant only to the external spectral branch, whose narrow target is

```
CDependentResidueWeightSpectralCancellationInput.
```

The conditional basis is unchanged:

```
(ActualNoncanonicalPrimitiveSourceConstructorFormulaAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The fully self-contained basis remains

```
ActualNoncanonicalPrimitiveSourceConstructorFormulaAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The clean-core reverse-provenance functor router then audits whether the older geometric and payment models can be used in the opposite direction. It records

$$
\begin{gathered}
\texttt{reverse\_provenance\_functor\_boundary\_closed=true},\\
\texttt{payment\_pushforward\_functoriality\_available=true},\\
\texttt{pushforward\_reverse\_uniqueness\_rejected=true},\\
\texttt{finite\_projection\_recovers\_gamma\_not\_source=true},\\
\texttt{constructor\_formula\_equivalent\_to\_registered\_fiber\_emitter=true},\\
\texttt{registered\_primitive\_prepushforward\_fiber\_emitter\_proved=false},\\
\texttt{actual\_noncanonical\_primitive\_constructor\_formula\_proved=false},\\
\texttt{row\_column\_unconditional\_closed=false}.
\end{gathered}
$$

The point is functorial rather than numerical. The actual payment graph Γ, its finite projections, cyclic arc restrictions, and transverse quotients are forward pushforwards or finite factors of a pre-Cauchy source. They preserve the registered formal unit and prevent source replacement, but they do not invert the pushforward: several primitive source disintegrations may have the same Γ. Thus one cannot recover a primitive constructor merely from the downstream payment graph.

Modulo the already closed return discipline, however, the constructor formula is equivalent to a more auditable input:

```
RegisteredPrimitivePrePushforwardFiberEmitterAndReturn.
```

Such an emitter must list, before Cauchy/dispersion, finite or polylogarithmically many primitive preimage summands over each positive-mass clean-core payment atom, with α/δ, branch keys, u/v maps, signs, local factors, formal-unit registration, and coefficient identities. Empty fibers, incompatible registrations, over-budget fibers, thin blocks, or unregistered sources must return through named exits. The latest conditional basis is therefore

```
(RegisteredPrimitivePrePushforwardFiberEmitterAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The fully self-contained basis becomes

```
RegisteredPrimitivePrePushforwardFiberEmitterAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The clean-core fiber-emitter field audit then separates the already available payment skeleton from the still missing signed coefficient lift. It records

```
fiber_emitter_field_audit_boundary_closed=true,
payment_fiber_skeleton_closed=true,
first_cover_payment_map_closed=true,
payment_count_identity_closed=true,
alpha_delta_coefficient_lift_proved=false,
polylog_branch_schema_proved=false,
registered_primitive_prepushforward_fiber_emitter_proved=false,
row_column_unconditional_closed=false.
```

The payment-level skeleton is now explicit: a completion y and a low hole c are sent by the first-cover rule $\mathrm{pay}(c, y)$ to a payment atom in Γ, and the count identity

$$\text{payment\_count} = \sum_{\text{phase}} M(\text{phase})\,|H_{\text{low}}(\text{phase})|$$

has been checked. ActualPaymentStitching also leaves no fourth payment-layer exit. What is not supplied by this skeleton is the signed clean-core primitive coefficient table. Thus the newest self-contained input is

```
ExactAlphaDeltaLiftForRegisteredPaymentFiberSkeletonAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The conditional basis is

```
(ExactAlphaDeltaLiftForRegisteredPaymentFiberSkeletonAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The alpha/delta disintegration router then rewrites this lift in its intrinsic form. It records

```
alpha_delta_disintegration_boundary_closed=true,
payment_base_map_closed=true,
lift_equivalent_to_signed_disintegration_dictionary=true,
canonical_decision_tree_template_scoped=true,
generic_or_unregistered_dictionary_blocked=true,
registered_alpha_delta_disintegration_dictionary_proved=false,
exact_alpha_delta_lift_proved=false,
row_column_unconditional_closed=false.
```

Let Φ denote the first-cover payment map from primitive source summands to the completion-hole payment skeleton. An exact alpha/delta lift is equivalent to a registered signed disintegration dictionary for the actual noncanonical source measure over Φ: one must list the signed preimage summands, branch keys, u/v maps, signs, local factors, variation/support budgets, and the push-forward identity. The canonical RIW/Buchstab decision tree is a model for such a dictionary only on the canonical branch; generic well-factorability or unregistered sources do not provide it. The latest self-contained basis is therefore

```
RegisteredAlphaDeltaDisintegrationDictionaryAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The conditional basis is

```
(RegisteredAlphaDeltaDisintegrationDictionaryAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Finally, the disintegration automaticity router removes a false hard point. It records

$$\texttt{disintegration\_automaticity\_boundary\_closed=true},$$
$$\texttt{discrete\_payment\_base\_map\_closed=true},$$
$$\texttt{signed\_fiber\_disintegration\_formal=true},$$
$$\texttt{pushforward\_identity\_is\_real\_gate=true},$$
$$\texttt{actual\_signed\_source\_measure\_phi\_compatibility\_budget\_proved=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

Once a signed source measure ν and the first-cover map Φ are given, the fiber decomposition

$$\nu = \sum_a \nu|_{\Phi^{-1}(a)}$$

is formal on the discrete payment skeleton. The real compatibility gate is not disintegration itself, but the proof that ν is the actual noncanonical alpha/delta source measure in the same formal unit, that $\Phi_*\nu$ equals the target payment-side coefficient, and that variation, absolute support, and branch-key budgets are registered. The latest self-contained basis is

```
ActualSignedSourceMeasurePhiCompatibilityBudgetAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The conditional basis is

```
(ActualSignedSourceMeasurePhiCompatibilityBudgetAndReturn
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This is a boundary refinement, not a proof of the signed source or budget identity.

The geometric Phi/budget bridge router then returns the line-covering, cylindrical, P-column anchor, layered wheel, and dynamic-capacity models to this same interface. It records

$$\texttt{geometric\_phi\_budget\_bridge\_boundary\_closed=true},$$
$$\texttt{geometric\_payment\_base\_available=true},$$
$$\texttt{geometry\_defines\_signed\_source\_measure=false},$$
$$\texttt{geometry\_proves\_phi\_pushforward\_identity=false},$$
$$\texttt{geometry\_supplies\_budget\_return\_shape=true},$$
$$\texttt{actual\_signed\_source\_measure\_phi\_compatibility\_budget\_proved=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

The geometric models provide the unsigned payment base map, candidate variation/support budgets, and named returns such as PDEC, SAE, ColumnCRT, and CleanKLS. They do not define the pre-Cauchy signed α/δ source measure, nor do they prove the coefficient identity $\Phi_*\nu =$ target payment-side coefficient. Hence the previous single input splits into two atoms:

```
ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND GeometricVariationBranchBudgetCertificateOrNamedReturn.
```

The latest conditional basis is therefore

```
((ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND GeometricVariationBranchBudgetCertificateOrNamedReturn)
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The fully self-contained basis is

```
ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND GeometricVariationBranchBudgetCertificateOrNamedReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This again refines the boundary rather than proving the two remaining atoms.

The geometric variation/branch budget router then attacks the second atom. It records

geometric_budget_attack_boundary_closed=true,
geometry_ledger_alphabet_closed=true,
dprc_analytic_capacity_bound_proved=false,
signed_variation_branch_lift_proved=false,
geometric_variation_branch_budget_certificate_proved=false,
row_column_unconditional_closed=false.

The line-cylinder, bottom-deficit, P-column anchor, dynamic promoted wheel, layered wheel, and input-atlas ledgers close the unsigned support and named-return alphabet. They do not control the total variation of the actual signed source, since mass may cancel after pushforward along Φ. The budget atom therefore splits as

```
GeometricVariationBranchBudgetCertificateOrNamedReturn
=> DPRCAlpha043CenteredDiscrepancyOrNamedLayerReturn
AND SignedGeometricLedgerVariationBranchLiftAndReturn.
```

The latest self-contained basis is

```
ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND DPRCAlpha043CenteredDiscrepancyOrNamedLayerReturn
AND SignedGeometricLedgerVariationBranchLiftAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The conditional basis is

```
((ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND DPRCAlpha043CenteredDiscrepancyOrNamedLayerReturn
AND SignedGeometricLedgerVariationBranchLiftAndReturn)
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This closes the budget-side interface, not the two new inputs.

The DPRC centered-discrepancy router then attacks the DPRC atom itself. It records

dprc_centered_discrepancy_boundary_closed=true,
rsm_identity_closed=true,
bes_compression_closed=true,
dprc_centered_discrepancy_input_proved=false,
row_column_unconditional_closed=false.

The relative-sieve margin identity reduces $T_Y < S_Y$ to an explicit model-gap/finite ledger and a centered positive-discrepancy bound. The BES compression reduces that bound to the exclusion of simultaneous high L^1 and high L^2 positive beta-bucket synchronization. If such a danger intersection is not directly excluded, the DLS route requires a named return:

```
PointLoad/ColumnCRT,  ShortWindow/SAE,  LowPhase/PDEC.
```

Thus

```
DPRCAlpha043CenteredDiscrepancyOrNamedLayerReturn
=> ExplicitModelGapAndFiniteDPRCLedger
AND BESDangerIntersectionExclusionOrDLSNamedReturn.
```

The latest self-contained basis is

```
ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND ExplicitModelGapAndFiniteDPRCLedger
AND BESDangerIntersectionExclusionOrDLSNamedReturn
AND SignedGeometricLedgerVariationBranchLiftAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The conditional basis is

```
((ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND ExplicitModelGapAndFiniteDPRCLedger
AND BESDangerIntersectionExclusionOrDLSNamedReturn
AND SignedGeometricLedgerVariationBranchLiftAndReturn)
OR CDependentResidueWeightSpectralCancellationInput)
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This closes the DPRC interface, not the finite/model-gap or BES-DLS proofs.

The BES-DLS named-return router then attacks the BES-DLS atom. It records

$$\texttt{bes\_dls\_named\_return\_boundary\_closed=true},$$
$$\texttt{dls\_kernel\_and\_danger\_algebra\_closed=true},$$
$$\texttt{bes\_dls\_named\_return\_input\_proved=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

The centered kernel, danger intersection, spike-bucket pigeonhole, and quadratic energy expansion are now registered. If the BES danger intersection occurs, it may no longer remain an unnamed synchronization failure. It must enter one of three exits:

```
PointLoad -> ColumnCRT/tail-anchor/displacement PDEC,
ShortWindow -> SAE/LocalSurvivor/persistent sparse PDEC,
LowPhase -> W-unit PDEC/new-layer PDEC/flat DLS.
```

Thus

```
BESDangerIntersectionExclusionOrDLSNamedReturn
=> DLSPointLoadColumnCRTBoundOrNamedReturn
AND DLSShortWindowSAEBoundOrNamedReturn
AND DLSLowPhasePDECNewLayerOrFlatDLSBound.
```

The latest self-contained basis is

```
ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND ExplicitModelGapAndFiniteDPRCLedger
AND DLSPointLoadColumnCRTBoundOrNamedReturn
AND DLSShortWindowSAEBoundOrNamedReturn
AND DLSLowPhasePDECNewLayerOrFlatDLSBound
AND SignedGeometricLedgerVariationBranchLiftAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This closes the named-return boundary, not the three microinputs.

The DLS LowPhase PDEC/flat-DLS router then attacks the LowPhase atom. It records

$$\texttt{dls\_lowphase\_boundary\_closed=true},$$
$$\texttt{lowphase\_input\_proved=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

LowPhase is not a single fixed-modulus law. If a fixed-wheel unit-class peak persists, it must return a W-unit PDEC. If a new Fourier layer concentrates in low dimension, it must return a new-layer PDEC. If both fixed-wheel and new-layer peaks dilute, the remaining mass is genuinely high-modulus and flat, so it must be absorbed by flat DLS/KLS. Thus

```
DLSLowPhasePDECNewLayerOrFlatDLSBound
=> DLSFixedWheelUnitPeakDilutionOrPDECReturn
AND DLSNewLayerFourierConcentrationPDECReturn
AND DLSFlatHighModLargeSieveAbsorption.
```

The latest self-contained basis is

```
ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND ExplicitModelGapAndFiniteDPRCLedger
AND DLSPointLoadColumnCRTBoundOrNamedReturn
AND DLSShortWindowSAEBoundOrNamedReturn
AND DLSFixedWheelUnitPeakDilutionOrPDECReturn
AND DLSNewLayerFourierConcentrationPDECReturn
AND DLSFlatHighModLargeSieveAbsorption
AND SignedGeometricLedgerVariationBranchLiftAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

This closes the LowPhase boundary, not the three new microinputs.

The new-layer external-lemma matching router then compares the most structural LowPhase microinput with the already registered external FullS–KLS and CleanKLS/DLS interfaces. It records

$$\texttt{direct\_external\_lemma\_closes\_newlayer=false},$$
$$\texttt{self\_contained\_newlayer\_closed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

The comparison is not a new numerical experiment. It is a parameter-level boundary check: external KLS/FullS lemmas estimate a prepared, completed, coefficient-given flat spectral block. The current new-layer atom sits one layer earlier. It must first prove that low-dimensional concentration on the added wheel fibre is an actual PDEC certificate, or that after deleting all such fibres the remaining object satisfies the clean flat-admission hypotheses. Hence

```
DLSNewLayerFourierConcentrationPDECReturn
=> ExactNewLayerFiberPDECProjectionMorphism
AND NewLayerNoConcentrationImpliesFlatAdmission.
```

The latest self-contained basis becomes

```
ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND ExplicitModelGapAndFiniteDPRCLedger
AND DLSPointLoadColumnCRTBoundOrNamedReturn
AND DLSShortWindowSAEBoundOrNamedReturn
AND DLSFixedWheelUnitPeakDilutionOrPDECReturn
AND ExactNewLayerFiberPDECProjectionMorphism
AND NewLayerNoConcentrationImpliesFlatAdmission
AND DLSFlatHighModLargeSieveAbsorption
AND SignedGeometricLedgerVariationBranchLiftAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

Thus the external theorem can attach only after a new-layer projection/admission morphism; it cannot be used directly to erase the self-contained new-layer obligation.

The new-layer PDEC projection router then attacks the first of these two atoms. It records

$$\texttt{newlayer\_projection\_slicer\_closed\_schema\_admission\_open},$$
$$\texttt{exact\_newlayer\_projection\_morphism\_closed=false},$$
$$\texttt{row\_column\_unconditional\_closed=false}.$$

Two deterministic sublemmas are closed. First, on the same full-period completion set C_P, lifting Q to rQ cannot create support outside the old projection. Second, a large $r \nmid h$ Fourier coefficient on the new fibre can be rotated and sliced by a layer-cake argument to produce a cyclic-arc or half-plane cap with comparable signed mass discrepancy. Hence

```
ExactNewLayerFiberPDECProjectionMorphism
=> RegisteredNewLayerPDECFormalUnitAndCapStableSchema.
```

The latest self-contained basis is therefore

```
ActualSignedSourceMeasurePhiIdentityForGeometricPaymentMapAndReturn
AND ExplicitModelGapAndFiniteDPRCLedger
AND DLSPointLoadColumnCRTBoundOrNamedReturn
AND DLSShortWindowSAEBoundOrNamedReturn
AND DLSFixedWheelUnitPeakDilutionOrPDECReturn
AND RegisteredNewLayerPDECFormalUnitAndCapStableSchema
AND NewLayerNoConcentrationImpliesFlatAdmission
AND DLSFlatHighModLargeSieveAbsorption
AND SignedGeometricLedgerVariationBranchLiftAndReturn
AND DStructureTailLog4FiniteRankinFullLedgerIndependentAcceptance.
```

The remaining obstruction is formal PDEC admission for the sliced cap: the same Ω, τ, w and phase map must be fixed, the cap must be non-tautological and cap-stable, and its lower and upper budgets must be computed in that same formal unit. □

Theorem 7.4 (Internal line-by-line audit)**.** *The author-side line-by-line audit has found no new item classified as a mathematical block. The remaining blocks are referee blocks: Prime Matrix final promotion requires independent acceptance of the D-structure/Tail-log4/finite-verification interface, and RH final promotion requires independent acceptance of all controlled exits in the RH contradiction-field manuscript.*

Proof. The internal audit matrix assigns one of four labels to each link. No link is a math-block.

The terminal entries PM-16 and RH-16 remain referee-blocks. An author-side check cannot replace independent referee verification. Therefore the internal audit is complete, but final unconditional theorem promotion is not. □

Remark 7.5 (No overclaim principle)**.** The monograph may state the shared contradiction-field method and the completed audit closures. It must not state unconditional proofs of RH or of the prime-matrix row/column theorem until the remaining structured inputs have independent acceptance.

Chapter 8

Current Status and Honest Boundary

The purpose of this monograph draft is to make the shared structure visible and auditable. It is not yet a final unconditional theorem manuscript. The precise statuses are maintained in `docs/monograph/claim-status-table.md`. Any future promotion to final theorem status requires independent verification of every reduced structured input and every controlled exit.